Hints and Solutions for A Gentle Introduction to the Art of Mathematics

《数学艺术入门》的提示和解答

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Chapter 1

Introduction and notation

1.1 Basic sets

Exercises — 1.1

1. 下表中每一行代表的量都属于至少一个列所代表的集合。如果该量属于该集合，请在表格对应条目中打勾。

	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	$\mathbb{C}$
17					
π					
$22/7$					
-6					
e^0					
$1+i$					
$\sqrt{3}$					
i^2					

请注意，这些集合是相互包含的，所以如果你确定一个数是自然数，那么它自动也是整数、有理数、实数和复数⋯⋯

2. 使用单向无限列表写出整数集 $\mathbb{Z}$。

   "单向无限列表"到底是什么意思？为了帮助你理解，请注意

   $$\ldots -3, -2, -1, 0, 1, 2, 3, \ldots$$

   是一个双向无限列表。

3. 判断下列各数是有理数还是无理数。

   (a) $5021.2121212121\ldots$

   (b) $0.2340000000\ldots$

   (c) $12.31331133311133331111\ldots$

   (d) π

   (e) $2.98765432198765432198765 4321\ldots$

   有理数，有理数，无理数，无理数，有理数

4. "看与说"序列的生成方法是：首先写下 1，然后迭代以下过程：观察前一项，说出其中每个整数的个数，然后写下你刚才说的内容。该序列的前几项是 $1, 11, 21, 1112, 3112, 211213, 312213, 212223, \ldots$。请评论由连接"看与说"序列得到的十进制数字所构成的数的有理性（或无理性）。

   $$0.1112111123112211213...$$

   实验一下！

   这个数若为有理数意味着什么？如果我们遇到"看与说"序列中的一个元素，它本身就是对自己的描述，那么从那一点开始，序列就会陷入一遍又一遍的重复（而通过连接"看与说"序列元素得到的数字将进入一个重复模式）。

5. 请描述十进制小数部分有限的有理数集合。(或者，你可以认为它们的小数部分以无限长的零串结尾。)

如果一个小数在 k 位后终止，你能想出如何从这个数得到一个整数吗？想想乘以某个数……

6. 找出 π, $3/7$, $\sqrt{2}$, $2/5$, $16/17$, $\sqrt{3}$, $1/2$ 和 $42/100$ 的前 20 位小数。将这些数的小数展开分类为：有限小数、循环小数或无明显模式。

对于这个问题，计算器通常是不够的。你应该尝试使用 CAS (计算机代数系统)。我推荐 Sage 计算机代数系统，因为它和这本书一样是免费的——你可以下载 sage 并在你自己的系统上运行，或者你也可以在不安装的情况下在线试用。请访问 www.sagemath.org。

你可以在 sage> 提示符下输入 `pi.N(digits=21)` 来让 sage 输出高精度的 π。

7. 考虑长除法过程。这个算法是否能让我们洞察为什么有理数的小数展开是有限或循环的？请解释。

你真的需要坐下来做一些长除法问题。在这个过程中，你是在什么时候突然意识到数字将要开始重复的？这个决定点是否总会出现？为什么？

8. 请论证为什么两个有理数的乘积仍然是有理数。

可以认为使用通常的两个分数相乘的法则是没有问题的：

$$\frac{a}{b} * \frac{c}{d} = \frac{ac}{bd}.$$

你如何知道结果确实是一个有理数？

9. 进行以下复数计算

   (a) $(4+3i)-(3+2i)$

   (b) $(1+i)+(1-i)$

   (c) $(1+i)\cdot(1-i)$

   (d) $(2-3i)\cdot(3-2i)$

   这些都很直接。如果你真的必须以某种方式验证它们，你可以使用像 Sage 这样的 CAS，或者你可以学习如何在你的图形计算器上输入复数。(在我的 TI-84 上，按 2nd 键然后按小数点键可以得到 i。)

10. 一个复数的共轭用上标星号表示，是通过将虚部取反形成的。因此，如果 $z=3+4i$，那么 z 的共轭是 $z^*=3-4i$。请论证为什么一个复数与其共轭的乘积是一个实数。(即，无论 z 是哪个复数，$z\cdot z^*$ 的虚部必然为 0。)

    这真的很容易，但一定要通用地去做。换句话说，不要只用例子——要用变量来表示复数的实部和虚部进行计算。

1.2 Definitions: Prime numbers

Exercises — 1.2

1. 找出下列整数的素因数分解。

   (a) 105

   (b) 414

   (c) 168

   (d) 1612

   (e) 9177

   先除掉明显的因数，以降低剩余问题的复杂性。第一个数能被 5 整除。接下来的三个数都是偶数。回想一下，一个数能被 3 整除，当且仅当它的各位数字之和能被 3 整除。

2. 使用埃拉托斯特尼筛法找出 100 以内的所有素数。

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

   在这种情况下，筛法中使用的素数只有 2、3、5 和 7。任何小于 100 且不是 2、3、5 或 7 的倍数的数在筛选过程中都不会被划掉。如果你对这个过程仍不清楚，可以尝试在网上搜索`"Sieve of Eratosthenes"+applet`，有几个交互式的小程序可以帮助你理解如何进行筛选。

3. 为了找出 400 以内的所有素数，需要用到的最大素数是多少?

   请记住，如果一个数分解为两个乘数，那么其中较小的那个将小于原数的平方根。

4. 描述完全平方数的素因数分解特征。

   写下一堆例子可能会有帮助。思考一下，一个数的素因数分解在平方后会发生怎样的变化。

5. 完成下表，该表与“当 p 是素数时，2^p-1 也是素数”这一猜想有关。

p	2^p-1	prime? (是素数吗?)	factors (因数)
2	3	yes (是)	1 and 3
3	7	yes (是)	1 and 7
5	31	yes (是)	
7	127		
11			

   你需要判断 $2^{11}-1=2047$ 是否是素数。如果你还不知道如何查阅第 15 页的素数表，这里有个提示：如果 2047 是素数，那么在第 20 行第 4 列的单元格中会有一个 7。

   要快速找到一个不太大的数的因数，可以使用图形计算器的“表格”功能。输入 y1=2047/X 并选择表格视图（2ND GRAPH）。然后，向下浏览条目，直到找到一个小数点后没有数字的。那个 X 就是能整除 2047 的数!

   一个更快的方法是在 Sage 中输入 `factor(2047)`。

6. 为“当 x 是自然数时，$x^2 - 31x + 257$ 的计算结果总是一个素数”这一猜想找一个反例。

   该多项式“产生素数的能力”之所以令人印象深刻，部分原因在于它会两次给出每个素数——一次在抛物线的下降分支上，一次在上升分支上。换句话说，该多项式在一组连续的自然数 0,1,2,...,N 上给出素数值，并且其图形抛物线的顶点正好位于该范围的中间。你可以通过考虑范围的另一端来算出 N 是多少：(-1)2 + 31 (-1) + 257 = 289（289 不是素数，你应该认出它是一个完全平方数。）

7. 使用“素数”的第二定义来证明 6 不是素数。换句话说，找出两个数（定义中出现的 a 和 b），使得 6 不是它们中任何一个的因数，但却是它们乘积的因数。

   嗯，我们知道 6 真的不是素数⋯⋯也许它的因数和这有点关系⋯⋯

8. 使用“素数”的第二定义来证明 35 不是素数。

   试试 $a = 2 \cdot 5$ 和 $b = 3 \cdot 7$。现在你来想出一对不同的数！

9. 一个被认为是正确的著名猜想（但尚无证明）是孪生素数猜想。如果一对素数相差 2，则称它们为孪生素数。例如，11 和 13 是孪生素数，431 和 433 也是。孪生素数猜想指出，存在无穷多对这样的孪生素数。请尝试论证为什么 3、5 和 7 是唯一的三生素数。

   这与其中一个数能被 3 整除有关。（为什么必须是这种情况？）如果那个数不是 3，那么你就知道它是合数。

10. 另一个著名的猜想，同样被认为是正确的——但尚未被证明，是哥德巴赫猜想。哥德巴赫猜想指出，每个大于 4 的偶数都是两个奇素

数之和。有一个定义在正整数上的函数 $g(n)$，称为哥德巴赫函数，它给出将一个给定数写成两个奇素数之和的不同方式的数量。例如 $g(10) = 2$，因为 $10 = 5 + 5 = 7 + 3$。因此，哥德巴赫猜想的另一个版本是，当 n 是大于 4 的偶数时，$g(n)$ 是正数。

画出 $6 \leq n \leq 20$ 时 $g(n)$ 的图像。

如果你不喜欢画图，一个包含 g(n) 值的表格就足够了。注意，我们不重复计算仅顺序不同的和。例如，16 = 13+3 和 11+5（以及 5+11 和 3+13），但 g(16)=2。

1.3 More scary notation

Exercises — 1.3

1. 下面这个句子中有多少个量词（以及是哪种量词)?“每个人都有某个朋友，认为自己了解一项运动的所有事情。”

   四个。

2. “每种金属元素在室温下都是固体”这句话是错误的。为什么?

   作为例外的元素的化学符号是 Hg，它代表“水银”，也被称为“液态银”或“快银”。

3. “对于每一对（不同的）实数，它们之间都存在另一个实数”这句话是正确的。为什么?

   想一想：有没有什么方法（用一个公式）可以找到一个位于另外两个数之间的数?

4. 写出你自己的包含四个量词的句子。一个句子中量词以 $(\forall\exists\forall\exists)$ 的顺序出现，另一个句子中以 $(\exists\forall\exists\forall)$ 的顺序出现。

   这里要靠你自己了。发挥创造力!

1.4 Definitions of elementary number theory

Exercises — 1.4

1. 如果一个整数 n 是偶数，并且因 n 是偶数而保证存在的整数 m 本身也是偶数，则称 n 是双偶数。0 是双偶数吗？前 3 个正的双偶数整数是什么？

   答案：是，0、4 和 8。

2. 在二进制表示法中，将一个整数除以二有一个有趣的解释：只需将数字向右移动一位。因此，$22 = 10110_2$ 除以二得到 1011_2，即 $8 + 2 + 1 = 11$。你如何从一个整数的二进制表示中识别出它是否是双偶数？

   偶数的个位是零。在双偶数的二进制表示中，还有哪一位数字也必须是零？

3. 整数的八进制表示法在位值计数中使用 8 的幂。八进制数的数字范围是 0 到 7，永远不会看到 8 或 9。你会如何用八进制数表示 8 和 9？紧跟在 777_8 后面的八进制数是什么？777_8 是哪个（十进制）数？

   八是 10_8，九是 11_8。问关于 777 的问题的要点是，（在八进制中）7 是类似于十进制中 9 的数字。因此 777_8 有点像 999_{10}，因为它们后面的数都写作 1000（尽管 1000_8 和 1000_{10} 肯定不相等！）

4. 一种从十进制转换为其他进制的方法叫做重复相除法。将数字除以目标进制并记录余数——然后将得到的商再除以该进制并记录余数。继续用该进制除以连续的商，直到商小于该进制。使用重复相除法将 3267 转换为 7 进制。通过 7 进制位值表示法的含义来检查你的答案。(例如 54321_7 表示 $5 \cdot 7^4 + 4 \cdot 7^3 + 3 \cdot 7^2 + 2 \cdot 7^1 + 1 \cdot 7^0$。)

$$3267 = 466 \cdot 7 + 5$$

$$466 = 66 \cdot 7 + 4$$

在每个阶段写出形如 $n = qd + r$ 的式子会很有帮助。前两个阶段应该看起来像这样

$$\begin{aligned} 3267 &= 466 \cdot 7 + 5 \\ 466 &= 66 \cdot 7 + 4 \end{aligned}$$

剩下的你来完成⋯⋯

5. 陈述一个关于偶数的八进制表示的定理。

一种可能性是模仿十进制的结果，即一个偶数总是以 0、2、4、6 或 8 结尾。

6. 在十六进制 (16 进制) 表示法中，需要 16 个 “数字”，普通数字用于 0 到 9，字母 A 到 F 用于表示 10 到 15 的单个符号。前 32 个自然数的十六进制表示为：1,2,3,4,5,6,7,8,9,A,B,C,D,E,F,10,11,12,13,14,15,16, 17,18,19,1A, 1B,1C,1D,1E,1F,20.

写出 AB 之后的 10 个十六进制数。

写出 FA 之后的 10 个十六进制数。

作为检验，AB 之后的第十个数是 B5。
FA 之后的第十个十六进制数是 104。

7. 在计算机科学最常用的三种进制之间转换时，我们可以将二进制作为“标准”进制，并使用查表法进行转换。每个八进制数字对应一个三位二进制数，每个十六进制数字对应一个四位二进制数。请完成下表。(作为检验，十六进制表格中 A 旁边的四位二进制数应该是 1010——这很好，因为 A 实际上是 10，所以如果你把它读作“ten-ten”，这是一个很好的助记方法。)

octal (八进制)	binary (二进制)
0	000
1	001
2	
3	
4	
5	
6	
7	

hexadecimal (十六进制)	binary (二进制)
0	0000
1	0001
2	0010
3	
4	
5	
6	
7	
8	
9	
A	
B	
C	
D	
E	
F	

这只是用二进制计数。记住一个健全性检查，即十六进制数字 A 在二进制中表示为 1010。($10_{10} = A_{16} = 1010_2$)

8. 使用上一题的表格进行以下转换。

   (a) 将 757_8 转换为二进制。

   (b) 将 1007_8 转换为十六进制。

   (c) 将 100101010110_2 转换为八进制。

   (d) 将 1111101000110101_2 转换为十六进制。

   (e) 将 $FEED_{16}$ 转换为二进制。

   (f) 将 $FFFFFF_{16}$ 转换为八进制。

前三个的答案:

$$757_8 = 111101111_2$$

$$1007_8 = 001000000111_2 = 001000000111_2 = 207_{16}$$

$$100101010110_2 = 4526_8$$

9. 尝试在不同数制之间进行以下转换:

   (a) 将 30（十进制）转换为二进制。

   (b) 将 69（十进制）转换为 5 进制。

   (c) 将 1222_3 转换为二进制。

   (d) 将 1234_7 转换为十进制。

   (e) 将 $EEED_{15}$ 转换为 12 进制。(使用 $\{1, 2, 3 \ldots 9, d, e\}$ 作为 12 进制的数字。)

   (f) 将 678_9 转换为十六进制。

Blah Blah Blah!!!!

天书!!!

10. 一个众所周知的事实是，如果一个数能被 3 整除，那么 3 也能整除该数（十进制）的各位数字之和。这个结论在 7 进制中成立吗？你认为这个结论在任何进制中都成立吗？

    这个效应是否与 10 比 9（3 的倍数）大 1 有关？

11. 假设必须将 340 磅的沙子装入容量为 50 磅的袋子中。请使用下取整或上取整符号写出所需袋子数量的表达式。

    七个 50 磅的袋子可以装 350 磅的沙子。它们也能装下 340 磅！

12. $$\left\lfloor \frac{n}{d} \right\rfloor < \left\lceil \frac{n}{d} \right\rceil$$

    对所有的整数 n 和 $d > 0$，该式成立吗？请支持你的主张。

    你必须尝试一堆例子。你应该确保你尝试的例子涵盖了所有可能性。提供反例（即证明该陈述通常是错误的）的数对相对稀少，所以要有系统地进行。

13. $\lceil \pi \rceil^2 - \lceil \pi^2 \rceil$ 的值是多少？

    $\pi^2 = 9.8696$

14. 假设符号 n, d, q, r 的含义如商余定理（见 GIAM 第 29 页）中所述。请用下取整和/或上取整符号，以 n 和 d 的形式写出 q 和 r 的表达式。

    我实在不忍心给你剧透这个，你真的需要自己解决这个问题。

15. 计算以下量:

    (a) $3 \mod 5$

    (b) $37 \mod 7$

    (c) $1000001 \mod 100000$

    (d) 6 div 6

    (e) 7 div 6

    (f) 1000001 div 2

    偶数题的答案是 2、1、500000。

16. 计算以下二项式系数:

    (a) $\binom{3}{0}$

    (b) $\binom{7}{7}$

    (c) $\binom{13}{5}$

    (d) $\binom{13}{8}$

    (e) $\binom{52}{7}$

    偶数题的答案是 1 和 1287。TI-84 可以计算二项式系数。使用的符号是 `nCr`(它被放在数字之间——即它是一个中缀运算符)。你可以在 `MATH` 下的 `PRB` 菜单中找到 `nCr` 作为第 3 项。在 sage 中,命令是 `binomial(n,k)`。

17. 一家冰淇淋店出售以下口味:巧克力、香草、草莓、咖啡、黄油山核桃、薄荷巧克力片和覆盆子。他们可以制作多少种不同的三球冰淇淋碗?

    你要从一个大小为七的集合中选择三样东西⋯⋯

1.5 Some algorithms of elementary number theory

Exercises — 1.5

1. 用输入 $n = 27$ 和 $d = 5$ 追踪除法算法的执行过程，每次遇到赋值语句时都将其写出来。

   这个特定的计算涉及多少次赋值?

   r=27

   q=0

   r=27-5=22

   q=0+1=1

   r=22-5=17

   q=1+1=2

   r=17-5=12

   q=2+1=3

   r=12-5=7

   q=3+1=4

   r=7-5=2

   q=4+1=5

   return r is 2 and q is 5.

   返回 r 为 2，q 为 5。

2. 找出下列数对的最大公约数（gcd）和最小公倍数（lcm）。

a	b	$\gcd(a,b)$	$\mathrm{lcm}(a,b)$
110	273		
105	42		
168	189		

对于这么小的数，你可以直接找出它们的素因数分解并加以利用，不过，通过追踪欧几里得算法来找出最大公约数，然后使用公式

$$\operatorname{lcm}((,a),b)=\frac{ab}{\gcd((,a),b)}$$

来练习你对该算法的理解，可能会很有用。

3. 在一般情况下（即因数分解可能包含素数的幂），用两个数的素因数分解来描述它们的最大公约数。

假设一个数的素因数分解包含 p^e，另一个包含 p^f，其中 $e<f$。p 的哪个次幂（p^e 还是 p^f）能同时整除这两个数?

4. 用输入 $a=3731$ 和 $b=2730$ 追踪欧几里得算法的执行过程，每次遇到调用除法算法的赋值语句时，都写出表达式 $a=qb+r$。(请使用实际数值!)

你得到的商应该在 1 和 2 之间交替出现。

1.6 Rational and irrational numbers

Exercises — 1.6

1. 有理逼近是一个被广泛研究的数学领域。其主要思想是找到非常接近给定无理数的有理数。例如，22/7 是一个众所周知的 π 的有理近似值。请找出 $\sqrt{2}, \sqrt{3}, \sqrt{5}$ 和 e 的良好有理近似值。

   一种方法是截断一个小数近似值，然后将其有理化。例如，$\sqrt{2}$ 大约是 1.4142，所以 14142/10000 是一个不错的近似值（尽管 7071/5000 自然更好，因为它涉及更小的数字）。

2. 我们在关于 n 进制的子章节中探讨的 n 进制表示法理论可以通过引入一个小数点（或许应根据进制重新命名）并在其右侧添加数字来扩展，以处理实数和有理数。例如，1.1011 是二进制表示法，代表 $1 \cdot 2^0 + 1 \cdot 2^{-1} + 0 \cdot 2^{-2} + 1 \cdot 2^{-3} + 1 \cdot 2^{-4}$ 或 $1 + \frac{1}{2} + \frac{1}{8} + \frac{1}{16} = 1\frac{11}{16}$。考虑二进制数 .1010010001000010000010000001...，这个数是有理数还是无理数？为什么？

   关于有理数具有有限或循环小数表示的规则是否也适用于其他进制？

3. 如果一个数 x 是偶数，很容易证明它的平方 x^2 也是偶数。本节中未被证明的引理要求我们从一个偶数的平方（x^2）出发，推断出未平方的数（x）也是偶数。进行一些数值实验来检验这个断言是否合理。你能给出一个证明它的论据吗?

   如果引理不成立会怎样? 你能想出如果我们有一个数 x，使得 x^2 是偶数但 x 本身是奇数，这意味着什么吗?

4. $\sqrt{2}$ 是无理数的证明可以被推广，以证明对于每一个素数 p，$\sqrt{p}$ 都是无理数。在这样的推广中，哪个陈述会等同于关于 x 和 x^2 奇偶性的引理?

   提示：说“x 是偶数”等同于说“x 能被 2 整除”。将 2 替换为 p，你就完成一半了……

5. 写一个证明，证明 $\sqrt{3}$ 是无理数。

   你基本上可以直接照搬 $\sqrt{2}$ 的论证过程。

1.7 Relations

Exercises — 1.7

1. 考虑从 1 到 10 的数字。请给出与整除关系相对应的这些数字的序偶集合。

   当第一个数“整除”第二个数时，一个序偶就“在”这个关系中。1“整除”任何数，2“整除”偶数，3“整除”3、6 和 9，等等。(另外，一个数总是“整除”它自己。)

2. 一个函数（或二元关系）的定义域是出现在第一坐标的数字集合。一个函数（或二元关系）的值域是出现在第二坐标的数字集合。考虑集合 $\{0,1,2,3,4,5,6\}$ 和函数 $f(x) = x^2 \pmod 7$。通过明确写出它所包含的有序对集合，将此函数表示为一个关系。这个函数的值域是什么?

$$\begin{aligned} f &= \{(0,0),(1,1),(2,4),(3,2),(4,2),(5,4),(6,1)\} \\ \mathrm{Rng}(f) &= \{0,1,2,4\} \end{aligned}$$

3. 下面这组有序对代表了 1 到 10 之间数字的什么关系?

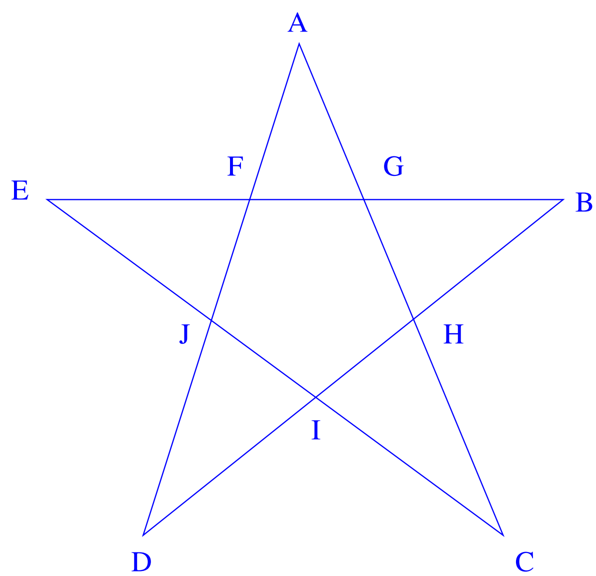
$$\begin{gathered}
\{(1,1),(1,2),(1,3),(1,4),(1,5),(1,6),(1,7),(1,8),(1,9),(1,10),\\
(2,2),(2,3),(2,4),(2,5),(2,6),(2,7),(2,8),(2,9),(2,10),\\
(3,3),(3,4),(3,5),(3,6),(3,7),(3,8),(3,9),(3,10),\\
(4,4),(4,5),(4,6),(4,7),(4,8),(4,9),(4,10),\\
(5,5),(5,6),(5,7),(5,8),(5,9),(5,10),\\
(6,6),(6,7),(6,8),(6,9),(6,10),\\
(7,7),(7,8),(7,9),(7,10),\\
(8,8),(8,9),(8,10),\\
(9,9),(9,10),\\
(10,10)\}
\end{gathered}$$

Less-than-or-equal-to

小于或等于

4. 画一个五角星，并标记所有 10 个点。这些标记中有 40 个三元组满足介于关系。请将它们列出来。

是的，嗯。四十个有点多... 我们来看看下图中水平线上的点（E、F、G 和 B）。涉及这四个点的三元组是：(E,F,G), (G,F,E), (E,F,B), (B,F,E), (E,G,B), (B,G,E), (F,G,B), (B,G,F)。



5. 绘制关系 $\{(x,y) \,|\, x,y \in \mathbb{R} \text{ and } y > x^2\}$ 的图像。

Is this the region above or below the curve $y = x^2$?

这是曲线 $y = x^2$ 上方的区域还是下方的区域?

6. 如果存在另一个函数 $g(x)$ 使得对于所有 x 的值都有 $g(f(x)) = x$，那么函数 $f(x)$ 就被称为可逆的。(通常，逆函数 $g(x)$ 会被记为 $f^{-1}(x)$。) 假设一个函数以关系的形式呈现给你——也就是说，你只得到一个序偶集合。你如何判断这个输入/输出序偶列表所代表的函数是否可逆？你如何（以有序对集合的形式）生成其逆函数？

   If f sends x to y, then we want f^{-1} to send y back to x. So the inverse just has the pairs in f reversed. When is the inverse going to fail to be a function?

   如果 f 将 x 映射到 y，那么我们希望 f^{-1} 将 y 映射回 x。所以逆函数只是将 f 中的序偶反转。在什么情况下，这个逆关系会不是一个函数？

7. What pairs are in “has color”?

   存在一个被称为“拥有颜色”的关系，它从集合 $F = \{\ ,\ ,\ ,\ \}$ 映射到集合 $C = \{\ ,\ ,\ ,\ \}$。哪些序偶属于“拥有颜色”这个关系？

   根据你个人对水果的经验水平，可能会有不同的答案。当然（橙子，橙色）会是其中一个序偶，但（橙子，绿色）也可能发生！

Chapter 2

Logic and quantifiers 逻辑与量词

2.1 Predicates and Logical Connectives 谓词与逻辑联结词

Exercises — 2.1

1. Design a digital logic circuit (using and, or & not gates) that implements an exclusive or.

   设计一个实现异或功能的数字逻辑电路（使用与门、或门和非门）。

   First, it's essential to know what is meant by the term ”exclusive or”. This is the interpretation that many people give to the word ”or” – where ”X or Y” means either X is true or Y is true, but that it isn't the case that both X and Y are true. This (wrong) understanding of what ”or” means is common because it is often the case that X and Y represent complimentary possibilities: old or new, cold or hot, right or wrong... The truth table for exclusive or (often written xor, pronounced ”ex-or”, symbolically it is usually $\oplus$) is

   首先，必须知道“异或”这个术语的含义。这是许多人对“或”这个词的解释——其中“X 或 Y”意味着 X 为真或 Y 为真，但 X 和 Y

不同时为真。这种对“或”的（错误）理解很普遍，因为 X 和 Y 通常代表互补的可能性：旧或新、冷或热、对或错……异或（常写作 xor，读作“ex-or”，符号通常是 $\oplus$）的真值表是：

X	Y	$X \oplus Y$
T	T	ϕ
T	ϕ	T
ϕ	T	T
ϕ	ϕ	ϕ

So it's true when one, or the other, but not both of its inputs are true. The upshot of the last sentence is that we can write $X \oplus Y \equiv (X \vee Y) \wedge \neg(X \wedge Y)$.

所以当它的一个输入为真，或另一个输入为真，但不是两个都为真时，它为真。最后一句话的要点是我们可以写成 $X \oplus Y \equiv (X \vee Y) \wedge \neg(X \wedge Y)$。

The above reformulation should help...

上面的重新表述应该会有帮助……

2. Consider the sentence "This is a sentence which does not refer to itself." which was given in the beginning of this chapter as an example. Is this sentence a statement? If so, what is its truth value?

   思考一下本章开头作为例子给出的句子"这是一个不指代自身的句子。"这个句子是一个命题吗？如果是，它的真值是什么？

   The only question in your mind, when deciding whether a sentence is a statement, should be "Does this thing have a definite truth value?" Well?

   Isn't it just plainly false?

   在判断一个句子是否是命题时，你脑中唯一的问题应该是"这个东西有确定的真值吗？"怎么样？它难道不就是明显错误的吗？

3. Consider the sentence "This sentence is false." Is this sentence a statement?

   思考一下句子"这句话是假的。"这个句子是一个命题吗？

   Try to justify why this sentence can't be either true or false.

   试着证明为什么这个句子既不能为真也不能为假。

4. Complete truth tables for each of the sentences $(A \wedge B) \vee C$ and $A \wedge (B \vee C)$. Does it seem that these sentences have the same logical content?

   完成句子 $(A \wedge B) \vee C$ 和 $A \wedge (B \vee C)$ 的真值表。这两个句子似乎有相同的逻辑内容吗?

   A tiny hint here: since the sentences involve 3 variables you'll need truth tables with 8 rows. Here's a template.

   这里有一个小提示：因为句子涉及 3 个变量，所以你需要 8 行的真值表。这是一个模板。

A	B	C	$(A \wedge B) \vee C$	$A \wedge (B \vee C)$
T	T	T		
T	T	ϕ		
T	ϕ	T		
T	ϕ	ϕ		
ϕ	T	T		
ϕ	T	ϕ		
ϕ	ϕ	T		
ϕ	ϕ	ϕ		

5. There are two other logical connectives that are used somewhat less commonly than $\vee$ and $\wedge$. These are the Scheffer stroke and the Peirce arrow – written $|$ and $\downarrow$, respectively — they are also known as NAND and NOR.

   还有另外两个逻辑联结词，它们的使用频率比 $\vee$ 和 $\wedge$ 稍低。它们是谢弗竖线和皮尔斯箭头——分别写作 $|$ 和 $\downarrow$——它们也被称为与非和或非。The truth tables for these connectives are:

   这些联结词的真值表是：

A	B	$A \mid B$
T	T	ϕ
T	ϕ	T
ϕ	T	T
ϕ	ϕ	T

and (和)

A	B	$A \downarrow B$
T	T	ϕ
T	ϕ	ϕ
ϕ	T	ϕ
ϕ	ϕ	T

   Find an expression for $(A \wedge \neg B) \vee C$ using only these new connectives (as well as negation and the variable symbols themselves).

   仅使用这些新的联结词（以及否定和变量符号本身）来找出 $(A \wedge \neg B) \vee C$ 的一个表达式。

   Sorry, I know this is probably the hardest problem in the chapter, but I'm (mostly) not going to help... Just one hint to help you get started: NAND and NOR are the negations of AND and OR (respectively) so, for example, $(X \wedge Y) \equiv \neg(A \mid B)$.

   抱歉，我知道这可能是本章最难的问题，但我（基本上）不打算帮忙……只给一个提示让你开始：与非（NAND）和或非（NOR）分别是与（AND）和或（OR）的否定，所以，例如，$(X \wedge Y) \equiv \neg(A \mid B)$。

6. The famous logician Raymond Smullyan devised a family of logical puzzles around a fictitious place he called "the Island of Knights and Knaves." The inhabitants of the island are either knaves, who always make false statements, or knights, who always make truthful statements. In the most famous knight/knave puzzle, you are in a room

which has only two exits. One leads to certain death and the other to freedom. There are two individuals in the room, and you know that one of them is a knight and the other is a knave, but you don't know which. Your challenge is to determine the door which leads to freedom by asking a single question.

著名的逻辑学家雷蒙德 · 斯穆里安围绕一个他称之为“骑士与无赖岛”的虚构地方设计了一系列逻辑谜题。岛上的居民要么是总是说谎的无赖，要么是总是说真话的骑士。在最著名的骑士/无赖谜题中，你身处一个只有两个出口的房间。一个通向死亡，另一个通向自由。房间里有两个人，你知道其中一个是骑士，另一个是无赖，但你不知道谁是谁。你的挑战是通过问一个问题来确定通向自由的门。

Ask one of them what the other one would say to do.

问其中一个人，另一个人会让你走哪扇门。

2.2 Implication 蕴涵

Exercises — 2.2

1. The transitive property of equality says that if $a = b$ and $b = c$ then $a = c$. Does the implication arrow satisfy a transitive property? If so, state it.

   等式的传递性表明，如果 $a = b$ 且 $b = c$，那么 $a = c$。蕴含箭头是否满足传递性？如果满足，请陈述之。

   I sometimes like to rephrase the implication $X \implies Y$ as "X's truth forces Y to be true." Does that help? If we know that X being true forces Y to be true, and we also know that Y being true will force Z to be true, what can we conclude?

   我有时喜欢将蕴涵 $X \implies Y$ 改写为"X 的真迫使 Y 为真。"这有帮助吗？如果我们知道 X 为真迫使 Y 为真，并且我们也知道 Y 为真将迫使 Z 为真，我们能得出什么结论？

2. Complete truth tables for the compound sentences $A \implies B$ and $\neg A \vee B$.

   完成复合句 $A \implies B$ 和 $\neg A \vee B$ 的真值表。

   You should definitely be able to do this one on your own, but anyway, here's an outline of the table:

   你绝对应该能自己完成这个，但无论如何，这里是表格的框架：

A	B	$A \implies B$	$\neg A \vee B$
T	T		
T	ϕ		
ϕ	T		
ϕ	ϕ		

3. Complete a truth table for the compound sentence $A \implies (B \implies C)$ and for the sentence $(A \implies B) \implies C$. What can you conclude about conditionals and the associative property?

   完成复合句 $A \implies (B \implies C)$ 和句子 $(A \implies B) \implies C$ 的真值表。关于条件句和结合律，你能得出什么结论?

   No help on this one other than to say that the associative property **does not** hold for implications.

   除了说明蕴涵**不**满足结合律之外，这个问题没有其他帮助。

4. Determine a sentence using the *and* connector ($\wedge$) that gives the negation of $A \implies B$.

   使用与联结词 ($\wedge$) 确定一个句子，该句子是 $A \implies B$ 的否定。

   Hmmm…This will seem like a strange hint, but if you were to hear a kid at the playground say “Oh yeah? Well, I did call your mom a fatty and you still haven’t clobbered me! Owww! OWWW!!! Stop hitting me!!”

   嗯……这提示可能有点奇怪，但如果你在操场上听到一个孩子说：“哦是吗？我确实叫了你妈妈胖子，但你还没揍我！哎哟！嗷！！！别打了！！”

   What conditional sentence was he attempting to negate?

   他试图否定的是哪个条件句？

5. Rewrite the sentence “Fix the toilet or I won’t pay the rent!” as a conditional.

   将句子“修好马桶，否则我不付房租！”改写为条件句。

   The way I see it there are eight possible ways to arrange ”You fix the toilet” and ”I’ll pay the rent” (or their respective negations) around an implication arrow. Here they all are. You decide which one sounds best. If you fix the toilet, then I’ll pay the rent.
   If you fix the toilet, then I won’t pay the rent.
   If you don’t fix the toilet, I’ll pay the rent.
   If you don’t fix the toilet, then I won’t pay the rent.
   If I payed the rent, then you must have fixed the toilet.
   If I payed the rent, then you must not have fixed the toilet.
   If I didn’t pay the rent, then you must have fixed the toilet.
   If I didn’t pay the rent, then you must not have fixed the toilet.

在我看来，有八种可能的方式可以将“你修好马桶”和“我会付房租”（或它们各自的否定）围绕一个蕴涵箭头排列。这里是所有的方式。你来决定哪一个听起来最好。如果你修好马桶，那么我会付房租。
如果你修好马桶，那么我不会付房租。
如果你不修好马桶，我会付房租。
如果你不修好马桶，那么我不会付房租。
如果我付了房租，那么你一定修好了马桶。
如果我付了房租，那么你一定没有修好马桶。
如果我没有付房租，那么你一定修好了马桶。
如果我没有付房租，那么你一定没有修好马桶。

Some of those are truly strange...

其中一些真的很奇怪⋯⋯

6. Why is it that the sentence “If pigs can fly, I am the king of Mesopotamia.” true?

   为什么“如果猪会飞，我就是美索不达米亚的国王”这句话是真的?

   Unless we’re talking about some celebrity bringing their pet Vietnamese pot-bellied pig into first class with them, or possibly a catapult of some type... The antecedent (the if part) is false, so Yay! I AM the king of Mesopotamia!! Whoo-hooh! What? I’m not? Oh. But the if-then sentence is true. Bummer.

   除非我们谈论的是某个名人把他们的宠物越南大肚猪带进头等舱，或者可能是某种弹射器⋯⋯前件（如果部分）是假的，所以耶！我就是美索不达米亚的国王！！哇哦！什么？我不是？哦。但这个如果-那么句子是真的。真扫兴。

7. Express the statement $A \implies B$ using the Peirce arrow and/or the Scheffer stroke. (See Exercise 5 in the previous section.)

使用皮尔斯箭头和/或谢弗竖线来表达陈述 $A \implies B$。（参见上一节的练习 5。）

You'll want to use $|$, the Scheffer stroke, aka NAND, because it's truth table contains three T's and one ϕ – you'll just need to figure out which of its inputs to negate so as to make that one ϕ occur in the second row of the table instead of the first.

你会想使用 $|$，即谢弗竖线，也就是与非门，因为它的真值表包含三个 T 和一个 ϕ——你只需要弄清楚应该否定哪个输入，以使那个唯一的 ϕ 出现在表的第二行而不是第一行。

8. Find the contrapositives of the following sentences.

   找出下列句子的逆否命题。

   (a) If you can't do the time, don't do the crime.

       如果你承担不了后果，就不要做坏事。

   (b) If you do well in school, you'll get a good job.

       如果你在学校表现好，你就会找到一份好工作。

   (c) If you wish others to treat you in a certain way, you must treat others in that fashion.

       如果你希望别人以某种方式对待你，你必须以那种方式对待别人。

   (d) If it's raining, there must be clouds.

       如果下雨，就一定有云。

   (e) If $a_n \leq b_n$, for all n and $\sum_{n=0}^{\infty} b_n$ is a convergent series, then $\sum_{n=0}^{\infty} a_n$ is a convergent series.

       如果对于所有的 n 都有 $a_n \leq b_n$，并且 $\sum_{n=0}^{\infty} b_n$ 是一个收敛级数，那么 $\sum_{n=0}^{\infty} a_n$ 也是一个收敛级数。

(a) If you do the crime, you must do the time.

如果你做了坏事，你就必须承担后果。

(b) If you don't have a good job, you must've done poorly in school.

如果你没有一份好工作，你一定是在学校表现不好。

(c) If you don't treat others in a certain way, you can't hope for others to treat you in that fashion,

如果你不以某种方式对待他人，你就不能指望他人以那种方式对待你。

(d) If there are no clouds, it can't be raining.

如果没有云，就不可能下雨。

(e) If $\sum_{n=0}^{\infty} a_n$ is not a convergent series, then either $a_n \leq b_n$, for some n or $\sum_{n=0}^{\infty} b_n$ is not a convergent series.

如果 $\sum_{n=0}^{\infty} a_n$ 不是一个收敛级数，那么要么对于某个 n 有 $a_n > b_n$[注意：原文此处应为 $>$], 要么 $\sum_{n=0}^{\infty} b_n$ 不是一个收敛级数。

9. What are the converse and inverse of "If you watch my back, I'll watch your back."?

"如果你掩护我，我也会掩护你" 的逆命题和否命题是什么?

The converse is "If I watch your back, then you'll watch my back." (Sounds a little dopey doesn't it – likes its sort of a wishful thinking...) The inverse is "If you don't watch my back, then I won't watch your back." (Sounds less vapid, but it means the same thing...)

逆命题是"如果我掩护你，那么你也会掩护我。"（听起来有点傻，不是吗——有点像一厢情愿……）否命题是"如果你不掩护我，那么我也不会掩护你。"（听起来不那么空洞，但意思是一样的……）

10. The integral test in Calculus is used to determine whether an infinite series converges or diverges: Suppose that $f(x)$ is a positive, decreasing, real-valued function with $\lim_{x\longrightarrow\infty} f(x) = 0$, if the improper integral $\int_0^\infty f(x)$ has a finite value, then the infinite series $\sum_{n=1}^\infty f(n)$ converges. The integral test should be envisioned by letting the series correspond to a right-hand Riemann sum for the integral, since the function is decreasing, a right-hand Riemann sum is an underestimate for the value of the integral, thus

    微积分中的积分判别法用于判断一个无穷级数是收敛还是发散：假设 $f(x)$ 是一个正的、递减的实值函数，且 $\lim_{x\longrightarrow\infty} f(x) = 0$，如果反常积分 $\int_0^\infty f(x)$ 有一个有限值，那么无穷级数 $\sum_{n=1}^\infty f(n)$ 收敛。积分判别法可以通过将级数对应于积分的右黎曼和来形象地理解，因为函数是递减的，右黎曼和是积分值的下估计，因此

$$\sum_{n=1}^{\infty} f(n) < \int_0^\infty f(x).$$

    Discuss the meanings of and (where possible) provide justifications for the inverse, converse and contrapositive of the conditional statement in the integral test.

    讨论积分判别法中条件陈述的否命题、逆命题和逆否命题的含义，并在可能的情况下为其提供理由。

    The inverse says – if the integral isn't finite, then the series doesn't converge. You can cook-up a function that shows this to be false by (for example) creating one with vertical asymptotes that occur in between the integer x-values. Even one such pole can be enough to make the integral go infinite. The converse says that if the series converges, the integral must be finite. The counter-example we just discussed would work here too.

    否命题说——如果积分不是有限的，那么级数就不收敛。你可以构造

一个函数来证明这是错误的，例如，创建一个在整数 x 值之间有垂直渐近线的函数。即使只有一个这样的极点也足以使积分变为无穷大。逆命题说，如果级数收敛，那么积分必须是有限的。我们刚才讨论的反例在这里也适用。

The contrapositive says that if the series doesn't converge, then the integral must not be finite. If we were allowed to use discontinuous functions, it isn't too hard to come up with an f that actually has zero area under it – just make f be identically zero except at the integer x-values where it will take the same values as the terms of the series. But wait, the function we just described isn't "decreasing" – which is probably why that hypothesis was put in there!

逆否命题说，如果级数不收敛，那么积分必定不是有限的。如果我们被允许使用不连续函数，不难想出一个其下方实际面积为零的 f ——只需让 f 在除了整数 x 值之外的地方恒为零，在整数 x 值处取与级数项相同的值即可。但是等等，我们刚才描述的函数不是"递减"的——这可能就是为什么把那个假设放在那里的原因！

11. On the Island of Knights and Knaves (see page 67) you encounter two individuals named Locke and Demosthenes. Locke says, "Demosthenes is a knave."
Demosthenes says "Locke and I are knights."

在骑士与无赖岛上（见第 67页），你遇到了两个名叫洛克和德摩斯梯尼的人。洛克说："德摩斯梯尼是个无赖。"
德摩斯梯尼说："洛克和我是骑士。"

Who is a knight and who a knave?

谁是骑士，谁是无赖？

Could Demosthenes be telling the truth?

德摩斯梯尼可能在说真话吗？

2.3 Logical equivalences 逻辑等价

Exercises — 2.3

1. There are 3 operations used in basic algebra (addition, multiplication and exponentiation) and thus there are potentially 6 different distributive laws. State all 6 "laws" and determine which 2 are actually valid. (As an example, the distributive law of addition over multiplication would look like $x + (y \cdot z) = (x + y) \cdot (x + z)$, this isn't one of the true ones.)

   基础代数中使用 3 种运算（加法、乘法和指数运算），因此可能存在 6 种不同的分配律。请陈述所有 6 种"定律"，并确定哪 2 种是实际有效的。（举个例子，加法对乘法的分配律看起来像 $x + (y \cdot z) = (x + y) \cdot (x + z)$，这不是一个成立的定律。）

   These "laws" should probably be layed-out in a big 3 by 3 table. Such a table would of course have 9 cells, but we won't be using the cells on the diagonal because they would involve an operation distributing over itself. (That can't happen, can it?) I'm going to put a few of the entries in, and you do the rest.

   这些"定律"或许应该在一个大的 3x3 表格中列出。这样的表格当然有 9 个单元格，但我们不会使用对角线上的单元格，因为那将涉及一个运算对自己进行分配。(那不可能发生，是吗?) 我将填入一些条目，剩下的由你来完成。

	$+$	$*$	$\wedge$
$+$	$\emptyset$	$x+(y*z)$ $=(x+y)*(x+z)$	$x+(y^z)$ $=(x+y)^{(x+z)}$
$*$	$x*(y+z)$ $=(x*y)+(x*z)$	$\emptyset$	
$\wedge$			$\emptyset$

2. Use truth tables to verify or disprove the following logical equivalences.

   使用真值表来验证或证伪以下逻辑等价式。

   (a) $(A \wedge B) \vee B \ \cong \ (A \vee B) \wedge B$

   (b) $A \wedge (B \vee \neg A) \ \cong \ A \wedge B$

   (c) $(A \wedge \neg B) \vee (\neg A \wedge \neg B) \cong (A \vee \neg B) \wedge (\neg A \vee \neg B)$

   (d) The absorption laws.

   吸收律。

   You should be able to do these on your own.

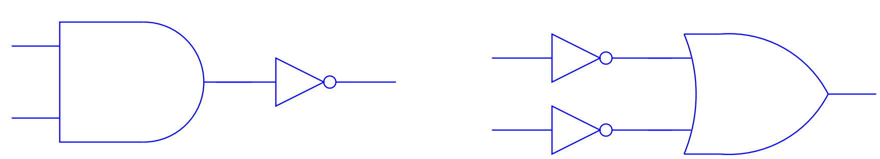
   你应该能自己完成这些。

3. Draw pairs of related digital logic circuits that illustrate DeMorgan's laws.

   画出相关的数字逻辑电路对来说明德摩根定律。

   Here's the pair that shows the negation of an AND is the same as the OR of the same inputs negated.

   这里有一对电路, 显示了一个"与"运算的否定等同于其输入的否定进



   行"或"运算。

4. Find the negation of each of the following and simplify as much as possible.

   找出下列各式子的否定，并尽可能地化简。

(a) $(A \vee B) \iff C$

(b) $(A \vee B) \implies (A \wedge B)$

Neither of these is particularly amenable to simplification. Nor, perhaps, is it readily apparent what "simplify" means in this context! My interpretation is that we should look for a logically equivalent expression using the fewest number of operators and if possible *not* using the more complicated operators ($\implies$ and $\iff$). However, if we try to rewrite the first statement's negation using only $\wedge$, $\vee$ and $\neg$ we get things that look a lot more complicated than $(A \vee B) \iff \neg C$ – the quick way to negate a biconditional is simply to negate one of its parts. The second statement's negation turns out to be the same thing as exclusive or, so a particularly simple response would be to write $A \oplus B$ although that feels a bit like cheating, so maybe we should answer with $(A \vee B) \wedge \neg(A \wedge B)$ – but that answer is what we would get by simply applying the rule for negating a conditional and doing no further simplification.

这两个式子都不太容易化简。而且，在这种情况下，"化简" 的含义可能也不是很明显！我的理解是，我们应该寻找一个使用最少运算符的逻辑等价表达式，并且如果可能的话，不使用更复杂的运算符（ $\implies$ 和 $\iff$ ）。然而，如果我们尝试只使用 $\wedge$、$\vee$ 和 $\neg$ 来重写第一个陈述的否定，我们会得到比 $(A \vee B) \iff \neg C$ 复杂得多的东西——否定一个双条件句的快捷方法是简单地否定它的其中一部分。第二个陈述的否定结果与异或相同，所以一个特别简单的回答是写成 $A \oplus B$，尽管这感觉有点像作弊，所以也许我们应该用 $(A \vee B) \wedge \neg(A \wedge B)$ 来回答——但这个答案只是简单应用否定条件句的规则而没有做进一步化简得到的结果。

5. Because a conditional sentence is equivalent to a certain disjunction, and because DeMorgan's law tells us that the negation of a disjunc-

tion is a conjunction, it follows that the negation of a conditional is a conjunction. Find denials (the negation of a sentence is often called its "denial") for each of the following conditionals.

因为一个条件句等价于某个析取，又因为德摩根定律告诉我们一个析取的否定是一个合取，所以一个条件句的否定是一个合取。请为以下每个条件句找出其否定形式（一个句子的否定通常被称为它的"denial"）。

(a) "If you smoke, you'll get lung cancer."
"如果你吸烟，你会得肺癌。"
(b) "If a substance glitters, it is not necessarily gold."
"如果一个物质闪闪发光，它不一定是金子。"
(c) "If there is smoke, there must also be fire."
"有烟必有火。"
(d) "If a number is squared, the result is positive."
"如果一个数被平方，结果是正数。"
(e) "If a matrix is square, it is invertible."
"如果一个矩阵是方阵，那么它是可逆的。"

(a) "You smoke and you haven't got lung cancer."
"你吸烟但你没有得肺癌。"
(b) "A substance glitters and it is necessarily gold."
"一个物质闪闪发光并且它必然是金子。"
(c) "There is smoke,and there isn't fire."
"有烟但没有火。"
(d) "A number is squared, and the result is not positive."
"一个数被平方，但结果不是正数。"

(e) “A matrix is square and it is not invertible.”

“一个矩阵是方阵但它不是可逆的。”

6. The so-called "ethic of reciprocity" is an idea that has come up in many of the world's religions and philosophies. Below are statements of the ethic from several sources. Discuss their logical meanings and determine which (if any) are logically equivalent.

   所谓的"互惠伦理"是在世界上许多宗教和哲学中都出现过的一个思想。以下是来自几个不同来源的关于该伦理的陈述。请讨论它们的逻辑含义，并确定哪些（如果有的话）在逻辑上是等价的。

   (a) "One should not behave towards others in a way which is disagreeable to oneself." Mencius Vii.A.4 (Hinduism)

   "己所不欲，勿施于人。" ——《孟子》Vii.A.4 (印度教)

   (b) "None of you [truly] believes until he wishes for his brother what he wishes for himself." Number 13 of Imam "Al-Nawawi's Forty Hadiths." (Islam)

   "你们中没有一个人 [真正] 信仰，直到他为他的兄弟所希望的，如同为他自己所希望的一样。" ——伊玛目"安-纳瓦维的四十段圣训"第 13 段 (伊斯兰教)

   (c) "And as ye would that men should do to you, do ye also to them likewise." Luke 6:31, King James Version. (Christianity)

   "你们愿意人怎样待你们，你们也要怎样待人。" ——《路加福音》6:31, 英王钦定本 (基督教)

   (d) "What is hateful to you, do not to your fellow man. This is the law: all the rest is commentary." Talmud, Shabbat 31a. (Judaism)

   "你所憎恶的事，不要对你的同胞做。这就是律法：其余的都是注释。" ——《塔木德》，安息日篇 31a (犹太教)

   (e) "An it harm no one, do what thou wilt" (Wicca)

   "只要不伤害任何人，就做你想做的事。" (威卡教)

   (f) "What you would avoid suffering yourself, seek not to impose on others." (the Greek philosopher Epictetus – first century A.D.)

"你不愿自己承受的痛苦，就不要强加于他人。" (希腊哲学家爱比克泰德——公元一世纪)

(g) "Do not do unto others as you expect they should do unto you. Their tastes may not be the same." (the Irish playwright George Bernard Shaw – 20th century A.D.)

"不要按照你期望别人对待你的方式去对待别人。他们的品味可能不一样。" (爱尔兰剧作家萧伯纳——公元二十世纪)

The ones from Wicca and George Bernard Shaw are just there for laughs. For the remainder, you may want to contrast how restrictive they seem. For example the Christian version is (in my opinion) a lot stronger than the one from the Talmud – "treat others as you would want to be treated" restricts your actions both in terms of what you would like done to you and in terms of what you wouldn't like done to you; "Don't treat your fellows in a way that would be hateful to you." is leaving you a lot more freedom of action, since it only prohibits you from doing those things you wouldn't want done to yourself to others. The Hindus, Epictetus and the Jews (and the Wiccans for that matter) seem to be expressing roughly the same sentiment – and promoting an ethic that is rather more easy for humans to conform to! From a logical perspective it might be nice to define open sentences:

来自威卡教和萧伯纳的引言只是为了博君一笑。对于其余的，你可能想对比一下它们的限制性有多强。例如，基督教的版本（在我看来）比《塔木德》的版本要强得多——"你希望别人怎样待你，你也要怎样待人"在你希望别人对你做的事和你希望别人不要对你做的事两方面都限制了你的行为；"不要以你所憎恶的方式对待你的同胞"则给了你更多的行动自由，因为它只禁止你做那些你不希望别人对自己做的事。印度教徒、爱比克泰德和犹太教徒（以及威卡教徒）似乎在表达大致相同的情感——并提倡一种人类更容易遵守的伦理！从逻辑的角度来看，定义以下开放句可能会很好：

$$W(x,y) \;=\; \text{“x would want y done to him.”}$$

$$W(x,y) \;=\; \text{“x 希望 y 这样对他。”}$$

$$N(x,y) \;=\; \text{“x would not want y done to him.”}$$

$$N(x,y) \;=\; \text{“x 不希望 y 这样对他。”}$$

$$D(x,y) \;=\; \text{“do y to x.”}$$

$$D(x,y) \;=\; \text{“对 x 做 y。”}$$

$$DD(x,y) \;=\; \text{“don't do y to x.”}$$

$$DD(x,y) \;=\; \text{“不对 x 做 y。”}$$

In which case, the aphorism from Luke would be

在这种情况下，来自路加福音的格言将是

$$(W(you,y) \implies D(others,y)) \wedge (N(you,y) \implies DD(others,y))$$

7. You encounter two natives of the land of knights and knaves. Fill in an explanation for each line of the proofs of their identities.

   你遇到了骑士与无赖之地的两位土著。请为证明他们身份的每一行填入解释。

(a) Natasha says, "Boris is a knave."
Boris says, "Natasha and I are knights."

娜塔莎说："鲍里斯是个无赖。"
鲍里斯说："娜塔莎和我是骑士。"

Claim: Natasha is a knight, and Boris is a knave.

主张： 娜塔莎是骑士，鲍里斯是无赖。

Proof: If Natasha is a knave, then Boris is a knight.
If Boris is a knight, then Natasha is a knight.
Therefore, if Natasha is a knave, then Natasha is a knight.
Hence Natasha is a knight.
Therefore, Boris is a knave.
如果娜塔莎是无赖，那么鲍里斯是骑士。
如果鲍里斯是骑士，那么娜塔莎是骑士。
因此，如果娜塔莎是无赖，那么娜塔莎是骑士。
所以娜塔莎是骑士。
因此，鲍里斯是无赖。

Q.E.D.

(b) Bonaparte says "I am a knight and Wellington is a knave."
Wellington says "I would tell you that B is a knight."
波拿巴说："我是骑士，威灵顿是无赖。"
威灵顿说："我会告诉你 B 是骑士。"

Claim: Bonaparte is a knight and Wellington is a knave.

主张： 波拿巴是骑士，威灵顿是无赖。

Proof: Either Wellington is a knave or Wellington is a knight.
If Wellington is a knight it follows that Bonaparte is a knight.
If Bonaparte is a knight then Wellington is a knave.
So, if Wellington is a knight then Wellington is a knave (which is impossible!)
Thus, Wellington is a knave.

Since Wellington is a knave, his statement “I would tell you that Bonaparte is a knight” is false.
So Wellington would in fact tell us that Bonaparte is a knave.
Since Wellington is a knave we conclude that Bonaparte is a knight.
Thus Bonaparte is a knight and Wellington is a knave (as claimed).

威灵顿要么是无赖，要么是骑士。
如果威灵顿是骑士，那么波拿巴是骑士。
如果波拿巴是骑士，那么威灵顿是无赖。
所以，如果威灵顿是骑士，那么威灵顿是无赖（这是不可能的！）。
因此，威灵顿是无赖。
既然威灵顿是无赖，他的陈述“我会告诉你波拿巴是骑士”是假的。
所以威灵顿实际上会告诉我们波拿巴是无赖。
既然威灵顿是无赖，我们得出结论波拿巴是骑士。
因此波拿巴是骑士，威灵顿是无赖（如主张所述）。

Q.E.D.

Here's the second one:

这是第二个：

Proof: Either Wellington is a knave or Wellington is a knight.

威灵顿要么是无赖，要么是骑士。

It's either one thing or the other!

非此即彼！

If Wellington is a knight it follows that Bonaparte is a knight.

如果威灵顿是骑士，那么波拿巴是骑士。

That's what he said he would tell us and if he's a knight we can trust him.

那是他说的他会告诉我们的，如果他是骑士，我们可以相信他。

If Bonaparte is a knight then Wellington is a knave.

如果波拿巴是骑士，那么威灵顿是无赖。

True, because that is one of the things Bonaparte states.

正确，因为那是波拿巴陈述的事情之一。

So, if Wellington is a knight then Wellington is a knave (which is impossible!)

所以，如果威灵顿是骑士，那么威灵顿是无赖（这是不可能的！）。

This is just summing up what was deduced above.

这只是总结了上面推导出的内容。

Thus, Wellington is a knave.

因此，威灵顿是无赖。

Because the other possibility leads to something *im*possible.

因为另一种可能性导致了不可能的事情。

Since Wellington is a knave, his statement "I would tell you that Bonaparte is a knight" is false.

既然威灵顿是无赖，他的陈述“我会告诉你波拿巴是骑士”是假的。

Knave's statements are always false!

无赖的陈述总是假的!

So Wellington would in fact tell us that Bonaparte is a knave.

所以威灵顿实际上会告诉我们波拿巴是无赖。

He was lying when he said he would tell us B is a knight.

当他说他会告诉我们 B 是骑士时，他在说谎。

Since Wellington is a knave we conclude that Bonaparte is a knight.

既然威灵顿是无赖，我们得出结论波拿巴是骑士。

Wait, now I'm confused...can you do this part?

等等，现在我糊涂了……你能做这部分吗?

Thus Bonaparte is a knight and Wellington is a knave (as claimed).

因此波拿巴是骑士，威灵顿是无赖（如主张所述）。

Just summarizing.

只是总结一下。

Q.E.D.

2.4 Two-column proofs 二列证明

Exercises — 2.4

Write two-column proofs that verify each of the following logical equivalences.

写出二列式证明来验证以下每一个逻辑等价式。

1. $A \vee (A \wedge B) \;\cong\; A \wedge (A \vee B)$
2. $(A \wedge \neg B) \vee A \;\cong\; A$
3. $A \vee B \;\cong\; A \vee (\neg A \wedge B)$
4. $\neg(A \vee \neg B) \vee (\neg A \wedge \neg B) \;\cong\; \neg A$
5. $A \;\cong\; A \wedge ((A \vee \neg B) \vee (A \vee B))$
6. $(A \wedge \neg B) \wedge (\neg A \vee B) \;\cong\; c$
7. $A \;\cong\; A \wedge (A \vee (A \wedge (B \vee C)))$
8. $\neg(A \wedge B) \wedge \neg(A \wedge C) \;\cong\; \neg A \vee (\neg B \wedge \neg C)$

Here's the last one:

这是最后一个:

Proof:

$\neg(A \wedge B) \wedge \neg(A \wedge C)$	
	DeMorgan's law (times 2) (德摩根定律 (两次))
$\equiv \quad (\neg A \vee \neg B) \wedge (\neg A \vee \neg C)$	
	Distributive law (分配律)
$\equiv \quad \neg A \vee (\neg B \wedge \neg C)$	

Q.E.D.

2.5 Quantified statements 量化陈述

Exercises — 2.5

1. There is a common variant of the existential quantifier, $\exists!$, if you write $\exists!\, x,\ P(x)$ you are asserting that there is a *unique* element in the universe that makes $P(x)$ true. Determine how to negate the sentence $\exists!\, x,\ P(x)$.

   存在量词有一个常见的变体，$\exists!$，如果你写 $\exists!\, x,\ P(x)$，你是在断言论域中存在一个唯一的元素使得 $P(x)$ 为真。请确定如何否定句子 $\exists!\, x,\ P(x)$。

   Unique existence is essentially saying that there is exactly 1 element of the universe of discourse that makes P(x) true. The negation of "there is exactly 1" is "there's either none, or at least 2". Is that enough of a hint?

   唯一存在本质上是说论域中恰好有 1 个元素使 P(x) 为真。“恰好有 1 个”的否定是“一个都没有，或者至少有 2 个”。这个提示够了吗?

2. The order in which quantifiers appear is important. Let $L(x, y)$ be the open sentence "x is in love with y." Discuss the meanings of the following quantified statements and find their negations.

   量词出现的顺序很重要。设 $L(x, y)$ 为开放句 “x 爱着 y”。讨论以下量化陈述的含义并找出它们的否定。

   (a) $\forall x\, \exists y\ L(x, y)$.

   (b) $\exists x\, \forall y\ L(x, y)$.

   (c) $\forall x\, \forall y\ L(x, y)$.

   (d) $\exists x\, \exists y\ L(x, y)$.

(a) $\forall x\, \exists y\; L(x, y)$.

This is a fairly optimistic statement "For everyone out there, there's somebody that they are in love with."

这是一个相当乐观的陈述："对于外面的每一个人，都有一个他们所爱的人。"

(b) $\exists x\, \forall y\; L(x, y)$.

This one, on the other hand, says something fairly strange: "There's someone who has fallen in love with every other human being." I don't know, maybe the Dalai Lama? Mother Theresa?... Anyway, do the last two for yourself.

另一方面，这个陈述说了一些相当奇怪的事情："有一个人爱上了所有其他的人。" 我不知道，也许是达赖喇嘛？特蕾莎修女？……无论如何，自己完成最后两个。

(c) $\forall x\, \forall y\; L(x, y)$.

(d) $\exists x\, \exists y\; L(x, y)$.

Here's a couple of bonus questions. Two of the statements above have different meanings if you just interchange the order that the quantifiers appear in. What do the following mean (in contrast to the ones above)?

这里有几个附加问题。上面两个陈述如果你交换量词出现的顺序，它们的含义就会不同。与上面的陈述相比，下面的陈述是什么意思？

(e) $\exists y\, \forall x\; L(x, y)$.

(f) $\forall y\, \exists x\; L(x, y)$.

3. Determine a useful denial of:

   $\forall \epsilon > 0 \, \exists \delta > 0 \, \forall x \, (|x - c| < \delta) \implies (|f(x) - l| < \epsilon)$. The denial above gives a criterion for saying $\lim_{x \to c} f(x) \neq l$.

   确定一个有用的否定形式：$\forall \epsilon > 0 \, \exists \delta > 0 \, \forall x \, (|x - c| < \delta) \implies (|f(x) - l| < \epsilon)$. 上面的否定给出了一个判断 $\lim_{x \to c} f(x) \neq l$ 的标准。

   This is asking you to put a couple of things together. The first thing is that in negating a quantified statement, we get a new statement with all the quantified variables occurring in the same order but with $\forall$'s and $\exists$'s interchanged. The second issue is that the logical statement that appears after all the quantifiers needs to be negated. Since, in this statement we have a conditional, you must remember to negate that properly (its negation is a conjunction). $\exists \epsilon > 0 \, \forall \delta > 0 \, \exists x \, (|x - c| < \delta) \land (|f(x) - l| \geq \epsilon)$.

   这要求你把几件事结合起来。第一件事是，在否定一个量化陈述时，我们会得到一个新陈述，其中所有量化变量的出现顺序相同，但 $\forall$ 和 $\exists$ 互换。第二个问题是，出现在所有量词之后的逻辑陈述需要被否定。由于在这个陈述中我们有一个条件句，你必须记住正确地否定它（它的否定是一个合取）。$\exists \epsilon > 0 \, \forall \delta > 0 \, \exists x \, (|x - c| < \delta) \land (|f(x) - l| \geq \epsilon)$.

4. A *Sophie Germain prime* is a prime number p such that the corresponding odd number $2p + 1$ is also a prime. For example 11 is a Sophie Germain prime since $23 = 2 \cdot 11 + 1$ is also prime. Almost all Sophie Germain primes are congruent to $5 \pmod 6$, nevertheless, there are exceptions – so the statement "There are Sophie Germain primes that are not 5 mod 6." is true. Verify this.

   一个索菲·热尔曼素数是一个素数 p，使得相应的奇数 $2p+1$ 也是一个素数。例如 11 是一个索菲·热尔曼素数，因为 $23 = 2 \cdot 11 + 1$ 也是素数。几乎所有的索菲·热尔曼素数都与 $5 \pmod 6$ 同余，然而，也

有例外——所以“存在不是模 6 余 5 的索菲 · 热尔曼素数”这个陈述是真的。请验证这一点。

The exceptions are very small prime numbers. You should be able to find them easily.

例外的是非常小的素数。你应该能轻易找到它们。

5. Alvin, Betty, and Charlie enter a cafeteria which offers three different entrees, turkey sandwich, veggie burger, and pizza; four different beverages, soda, water, coffee, and milk; and two types of desserts, pie and pudding. Alvin takes a turkey sandwich, a soda, and a pie. Betty takes a veggie burger, a soda, and a pie. Charlie takes a pizza and a soda. Based on this information, determine whether the following statements are true or false.

   Alvin、Betty 和 Charlie 进入一家自助餐厅，该餐厅提供三种不同的主菜：火鸡三明治、素食汉堡和披萨；四种不同的饮料：苏打水、水、咖啡和牛奶；以及两种甜点：派和布丁。Alvin 拿了一个火鸡三明治、一杯苏打水和一个派。Betty 拿了一个素食汉堡、一杯苏打水和一个派。Charlie 拿了一个披萨和一杯苏打水。根据这些信息，判断以下陈述的真假。

   (a) $\forall$ people p, $\exists$ dessert d such that p took d.

   $\forall$ 人 p, $\exists$ 甜点 d 使得 p 拿了 d。

   false (假)

   (b) $\exists$ person p such that $\forall$ desserts d, p did not take d.

   $\exists$ 人 p 使得 $\forall$ 甜点 d, p 没有拿 d。

   true (真)

   (c) $\forall$ entrees e, $\exists$ person p such that p took e.

   $\forall$ 主菜 e, $\exists$ 人 p 使得 p 拿了 e。

   true (真)

(d) $\exists$ entree e such that $\forall$ people p, p took e.

$\exists$ 主菜 e 使得 $\forall$ 人 p, p 拿了 e。

false (假)

(e) $\forall$ people p, p took a dessert $\iff$ p did not take a pizza.

$\forall$ 人 p, p 拿了甜点 $\iff$ p 没有拿披萨。

true (真)

(f) Change one word of statement 5d so that it becomes true.

修改陈述5d中的一个词，使其为真。

entree $\longrightarrow$ beverage (主菜 $\longrightarrow$ 饮料)

(g) Write down the negation of 5a and compare it to statement 5b. Hopefully you will see that they are the same! Does this make you want to modify one or both of your answers to 5a and 5b?

写下5a的否定，并将其与陈述5b进行比较。希望你会发现它们是相同的！这是否让你想修改你对5a和5b的一个或两个答案？

$\exists$ person p such that $\forall$ desserts d, p did not take d. Yes I do. No, I got them right in the first place!

$\exists$ 人 p 使得 $\forall$ 甜点 d, p 没有拿 d。是的，我想修改。不，我一开始就答对了！

2.6 Deductive reasoning and Argument forms 演绎推理与论证形式

Exercises — 2.6

1. In the movie "Monty Python and the Holy Grail" we encounter a medieval villager who (with a bit of prompting) makes the following argument.

   If she weighs the same as a duck, then she's made of wood.
   If she's made of wood then she's a witch.
   Therefore, if she weighs the same as a duck, she's a witch.

   在电影《巨蟒与圣杯》中，我们遇到一位中世纪的村民，他（在一些提示下）提出了以下论证。

   如果她的体重和鸭子一样，那么她就是木头做的。
   如果她是木头做的，那么她就是个女巫。
   因此，如果她的体重和鸭子一样，她就是个女巫。

   Which rule of inference is he using?

   他用的是哪个推理规则?

   This is what many people refer to as the transitive rule of implication. As an argument form it's known as "hypothetical syllogism."

   这就是许多人所说的蕴涵的传递规则。作为一种论证形式，它被称为"假言三段论"。

2. In constructive dilemma, the antecedent of the conditional sentences are usually chosen to represent opposite alternatives. This allows us to introduce their disjunction as a tautology. Consider the following proof that there is never any reason to worry (found on the walls of an Irish pub).

> Either you are sick or you are well.
> If you are well there's nothing to worry about.
> If you are sick there are just two possibilities:
> Either you will get better or you will die.
> If you are going to get better there's nothing to worry about.
> If you are going to die there are just two possibilities:
> Either you will go to Heaven or to Hell.
> If you go to Heaven there is nothing to worry about. If you go to Hell, you'll be so busy shaking hands with all your friends there won't be time to worry …

在构造性二难推理中，条件句的前件通常被选择来代表相反的备选项。这使得我们可以将其析取作为重言式引入。思考以下证明，说明永远没有理由担心（发现于一家爱尔兰酒吧的墙上）。

> 你要么生病，要么健康。
> 如果你健康，就没什么好担心的。
> 如果你生病了，只有两种可能：
> 你要么会好起来，要么会死。
> 如果你会好起来，就没什么好担心的。
> 如果你会死，只有两种可能：
> 你要么上天堂，要么下地狱。
> 如果你上天堂，就没什么好担心的。如果你下地狱，你会忙着和所有的朋友握手，没时间担心……

Identify the three tautologies that are introduced in this "proof."

找出这个"证明"中引入的三个重言式。

Look at the lines that start with the word "Either."

看那些以"要么"开头的句子。

3. For each of the following arguments, write it in symbolic form and determine which rules of inference are used.

对于以下每个论证，请用符号形式写出，并确定使用了哪些推理规则。

(a) You are either with us, or you're against us. And you don't appear to be with us. So, that means you're against us!

你要么和我们站在一起，要么就是反对我们。而你看起来不和我们站在一起。所以，这意味着你反对我们！

$$\begin{array}{rl} & W \vee A \\ & \neg W \\ \hline \therefore & A \end{array}$$

This is "disjunctive syllogism."

这是"选言三段论"。

(b) All those who had cars escaped the flooding. Sandra had a car – therefore, Sandra escaped the flooding.

所有有车的人都逃脱了洪水。桑德拉有车——因此，桑德拉逃脱了洪水。

Let $C(x)$ be the open sentence "x has a car" and let $E(x)$ be the open sentence "x escaped the flooding." This argument is actually the particular form of universal modus ponens: (See the final question in the next set of exercises.)

设 $C(x)$ 为开放句"x 有车"，设 $E(x)$ 为开放句"x 逃脱了洪水"。这个论证实际上是全称肯定前件式的特称形式：（见下一组练习的最后一个问题。）

$$\begin{array}{rl} & \forall x, C(x) \implies E(x) \\ & C(\text{Sandra}) \\ \hline \therefore & E(\text{Sandra}) \end{array}$$

At this stage in the game it would be perfectly fine to just identify this as modus ponens and not worry about the quantifiers that appear.

在现阶段，将其识别为肯定前件式而不用担心出现的量词是完全可以的。

(c) When Johnny goes to the casino, he always gambles 'til he goes broke. Today, Johnny has money, so Johnny hasn't been to the casino recently.

当约翰尼去赌场时，他总是赌到破产为止。今天，约翰尼有钱，所以约翰尼最近没去过赌场。

(d) (A non-constructive proof that there are irrational numbers a and b such that a^b is rational.) Either $\sqrt{2}^{\sqrt{2}}$ is rational or it is irrational. If $\sqrt{2}^{\sqrt{2}}$ is rational, we let $a = b = \sqrt{2}$. Otherwise, we let $a = \sqrt{2}^{\sqrt{2}}$ and $b = \sqrt{2}$. (Since $\sqrt{2}^{\sqrt{2}^{\sqrt{2}}} = 2$, which is rational.) It follows that in either case, there are irrational numbers a and b such that a^b is rational.

(一个非构造性证明，证明存在无理数 a 和 b 使得 a^b 是有理数。) $\sqrt{2}^{\sqrt{2}}$ 要么是有理数，要么是无理数。如果 $\sqrt{2}^{\sqrt{2}}$ 是有理数，我们令 $a = b = \sqrt{2}$。否则，我们令 $a = \sqrt{2}^{\sqrt{2}}$ 且 $b = \sqrt{2}$。(因为 $\sqrt{2}^{\sqrt{2}^{\sqrt{2}}} = 2$，这是有理数。) 因此，在任何一种情况下，都存在无理数 a 和 b 使得 a^b 是有理数。

I'm leaving the last two for you to do. One small hint: both are valid forms.

最后两个留给你做。一个小提示：两种形式都有效。

2.7 Validity of arguments and common errors 论证的有效性与常见错误

Exercises — 2.7

1. Determine the logical form of the following arguments. Use symbols to express that form and determine whether the form is valid or invalid. If the form is invalid, determine the type of error made. Comment on the soundness of the argument as well, in particular, determine whether any of the premises are questionable.

   确定以下论证的逻辑形式。使用符号表达该形式，并判断其有效性。如果形式无效，请确定所犯错误的类型。同时评论论证的可靠性，特别是确定是否有任何前提值得怀疑。

   (a) All who are guilty are in prison.
   George is not in prison.
   Therefore, George is not guilty.
   所有有罪的人都在监狱里。
   乔治不在监狱里。
   因此，乔治是无罪的。

   This looks like modus tollens. Let G refer to "guilt" and P refer to "in prison"

   这看起来像是否定后件式。设 G 指代 "有罪"，P 指代 "在监狱里"。

$$\begin{array}{ll} & \forall x, G(x) \implies P(x) \\ & \neg P(\text{George}) \\ \hline \therefore & \neg G(\text{George}) \end{array}$$

   You should note that while the form is valid, there is something terribly wrong with this argument. Is it really true that everyone who is guilty of a crime is in prison?

你应该注意到，虽然形式有效，但这个论证有严重的问题。所有犯了罪的人真的都在监狱里吗?

(b) If one eats oranges one will have high levels of vitamin C.
You do have high levels of vitamin C.
Therefore, you must eat oranges.
如果一个人吃橙子，他将会有高水平的维生素 C。
你确实有高水平的维生素 C。
因此，你一定吃橙子。

(c) All fish live in water.
The mackerel is a fish.
Therefore, the mackerel lives in water.
所有鱼都生活在水中。
鲭鱼是一种鱼。
因此，mackerel 鱼生活在水中。

(d) If you're lazy, don't take math courses.
Everyone is lazy.
Therefore, no one should take math courses.
如果你懒，就不要上数学课。
每个人都懒。
因此，没有一个人应该上数学课。

(e) All fish live in water.
The octopus lives in water.
Therefore, the octopus is a fish.
所有鱼都生活在水中。
章鱼生活在水中。
因此，章鱼是一种鱼。

(f) If a person goes into politics, they are a scoundrel.
Harold has gone into politics.
Therefore, Harold is a scoundrel.

如果一个人从政，他就是个无赖。
哈罗德已经从政了。
因此，哈罗德是个无赖。

2. Below is a rule of inference that we call extended elimination.

下面是我们称之为扩展消去法的推理规则。
$$\begin{array}{ll} & (A \vee B) \vee C \\ & \neg A \\ & \neg B \\ \hline \therefore & C \end{array}$$

Use a truth table to verify that this rule is valid.

使用真值表来验证此规则的有效性。

In the following truth table the predicate variables occupy the first 3 columns, the argument's premises are in the next three columns and the conclusion is in the right-most column. The truth values have already been filled-in. You only need to identify the critical rows and verify that the conclusion is true in those rows.

在下面的真值表中，谓词变量占据前 3 列，论证的前提在接下来的三列中，结论在最右边的一列。真值已经填好。你只需要识别出关键行并验证在这些行中结论为真。

A	B	C	$(A \vee B) \vee C$	$\neg A$	$\neg B$	C
T	T	T	T	ϕ	ϕ	T
T	T	ϕ	T	ϕ	ϕ	ϕ
T	ϕ	T	T	ϕ	T	T
T	ϕ	ϕ	T	ϕ	T	ϕ
ϕ	T	T	T	T	ϕ	T
ϕ	T	ϕ	T	T	ϕ	ϕ
ϕ	ϕ	T	T	T	T	T
ϕ	ϕ	ϕ	ϕ	T	T	ϕ

3. If we allow quantifiers and open sentences in an argument form we get a couple of new argument forms. Arguments involving existentially quantified premises are rare – the new forms we are speaking of are called "universal modus ponens" and "universal modus tollens." The minor premises may also be quantified or they may involve particular elements of the universe of discourse – this leads us to distinguish argument subtypes that are termed "universal" and "particular."

如果我们在论证形式中允许量词和开放句，我们会得到几个新的论证形式。涉及存在量化前提的论证很少见——我们所说的新形式被称为"全称肯定前件式"和"全称否定后件式"。小前提也可能被量化，或者它们可能涉及论域的特定元素——这导致我们区分被称为"全称"和"特称"的论证子类型。

For example $\begin{array}{ll} & \forall x, A(x) \implies B(x) \\ & A(p) \\ \hline \therefore & B(p) \end{array}$ is the particular form of universal modus ponens (here, p is not a variable – it stands for some particular element of the universe of discourse) and $\begin{array}{ll} & \forall x, A(x) \implies B(x) \\ & \forall x, \neg B(x) \\ \hline \therefore & \forall x, \neg A(x) \end{array}$ is the universal form of (universal) modus tollens.

例如 $\begin{array}{ll} & \forall x, A(x) \implies B(x) \\ & A(p) \\ \hline \therefore & B(p) \end{array}$ 是全称肯定前件式的特称形式（这里，p 不是一个变量——它代表论域中的某个特定元素）而 $\begin{array}{ll} & \forall x, A(x) \implies B(x) \\ & \forall x, \neg B(x) \\ \hline \therefore & \forall x, \neg A(x) \end{array}$ 是（全称）否定后件式的全称形式。

Reexamine the arguments from problem (1), determine their forms (including quantifiers) and whether they are universal or particular.

重新检查问题 (1) 中的论证，确定它们的形式（包括量词）以及它们是全称的还是特称的。

Hint: All of them except for one are the particular form – number

4 is the exception. Here's an analysis of number 5:

提示：除了一个之外，它们都是特称形式——第 4 个是例外。以下是对第 5 个的分析：

All fish live in water.
The octopus lives in water.
Therefore, the octopus is a fish.

所有鱼都生活在水中。
章鱼生活在水中。
因此，章鱼是鱼。

Let $F(x)$ be the open sentence "x is a fish" and let $W(x)$ be the open sentence "x lives in water."

设 $F(x)$ 为开放句 "x 是鱼"，设 $W(x)$ 为开放句 "x 生活在水中"。

Our argument has the form

我们的论证形式为

$$\begin{array}{ll} & \forall x, F(x) \implies W(x) \\ & W(\text{the octopus}) \\ \hline \therefore & F(\text{the octopus}) \end{array}$$

Clearly something is wrong – a converse error has been made – if everything that lived in water was necessarily a fish the argument would be OK (in fact it would then be the particular form of universal modus ponens). But that is the converse of the major premise given.

显然有问题——犯了逆命题错误——如果所有生活在水里的都必然是鱼，那么这个论证就没问题（实际上，它将是全称肯定前件式的特称形式）。但那是给定大前提的逆命题。

4. Identify the rule of inference being used.

   识别所使用的推理规则。

   (a) The Buley Library is very tall.
   Therefore, either the Buley Library is very tall or it has many levels underground.
   Buley 图书馆非常高。
   因此，Buley 图书馆要么非常高，要么地下有很多层。
   disjunctive addition (析取附加)

   (b) The grass is green.
   The sky is blue.
   Therefore, the grass is green and the sky is blue.
   草是绿的。
   天是蓝的。
   因此，草是绿的，天是蓝的。
   conjunctive addition (合取附加)

   (c) g has order 3 or it has order 4.
   If g has order 3, then g has an inverse.
   If g has order 4, then g has an inverse.
   Therefore, g has an inverse.
   g 的阶是 3 或 4。
   如果 g 的阶是 3，那么 g 有逆元。
   如果 g 的阶是 4，那么 g 有逆元。
   因此，g 有逆元。
   constructive dilemma (构造性二难)

   (d) x is greater than 5 and x is less than 53.
   Therefore, x is less than 53.
   x 大于 5 且 x 小于 53。
   因此，x 小于 53。

conjunctive simplification (合取简化)

(e) If $a|b$, then a is a perfect square.
If $a|b$, then b is a perfect square.
Therefore, if $a|b$, then a is a perfect square and b is a perfect square.
如果 $a|b$，那么 a 是一个完全平方数。
如果 $a|b$，那么 b 是一个完全平方数。
因此，如果 $a|b$，那么 a 是一个完全平方数且 b 是一个完全平方数。

Note that the conclusion could be re-expressed as the conjunction of the two conditionals that are found in the premises. This is conjunctive addition with a bit of "window dressing."

注意，结论可以重新表述为前提中两个条件句的合取。这是带有少许"修饰"的合取附加。

5. Read the following proof that the sum of two odd numbers is even. Discuss the rules of inference used.

阅读以下关于两个奇数之和为偶数的证明。讨论所使用的推理规则。

Proof: Let x and y be odd numbers. Then $x = 2k + 1$ and $y = 2j + 1$ for some integers j and k. By algebra,

$$x + y = 2k + 1 + 2j + 1 = 2(k + j + 1).$$

Note that $k+j+1$ is an integer because k and j are integers. Hence $x + y$ is even.

设 x 和 y 为奇数。那么对于某些整数 j 和 k，$x = 2k + 1$ 且 $y = 2j + 1$。根据代数，

$$x + y = 2k + 1 + 2j + 1 = 2(k + j + 1).$$

注意 $k+j+1$ 是一个整数，因为 k 和 j 是整数。因此 $x+y$ 是偶数。

Q.E.D.

The definition for "odd" only involves the oddness of a single integer, but the first line of our proof is a conjunction claiming that x and y are both odd. It seems that two conjunctive simplifications, followed by applications of the definition, followed by a conjunctive addition have been used in order to go from the first sentence to the second.

"奇数" 的定义只涉及单个整数的奇偶性，但我们证明的第一行是一个声称 x 和 y 都是奇数的合取。为了从第一句到第二句，似乎使用了两次合取简化，接着是定义的应用，然后是一次合取附加。

6. Sometimes in constructing a proof we find it necessary to "weaken" an inequality. For example, we might have already deduced that $x < y$ but what we need in our argument is that $x \leq y$. It is okay to deduce $x \leq y$ from $x < y$ because the former is just shorthand for $x < y \lor x = y$. What rule of inference are we using in order to deduce that $x \leq y$ is true in this situation?

   有时在构建证明时，我们发现有必要 "弱化" 一个不等式。例如，我们可能已经推断出 $x < y$，但我们在论证中需要的是 $x \leq y$。从 $x < y$ 推断出 $x \leq y$ 是可以的，因为前者只是 $x < y \lor x = y$ 的简写。在这种情况下，我们使用什么推理规则来推断 $x \leq y$ 为真?

   disjunctive addition (析取附加)

Chapter 3

Proof techniques I — Standard methods 证明技巧 I — 标准方法

As a convenience, the table containing the definitions of elementary number theory is reproduced on the following page.

为方便起见，下页转载了包含初等数论定义的表格。

Even (偶数)

$\forall n \in \mathbb{Z}$,

$$n \text{ is even} \iff \exists k \in \mathbb{Z},\ n = 2k$$

Odd (奇数)

$\forall n \in \mathbb{Z}$,

$$n \text{ is odd} \iff \exists k \in \mathbb{Z},\ n = 2k + 1$$

Divisibility (整除性)

$\forall n \in \mathbb{Z}, \forall \quad d > 0 \in \mathbb{Z}$,

$$d \,|\, n \iff \exists k \in \mathbb{Z},\ n = kd$$

Floor (下取整)

$\forall x \in \mathbb{R}$,

$$y = \lfloor x \rfloor \iff y \in \mathbb{Z} \ \wedge \ y \leq x < y + 1$$

Ceiling (上取整)

$\forall x \in \mathbb{R}$,

$$y = \lceil x \rceil \iff y \in \mathbb{Z} \ \wedge \ y - 1 < x \leq y$$

Quotient-remainder theorem, Div and Mod (商余定理, Div 和 Mod)

$\forall n, d > 0 \in \mathbb{Z}$,

$$\exists! q, r \in \mathbb{Z},\ n = qd + r \ \wedge \ 0 \leq r < d$$

$$n \text{ div } d = q$$

$$n \text{ mod } d = r$$

Prime (素数)

$\forall p \in \mathbb{Z}$

$$p \text{ is prime} \iff$$

$$(p > 1) \quad \wedge \quad (\forall x, y \in \mathbb{Z}^+,\ p = xy \implies x = 1 \vee y = 1)$$

3.1 Direct proofs of universal statements 全称陈述的直接证明

Exercises — 3.1

1. Every prime number greater than 3 is of one of the two forms $6k+1$ or $6k+5$. What statement(s) could be used as hypotheses in proving this theorem?

   每个大于 3 的素数都具有 $6k+1$ 或 $6k+5$ 两种形式之一。在证明这个定理时，可以用哪些陈述作为假设?

   Fill in the blanks:

   填空:

   - p is a __________________ number, and

     p 是一个 __________________ 数，并且
   - p is greater than ____________.

     p 大于 ____________。

2. Prove that 129 is odd.

   证明 129 是奇数。

   All you have to do to show that some number is odd, is produce the integer k that the definition of "odd" says has to exist. Hint: the same number could be used to prove that 128 is even.

   要证明某个数是奇数，你所要做的就是找出"奇数"定义中所说的必须存在的那个整数 k。提示：同一个数也可以用来证明 128 是偶数。

3. Prove that the sum of two rational numbers is a rational number.

   证明两个有理数的和是一个有理数。

You want to argue about the sum of two generic rational numbers. Maybe call them a/b and c/d. The definition of "rational number" then tells you that a, b, c and d are integers and that neither b nor d are zero. You add these generic rational numbers in the usual way – put them over a common denominator and then add the numerators. One possible common denominator is bd, so we can express the sum as $(ad + bc)/(bd)$. You can finish off the argument from here: you need to show that this expression for the sum satisfies the definition of a rational number (quotient of integers w/ non-zero denominator). Also, write it all up a bit more formally…

你需要论证两个一般有理数的和。可以称它们为 a/b 和 c/d。"有理数"的定义告诉你 a, b, c, d 都是整数，且 b 和 d 都不为零。你用通常的方式将这两个有理数相加——将它们通分，然后将分子相加。一个可能的公分母是 bd，所以我们可以将和表示为 $(ad + bc)/(bd)$。你可以从这里完成论证：你需要证明这个和的表达式满足有理数的定义（非零分母的整数商）。另外，把整个过程写得更正式一些……

4. Prove that the sum of an odd number and an an even number is odd.

   证明一个奇数和一个偶数的和是奇数。

   Proof: Suppose that x is an odd number and y is an even number. Since x is odd there is an integer k such that $x = 2k+1$. Furthermore, since y is even, there is an integer m such that $y = 2m$. By substitution, we can express the sum $x+y$ as $x+y = (2k+1)+(2m) = 2(k+m)+1$. Since $k+m$ is an integer (the sum of integers is an integer) it follows that $x+y$ is odd.

   假设 x 是一个奇数，y 是一个偶数。因为 x 是奇数，所以存在一个整数 k 使得 $x = 2k+1$。此外，因为 y 是偶数，所以存在一个整数 m 使得 $y = 2m$。通过代换，我们可以将和 $x+y$ 表示为 $x+y = (2k+1)+(2m) = 2(k+m)+1$。因为 $k+m$ 是一个整数（整数之和是整数），所以 $x+y$ 是奇数。

   Q.E.D.

5. Prove that if the sum of two integers is even, then so is their difference.

   证明如果两个整数的和是偶数，那么它们的差也是偶数。

   Hint: If we write $x+y$ for the sum of two integers that is even (so $x+y = 2k$ for some integer k), then we could subtract ____________ from it in order to obtain $x-y$. Once you fill in that blank properly the flow of the argument should become apparent to you.

   提示：如果我们用 $x+y$ 表示两个整数的和，且其为偶数（所以对于某个整数 k，$x+y = 2k$），那么我们可以从中减去 ____________ 来得到 $x-y$。一旦你正确地填上这个空，论证的流程对你来说应该就显而易见了。

6. Prove that for every real number x, $\frac{2}{3} < x < \frac{3}{4} \implies \lfloor 12x \rfloor = 8$.

   证明对于每一个实数 x，如果 $\frac{2}{3} < x < \frac{3}{4}$，则 $\lfloor 12x \rfloor = 8$。

   Begin your proof like so:

   像这样开始你的证明:

   "Suppose that x is a real number such that $\frac{2}{3} < x < \frac{3}{4}$."

   "假设 x 是一个满足 $\frac{2}{3} < x < \frac{3}{4}$ 的实数。"

   You need to multiply all three parts of the inequality by something in order to "clear" the fractions. What should that be?

   你需要将不等式的三部分都乘以某个数来"消去"分数。这个数应该是什么?

   The definition for the floor of $12x$ will be satisfied if $8 \leq 12x < 9$ but unfortunately the work done previously will have deduced that $8 < 12x < 9$ is true. Don't just gloss over this discrepancy. Explain why one of these inequalities is implied by the other.

   如果 $8 \leq 12x < 9$，那么 $12x$ 的下取整的定义就满足了，但不幸的是，前面的工作会推导出 $8 < 12x < 9$ 为真。不要忽略这个差异。请解释为什么其中一个不等式可以推出另一个。

7. Prove that if x is an odd integer, then x^2 is of the form $4k+1$ for some integer k.

   证明如果 x 是一个奇数，那么 x^2 具有 $4k+1$ 的形式，其中 k 是某个整数。

   You may be tempted to write "Since x is odd, it can be expressed as $x = 2k+1$ where k is an integer." This is slightly wrong since the variable k is already being used in the statement of the theorem. But, except for replacing k with some other variable (maybe m or j?) that *is* a good way to get started. From there it's really just algebra until, eventually, you'll find out what k really is.

   你可能会想写"因为 x 是奇数，它可以表示为 $x = 2k+1$，其中 k 是一个整数。"这有点问题，因为变量 k 已经在定理的陈述中被使用了。但是，除了将 k 替换为其他某个变量（比如 m 或 j?），这是一个很好的开始方式。从那里开始，基本上就是代数运算，直到最后，你会发现 k 到底是什么。

8. Prove that for all integers a and b, if a is odd and $6 \mid (a+b)$, then b is odd.

   证明对于所有整数 a 和 b，如果 a 是奇数且 $6 \mid (a+b)$，那么 b 是奇数。

   The premise that $6 \mid (a+b)$ is a bit of a red herring (a clue that is designed to mislead). The premise that you really need is that $a+b$ is even. Can you deduce that from what's given?

   前提 $6 \mid (a+b)$ 有点像障眼法（一个旨在误导的线索）。你真正需要的前提是 $a+b$ 是偶数。你能从给定的条件中推导出这一点吗？

9. Prove that $\forall x \in \mathbb{R}, x \notin \mathbb{Z} \implies \lfloor x \rfloor + \lfloor -x \rfloor = -1$.

   证明 $\forall x \in \mathbb{R}, x \notin \mathbb{Z} \implies \lfloor x \rfloor + \lfloor -x \rfloor = -1$。

Proof: Suppose that x is a real number and $x \notin \mathbb{Z}$. Let $a = \lfloor x \rfloor$. By the definition of the floor function we have $a \in \mathbb{Z}$ and $a \leq x < a+1$. Since $x \notin \mathbb{Z}$ we know that $x \neq a$ so we may strengthen the inequality to $a < x < a+1$. Multiplying this inequality by -1 we obtain $-a > -x > -a-1$. This inequality may be weakened to $-a > -x \geq -a-1$. Finally, note that (since $-a-1 \in \mathbb{Z}$ and $-a = (-a-1)+1$ we have shown that $\lfloor -x \rfloor = -a-1$. Thus, by substitution we have $\lfloor x \rfloor + \lfloor -x \rfloor = a + (-a-1) = -1$ as desired.

假设 x 是一个实数且 $x \notin \mathbb{Z}$。令 $a = \lfloor x \rfloor$。根据下取整函数的定义，我们有 $a \in \mathbb{Z}$ 且 $a \leq x < a+1$。因为 $x \notin \mathbb{Z}$，我们知道 $x \neq a$，所以我们可以将不等式加强为 $a < x < a+1$。将这个不等式乘以-1，我们得到 $-a > -x > -a-1$。这个不等式可以弱化为 $-a > -x \geq -a-1$。最后，注意到（因为 $-a-1 \in \mathbb{Z}$ 且 $-a = (-a-1)+1$）我们已经证明了 $\lfloor -x \rfloor = -a-1$。因此，通过代换我们得到 $\lfloor x \rfloor + \lfloor -x \rfloor = a + (-a-1) = -1$，正如所求。

Q.E.D.

10. Define the *evenness* of an integer n by:

    定义一个整数 n 的偶性为：

$$\text{evenness}(n) = k \iff 2^k \mid n \wedge 2^{k+1} \nmid n$$

    State and prove a theorem concerning the evenness of products.

    陈述并证明一个关于乘积偶性的定理。

    Well, the statement is that the evenness of a product is the sum of the evennesses of the factors...

    嗯，这个陈述是：一个乘积的偶性是其各因数偶性之和……

11. Suppose that a, b and c are integers such that $a \mid b$ and $b \mid c$. Prove that $a \mid c$.

    假设 a, b, c 是整数，使得 $a \mid b$ 且 $b \mid c$。证明 $a \mid c$。

    This one is pretty straightforward. Be sure to not reuse any variables. Particularly, the fact that $a \mid b$ tells us (because of the definition of divisibility) that there is an integer k such that $b = ak$. It is not okay to also use k when converting the statement "$b \mid c$."

    这个问题相当直接。确保不要重复使用任何变量。特别是，从 $a \mid b$ 这个事实（根据整除的定义）我们知道存在一个整数 k 使得 $b = ak$。在转换陈述 "$b \mid c$" 时，再使用 k 是不行的。

12. Suppose that a, b, c and d are integers with $a \neq c$. Further, suppose that x is a real number satisfying the equation

    假设 a, b, c, d 是整数且 $a \neq c$。再假设 x 是一个满足方程的实数：

$$\frac{ax + b}{cx + d} = 1.$$

    Show that x is rational. Where is the hypothesis $a \neq c$ used?

证明 x 是有理数。假设 $a \neq c$ 在哪里被用到了？

Cross multiply and solve for x. If you need to divide by an expression, it had better be non-zero!

交叉相乘并解出 x。如果你需要除以一个表达式，它最好是非零的！

13. Show that if two positive integers a and b satisfy $a \mid b$ *and* $b \mid a$ then they are equal.

证明如果两个正整数 a 和 b 满足 $a|b$ 且 $b|a$，那么它们相等。

From the definition of divisibility, you get two integers j and k, such that $a = jb$ and $b = ka$. Substitute one of those into the other and ask yourself what the resulting equation says about j and k. Can they be any old integers? Or, are there restrictions on their values?

从整除的定义，你可以得到两个整数 j 和 k，使得 $a = jb$ 且 $b = ka$。将其中一个代入另一个，然后问问自己得到的方程对 j 和 k 意味着什么。它们可以是任意的整数吗？还是它们的值有限制？

3.2 More direct proofs 更多直接证明

Exercises — 3.2

1. Suppose you have a savings account which bears interest compounded monthly. The July statement shows a balance of \$ 2104.87 and the September statement shows a balance \$ 2125.97. What would be the balance on the (missing) August statement?

   假设你有一个按月复利计息的储蓄账户。七月份的账单显示余额为 \$2104.87，九月份的账单显示余额为 \$2125.97。那么（缺失的）八月份账单上的余额会是多少?

   A savings account where we are not depositing or withdrawing funds has a balance that is growing geometrically.

   一个我们没有存入或取出资金的储蓄账户，其余额是按几何级数增长的。

2. Recall that a quadratic equation $ax^2+bx+c=0$ has two real solutions if and only if the discriminant b^2-4ac is positive. Prove that if a and c have different signs then the quadratic equation has two real solutions.

   回想一下，一个二次方程 $ax^2+bx+c=0$ 有两个实数解，当且仅当判别式 b^2-4ac 为正。请证明如果 a 和 c 符号相反，那么该二次方程有两个实数解。

   You don't need all the hypotheses. If a and c have different signs, then ac is a negative quantity

   你不需要所有的假设。如果 a 和 c 符号相反，那么 ac 是一个负数。

3. Prove that if x^3-x^2 is negative then $3x+4<7$.

   证明如果 x^3-x^2 是负数，那么 $3x+4<7$。

This follows very easily by the method of working backwards from the conclusion. Remember that when multiplying or dividing both sides of an inequality by some number, the direction of the inequality may reverse (unless we know the number involved is positive). Also, remember that we can't divide by zero, so if we are (just for example, don't know why I'm mentioning it really…) dividing both sides of an inequality by x^2 then we must treat the case where $x = 0$ separately.

这可以通过从结论倒推的方法非常容易地得出。记住，当不等式两边同时乘以或除以某个数时，不等号的方向可能会改变（除非我们知道所涉及的数是正数）。另外，记住我们不能除以零，所以如果我们（举个例子，真的不知道我为什么要提这个⋯⋯）将不等式两边同时除以 x^2，那么我们必须单独处理 $x = 0$ 的情况。

4. Prove that for all integers a, b, and c, if $a|b$ and $a|(b + c)$, then $a|c$.

   证明对于所有整数 a, b, 和 c，如果 $a|b$ 且 $a|(b + c)$，那么 $a|c$。

5. Show that if x is a positive real number, then $x + \frac{1}{x} \geq 2$.

   证明如果 x 是一个正实数，那么 $x + \frac{1}{x} \geq 2$。

If you work backwards from the conclusion on this one, you should eventually come to the inequality $(x - 1)^2 \geq 0$. Notice that this inequality is always true – all squares are non-negative. When you go to write-up your proof (writing things in the forward direction), you'll want to acknowledge this truth. Start with something like "Regardless of the value of x, the quantity $(x - 1)^2$ is greater than or equal to zero as it is a perfect square."

如果你从结论倒推，你最终应该会得到不等式 $(x - 1)^2 \geq 0$。注意这个不等式总是成立的——所有的平方都是非负的。当你开始写你的证明时（按正向顺序写），你会想要承认这个事实。可以这样开始："无论 x 的值是多少，量 $(x - 1)^2$ 都大于或等于零，因为它是一个完全平方数。"

6. Prove that for all real numbers a, b, and c, if $ac < 0$, then the quadratic equation $ax^2 + bx + c = 0$ has two real solutions.

   Hint: The quadratic equation $ax^2 + bx + c = 0$ has two real solutions if and only if $b^2 - 4ac > 0$ and $a \neq 0$.

   证明对于所有实数 a, b, 和 c, 如果 $ac < 0$, 那么二次方程 $ax^2 + bx + c = 0$ 有两个实数解。

   提示: 二次方程 $ax^2 + bx + c = 0$ 有两个实数解，当且仅当 $b^2 - 4ac > 0$ 且 $a \neq 0$。

   This is very similar to problem 2.

   这与问题 2 非常相似。

7. Show that $\binom{n}{k} \cdot \binom{k}{r} = \binom{n}{r} \cdot \binom{n-r}{k-r}$ (for all integers r, k and n with $r \leq k \leq n$).

   证明 $\binom{n}{k} \cdot \binom{k}{r} = \binom{n}{r} \cdot \binom{n-r}{k-r}$（对于所有满足 $r \leq k \leq n$ 的整数 r, k 和 n）。

   Use the definition of the binomial coefficients as fractions involving factorials:

   使用二项式系数作为包含阶乘的分数的定义:

   E.g. $\binom{n}{k} = \dfrac{n!}{k!(n-k)!}$

   例如: $\binom{n}{k} = \dfrac{n!}{k!(n-k)!}$

   Write down the definitions, both of the left hand side and the right hand side and consider how you can convert one into the other.

   写下左边和右边的定义，并考虑如何将一个转换成另一个。

8. In proving the *product rule* in Calculus using the definition of the derivative, we might start our proof with:

   在微积分中使用导数的定义来证明乘法法则时，我们可能会这样开始我们的证明:

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(f(x)\cdot g(x)\right)$$

$$= \lim_{h\longrightarrow 0}\frac{f(x+h)\cdot g(x+h)-f(x)\cdot g(x)}{h}$$

The last two lines of our proof should be:

我们证明的最后两行应该是：

$$= \lim_{h\longrightarrow 0}\frac{f(x+h)-f(x)}{h}\cdot g(x) \ + \ f(x)\cdot\lim_{h\longrightarrow 0}\frac{g(x+h)-g(x)}{h}$$

$$= \frac{\mathrm{d}}{\mathrm{d}x}\left(f(x)\right)\cdot g(x) \ + \ f(x)\cdot\frac{\mathrm{d}}{\mathrm{d}x}\left(g(x)\right)$$

Fill in the rest of the proof.

填写证明的其余部分。

The critical step is to subtract and add the same thing: $f(x)g(x+h)$ in the numerator of the fraction in the limit which gives the definition of $\frac{\mathrm{d}}{\mathrm{d}x}\left(f(x)\cdot g(x)\right)$. Also, you'll need to recall the laws of limits (like "the limit of a product is the product of the limits – provided both exist")

关键步骤是在给出 $\frac{\mathrm{d}}{\mathrm{d}x}\left(f(x)\cdot g(x)\right)$ 定义的极限中的分数的分子上，减去并加上同一个东西：$f(x)g(x+h)$。另外，你还需要回想一下极限的法则（比如"积的极限是极限的积——前提是两者都存在"）。

3.3 Indirect proofs: contradiction and contraposition 间接证明：矛盾法与逆否证法

Exercises — 3.3

1. Prove that if the cube of an integer is odd, then that integer is odd.

   证明如果一个整数的立方是奇数，那么这个整数也是奇数。

   The best hint for this problem is simply to write down the contrapositive statement. It is trivial to prove!

   对这个问题最好的提示就是写下其逆否命题。证明它易如反掌！

2. Prove that whenever a prime p does not divide the square of an integer, it also doesn't divide the original integer. $(p \nmid x^2 \implies p \nmid x)$

   证明只要一个素数 p 不能整除一个整数的平方，它也不能整除这个整数本身。$(p \nmid x^2 \implies p \nmid x)$

   The contrapositive is $(p \mid x) \implies (p \mid x^2)$.

   其逆否命题是 $(p \mid x) \implies (p \mid x^2)$。

3. Prove (by contradiction) that there is no largest integer.

   用反证法证明不存在最大的整数。

   Well, if there was a largest integer – let's call it L (for largest) – then isn't $L+1$ an integer, and isn't it bigger? That's the main idea. A more formal proof might look like this:

   嗯，如果存在一个最大的整数——我们称之为 L（代表最大）——那么 $L+1$ 不也是一个整数，并且它不是更大吗？这就是主要思想。一个更正式的证明可能如下：

   > *Proof:* Suppose (by way of contradiction) that there is a largest integer L. Then $L \in \mathbb{Z}$ and $\forall z \in \mathbb{Z}, L \geq z$. Consider

the quantity $L+1$. Clearly $L+1$ is an integer (because it is the sum of two integers) and also $L+1 > L$. This is a contradiction so the original supposition is false. Hence there is no largest integer.

假设（通过反证法）存在一个最大的整数 L。那么 $L \in \mathbb{Z}$ 且 $\forall z \in \mathbb{Z}, L \geq z$。考虑量 $L+1$。显然 $L+1$ 是一个整数（因为它是两个整数的和），并且 $L+1 > L$。这是一个矛盾，所以最初的假设是错误的。因此，不存在最大的整数。

Q.E.D.

4. Prove (by contradiction) that there is no smallest positive real number.

   用反证法证明不存在最小的正实数。

   Assume there was a smallest positive real number – might as well call it s (for smallest) – what can we do to produce an even smaller number? (But be careful that it needs to remain positive – for instance $s-1$ won't work.)

   假设存在一个最小的正实数——不妨称之为 s（代表最小）——我们能做什么来产生一个更小的数？（但要小心，它需要保持为正——例如 $s-1$ 就行不通。）

5. Prove (by contradiction) that the sum of a rational and an irrational number is irrational.

   用反证法证明一个有理数和一个无理数的和是无理数。

   Suppose that x is rational and y is irrational and their sum (let's call it z) is also rational. Do some algebra to solve for y, and you will see that y (which is, by presumption, irrational) is also the difference of two rational numbers (and hence, rational – a contradiction.)

   假设 x 是有理数，y 是无理数，它们的和（我们称之为 z）也是有理数。做一些代数运算来解出 y，你会发现 y（根据假设，是无理数）也

是两个有理数的差（因此，是有理数——这是一个矛盾）。

6. Prove (by contraposition) that for all integers x and y, if $x+y$ is odd, then $x \neq y$.

   用逆否证法证明对于所有整数 x 和 y，如果 $x+y$ 是奇数，那么 $x \neq y$。

   Well, the problem says to do this by contraposition, so let's write down the contrapositive:

   嗯，题目要求用逆否证法来做，所以我们先写下逆否命题：

$$\forall x, y \in \mathbb{Z},\ x = y \implies x + y \text{ is even.}$$

   But proving that is obvious!

   但证明那个是显而易见的！

7. Prove (by contraposition) that for all real numbers a and b, if ab is irrational, then a is irrational or b is irrational.

   用逆否证法证明对于所有实数 a 和 b，如果 ab 是无理数，那么 a 是无理数或 b 是无理数。

   The contrapositive would be:

   逆否命-题将是：

$$\forall a, b \in \mathbb{R},\ (a \in \mathbb{Q} \wedge b \in \mathbb{Q}) \implies ab \in \mathbb{Q}.$$

   Wow! Haven't we proved that before?

   哇！我们以前不是证明过这个吗？

8. A *Pythagorean triple* is a set of three natural numbers, a, b and c, such that $a^2 + b^2 = c^2$. Prove that, in a Pythagorean triple, at least one

of a and b is even. Use either a proof by contradiction or a proof by contraposition.

一个勾股数是一组三个自然数 a, b, c，使得 $a^2 + b^2 = c^2$。证明在一个勾股数中，a 和 b 至少有一个是偶数。使用反证法或逆否证法。

If both a and b are odd then their squares will be 1 mod 4 – so the sum of their squares will be 2 mod 4. But c^2 can only be 0 or 1 mod 4, which gives us a contradiction.

如果 a 和 b 都是奇数，那么它们的平方将是模 4 余 1——所以它们的平方和将是模 4 余 2。但是 c^2 只能是模 4 余 0 或 1，这就产生了一个矛盾。

9. Suppose you have 2 pairs of positive real numbers whose products are 1. That is, you have (a, b) and (c, d) in $\mathbb{R}^2$ satisfying $ab = cd = 1$. Prove that $a < c$ implies that $b > d$.

   假设你有两对乘积为 1 的正实数。也就是说，你有 (a, b) 和 (c, d) 在 $\mathbb{R}^2$ 中满足 $ab = cd = 1$。证明 $a < c$ 蕴涵 $b > d$。

   Proof: Suppose by way of contradiction that $a, b, c, d \in \mathbb{R}$ satisfy $ab = cd = 1$ and that $a < c$ and $b \leq d$. By multiplying the inequalities we get that $ab < cd$ which contradicts the assumption that both products are equal to 1 (and so must be equal to one another).

   假设（通过反证法）$a, b, c, d \in \mathbb{R}$ 满足 $ab = cd = 1$ 且 $a < c$ 和 $b \leq d$。将这两个不等式相乘，我们得到 $ab < cd$，这与两个乘积都等于 1（因此必须彼此相等）的假设相矛盾。

   Q.E.D.

3.4 Disproofs 反证

Exercises — 3.4

1. Find a polynomial that assumes only prime values for a reasonably large range of inputs.

   找一个在一个相当大的输入范围内只取素数值的多项式。

   It sort of depends on what is meant by "a reasonably large range of inputs." For example the polynomial $p(x) = 2x + 1$ gives primes three times in a row (at $x = 1, 2$ and 3). See if you can do better than that.

   这有点取决于"一个相当大的输入范围"是什么意思。例如，多项式 $p(x) = 2x + 1$ 连续三次（在 $x = 1, 2$ 和 3 时）给出素数。看看你能不能做得更好。

2. Find a counterexample to the conjecture that $\forall a, b, c \in \mathbb{Z}, a \mid bc \implies a \mid b \lor a \mid c$ using only powers of 2.

   仅使用 2 的幂次，为"对于所有整数 a, b, c，如果 $a \mid bc$，那么 $a \mid b$ 或 $a \mid c$"这一猜想找一个反例。

   The intent of the problem is that you find three numbers, a, b and c, that are all powers of 2 and such that a divides the product bc, but neither of the factors separately. For instance, if you pick $a = 16$, then you would need to choose b and c so that 16 doesn't divide evenly into them (they would need to be less than 16…) but so that their product *is* divisible by 16.

   这个问题的意图是让你找到三个数 a, b, c，它们都是 2 的幂，并且 a 能整除乘积 bc，但不能整除任何一个因子。例如，如果你选择 $a = 16$，那么你需要选择 b 和 c，使得 16 不能整除它们（它们需要小于 16⋯⋯），但它们的乘积可以被 16 整除。

3. The alternating sum of factorials provides an interesting example of a sequence of integers.

阶乘的交错和提供了一个有趣的整数序列的例子。

$$1! = 1$$

$$2! - 1! = 1$$

$$3! - 2! + 1! = 5$$

$$4! - 3! + 2! - 1! = 19$$

et cetera (等等)

Are they all prime? (After the first two 1's.)

(在头两个 1 之后) 它们都是素数吗?

Here's some Sage code that would test this conjecture:

这里有一些可以测试这个猜想的 Sage 代码:

```
n=1
for i in [2..8]:
   n = factorial(i) - n
   show(factor(n))
```

Of course it turns out that going out to 8 isn't quite far enough...

当然,事实证明,算到 8 还不够远⋯⋯

4. It has been conjectured that whenever p is prime, $2^p - 1$ is also prime. Find a minimal counterexample.

   有人猜想,只要 p 是素数,$2^p - 1$ 也是素数。请找一个最小的反例。

   I would definitely seek help at your friendly neighborhood CAS. In Sage you can loop over the first several prime numbers using the following syntax. `for p in [2,3,5,7,11,13]:`

我肯定会向你友好的邻居 CAS（计算机代数系统）求助。在 Sage 中，你可以使用以下语法遍历前几个素数。`for p in [2,3,5,7,11,13]:`

If you want to automate that somewhat, there is a Sage function that returns a list of all the primes in some range. So the following does the same thing.

如果你想在某种程度上自动化这个过程，有一个 Sage 函数可以返回某个范围内的所有素数列表。所以下面这个做的是同样的事情。

```
for p in primes(2,13):
```

5. True or false: The sum of any two irrational numbers is irrational. Prove your answer.

   真或假：任意两个无理数的和是无理数。证明你的答案。

   This statement and the next are negations of one another. Your answers should reflect that.

   这个陈述和下一个陈述互为否定。你的答案应该反映出这一点。

6. True or false: There are two irrational numbers whose sum is rational. Prove your answer.

   真或假：存在两个无理数，它们的和是有理数。证明你的答案。

   If a number is irrational, isn't its negative also irrational? That's actually a pretty huge hint.

   如果一个数是无理数，它的相反数不也是无理数吗？这其实是一个非常大的提示。

7. True or false: The product of any two irrational numbers is irrational. Prove your answer.

   真或假：任意两个无理数的乘积是无理数。证明你的答案。

   This one and the next are negations too. Aren't they?

   这个和下一个也是互为否定的。不是吗？

8. True or false: There are two irrational numbers whose product is rational. Prove your answer.

   真或假：存在两个无理数，它们的乘积是有理数。证明你的答案。

   The two numbers *could* be equal couldn't they?

   这两个数可以是相等的，不是吗？

9. True or false: Whenever an integer n is a divisor of the square of an integer, m^2, it follows that n is a divisor of m as well. (In symbols, $\forall n \in \mathbb{Z}, \forall m \in \mathbb{Z}, n \mid m^2 \implies n \mid m$.) Prove your answer.

   真或假：只要整数 n 是整数 m^2 的一个约数，那么 n 也是 m 的一个约数。（用符号表示为，$\forall n \in \mathbb{Z}, \forall m \in \mathbb{Z}, n \mid m^2 \implies n \mid m$。）证明你的答案。

   Hint: List all of the divisors of $36 = (2 \cdot 3)^2$. See if any of them are bigger than 6.

   提示：列出 $36 = (2 \cdot 3)^2$ 的所有约数。看看其中是否有比 6 大的。

10. In an exercise in Section 3.2 we proved that the quadratic equation $ax^2 + bx + c = 0$ has two solutions if $ac < 0$. Find a counterexample which shows that this implication cannot be replaced with a biconditional.

    在第 3.2节的一个练习中，我们证明了如果 $ac < 0$，二次方程 $ax^2 + bx + c = 0$ 有两个解。请找一个反例，说明这个蕴涵不能被替换为双条件句。

    We'd want ac to be positive (so a and c have the same sign) but nevertheless have $b^2 - 4ac > 0$. It seems that if we make b sufficiently large that could happen.

    我们希望 ac 是正的（所以 a 和 c 同号），但同时有 $b^2 - 4ac > 0$。似乎如果我们让 b 足够大，这就可以发生。

3.5 Even more direct proofs: By cases and By exhaustion 更多直接证明：分情况讨论与穷举法

Exercises — 3.5

1. Prove that if n is an odd number then $n^4 \pmod{16} = 1$.

   证明如果 n 是一个奇数，那么 $n^4 \pmod{16} = 1$。

   While one could perform fairly complicated arithmetic, expanding expression like $(16k + 13)^4$ and then regrouping to put it in the form $16q + 1$ (and one would need to do that work for each of the odd remainders modulo 16), that would be missing out on the true power of modular notation. In a " $\pmod{16}$" calculation one can simply ignore summands like $16k$ because they are $0 \pmod{16}$. Thus, for example,

   虽然可以进行相当复杂的算术运算，比如展开像 $(16k + 13)^4$ 这样的表达式，然后重新组合成 $16q + 1$ 的形式（并且需要对模 16 的每个奇数余数都进行这项工作），但这会错过模符号的真正威力。在“ $\pmod{16}$”计算中，可以简单地忽略像 $16k$ 这样的加数，因为它们是 $0 \pmod{16}$。因此，例如，

   $$(16k + 7)^4 \pmod{16} = 7^4 \pmod{16} = 2401 \pmod{16} = 1.$$

   So, essentially one just needs to compute the 4th powers of $1, 3, 5, 7, 9, 11, 13$ and 15, and then reduce them modulo 16. An even greater economy is possible if one notes that (modulo 16) many of those cases are negatives of one another – so their 4th powers are equal.

所以，基本上只需要计算 $1, 3, 5, 7, 9, 11, 13$ 和 15 的 4 次方，然后将它们对 16 取模。如果注意到（在模 16 下）许多这些情况互为相反数——所以它们的 4 次方相等，那么可以更加简化计算。

2. Prove that every prime number other than 2 and 3 has the form $6q+1$ or $6q+5$ for some integer q. (Hint: this problem involves thinking about cases as well as contrapositives.)

   证明除 2 和 3 之外的每个素数都具有 $6q+1$ 或 $6q+5$ 的形式，其中 q 是某个整数。（提示：这个问题涉及分情况讨论以及逆否命题的思考。）

   It is probably obvious that the ”cases” will be the possible remainders mod 6. Numbers of the form 6q+0 will be multiples of 6, so clearly not prime. The other forms that need to be eliminated are 6q+2, 6q+3, and 6q+4.

   很可能显而易见，“情况”将是模 6 的可能余数。形式为 6q+0 的数是 6 的倍数，所以显然不是素数。需要排除的其他形式是 6q+2、6q+3 和 6q+4。

3. Show that the sum of any three consecutive integers is divisible by 3.

   证明任意三个连续整数的和能被 3 整除。

   Write the sum as $n + (n + 1) + (n + 2)$.

   将和写成 $n + (n + 1) + (n + 2)$。

4. There is a graph known as K_4 that has 4 nodes and there is an edge between every pair of nodes. The pebbling number of K_4 has to be at least 4 since it would be possible to put one pebble on each of 3 nodes and not be able to reach the remaining node using pebbling moves. Show that the pebbling number of K_4 is actually 4.

   有一个被称为 K_4 的图，它有 4 个节点，并且每对节点之间都有一条边。K_4 的配石数至少为 4，因为可以在 3 个节点上各放一颗石子，而无法通过配石移动到达剩下的节点。证明 K_4 的配石数实际上是 4。

If there are two pebbles on any node we will be able to reach all the other nodes using pebbling moves (since every pair of nodes is connected).

如果任何节点上有两颗石子，我们将能够通过配石移动到达所有其他节点（因为每对节点都是相连的）。

5. Find the pebbling number of a graph whose nodes are the corners and whose edges are the, uhmm, edges of a cube.

   找出一个图的配石数，该图的节点是立方体的顶点，边是立方体的……呃……棱。

   It should be clear that the pebbling number is at least 8 – 7 pebbles could be distributed, one to a node, and the 8th node would be unreachable. It will be easier to play around with this if you figure out how to draw the cube graph "flattened-out" in the plane.

   应该很清楚，配石数至少是 8——可以分布 7 颗石子，每个节点一颗，这样第 8 个节点就无法到达。如果你能想出如何在平面上画出"展开"的立方体图，玩起来会更容易。

6. A *vampire number* is a $2n$ digit number v that factors as $v = xy$ where x and y are n digit numbers and the digits of v are precisely the digits in x and y in some order. The numbers x and y are known as the "fangs" of v. To eliminate trivial cases, both fangs can't end with 0.

   一个吸血鬼数是一个 $2n$ 位数 v，它可以分解为 $v = xy$，其中 x 和 y 是 n 位数，并且 v 的各位数字恰好是 x 和 y 中数字的某种排列。数 x 和 y 被称为 v 的"尖牙"。为消除平凡情况，两个尖牙都不能以 0 结尾。

   Show that there are no 2-digit vampire numbers. Show that there are seven 4-digit vampire numbers.

   证明不存在 2 位的吸血鬼数。证明存在 7 个 4 位的吸血鬼数。

The 2-digit challenge is do-able by hand (just barely). The 4 digit question certainly requires some computer assistance!

2 位数的问题用手算是可以完成的（勉强可以）。4 位数的问题肯定需要一些计算机辅助！

7. Lagrange's theorem on representation of integers as sums of squares says that every positive integer can be expressed as the sum of at most 4 squares. For example, $79 = 7^2 + 5^2 + 2^2 + 1^2$. Show (exhaustively) that 15 can not be represented using fewer than 4 squares.

   拉格朗日关于整数表示为平方和的定理指出，每个正整数都可以表示为至多 4 个平方数之和。例如，$79 = 7^2 + 5^2 + 2^2 + 1^2$。请（用穷举法）证明 15 不能用少于 4 个平方数的和来表示。

   Note that $15 = 3^2 + 2^2 + 1^2 + 1^2$. Also, if 15 were expressible as a sum of fewer than 4 squares, the squares involved would be 1, 4 and 9. It's really not that hard to try all the possibilities.

   注意 $15 = 3^2 + 2^2 + 1^2 + 1^2$。另外，如果 15 能表示为少于 4 个平方数的和，那么涉及的平方数将是 1、4 和 9。尝试所有可能性其实并不难。

8. Show that there are exactly 15 numbers x in the range $1 \leq x \leq 100$ that can't be represented using fewer than 4 squares.

   证明在 $1 \leq x \leq 100$ 的范围内，恰好有 15 个数 x 不能用少于 4 个平方数的和来表示。

   The following Sage code generates all the numbers up to 100 that *can* be written as the sum of at most 3 squares.

   下面的 Sage 代码生成了 100 以内所有可以写成至多 3 个平方数之和的数。`var('x y z')`

```
a=[s^2 for s in [1..10]]
b=[s^2 for s in [0..10]]
```

```
s = []
for x in a:
    for y in b:
        for z in b:
            s = union(s,[x+y+z])
s = Set(s)
H=Set([1..100])
show(H.intersection(s))
```

9. The *trichotomy property* of the real numbers simply states that every real number is either positive or negative or zero. Trichotomy can be used to prove many statements by looking at the three cases that it guarantees. Develop a proof (by cases) that the square of any real number is non-negative.

   实数的三分性简单地陈述了每个实数要么是正数，要么是负数，要么是零。三分性可以通过考察它所保证的三种情况来证明许多陈述。请（通过分情况讨论）给出一个证明，证明任何实数的平方都是非负的。

   By trichotomy, x is either zero, negative, or positive. If x is zero, its square is zero. If x is negative, its square is positive. If x is positive, its square is also positive.

   根据三分性，x 要么是零，要么是负数，要么是正数。如果 x 是零，它的平方是零。如果 x 是负数，它的平方是正数。如果 x 是正数，它的平方也是正数。

10. Consider the game called "binary determinant tic-tac-toe" which is played by two players who alternately fill in the entries of a 3×3 array. Player One goes first, placing 1's in the array and player Zero goes second, placing 0's. Player One's goal is that the final array have determinant 1, and player Zero's goal is that the determinant be 0. The determinant calculations are carried out mod 2.

    考虑一个名为"二进制行列式井字棋"的游戏，由两名玩家轮流填充一个 3×3 矩阵的项。玩家一先走，在矩阵中放入 1，玩家零后走，放入 0。玩家一的目标是使最终矩阵的行列式为 1，玩家零的目标是使行列式为 0。行列式计算在模 2 下进行。

    Show that player Zero can always win a game of binary determinant tic-tac-toe by the method of exhaustion.

    用穷举法证明玩家零总能赢得二进制行列式井字棋游戏。

    If you know something about determinants it would help here. The determinant will be 0 if there are two identical rows (or columns) in the finished array. Also, if there is a row or column that is all zeros, player Zero wins too. Also, cyclically permuting either rows or columns has no effect on the determinant of a binary array. This means we lose no generality in assuming player One's first move goes (say) in the upper-left corner.

    如果你了解一些关于行列式的知识，这会有帮助。如果最终的矩阵中有两行（或两列）相同，行列式将为 0。另外，如果有一行或一列全为零，玩家零也获胜。此外，对二进制矩阵的行或列进行循环置换不影响其行列式的值。这意味着我们可以不失一般性地假设玩家一的第一步棋走在（比如）左上角。

3.6 Proofs and disproofs of existential statements 存在性陈述的证明与反证

Exercises — 3.6

1. Show that there is a perfect square that is the sum of two perfect squares.

   证明存在一个完全平方数，它是两个完全平方数之和。

   Can you say "Pythagorean triple"? I thought you could.

   你会说“勾股数”吗？我想你会的。

2. Show that there is a perfect cube that is the sum of three perfect cubes.

   证明存在一个完全立方数，它是三个完全立方数之和。

   Hint: 6^3 can be expressed as such a sum.

   提示：6^3 可以表示为这样的和。

3. Show that the WOP doesn't hold in the integers. (This is an existence proof, you show that there is a subset of $\mathbb{Z}$ that doesn't have a smallest element.)

   证明良序原则在整数中不成立。（这是一个存在性证明，你要证明存在一个没有最小元素的 $\mathbb{Z}$ 子集。）

   How about even integers? Is there a smallest one? That's my example! You come up with a different one!

   偶数怎么样？有最小的偶数吗？那是我的例子！你来想一个不同的！

4. Show that the WOP doesn't hold in $\mathbb{Q}^+$.

   证明良序原则在 $\mathbb{Q}^+$ 中不成立。

   Consider the set $\{1, 1/2, 1/4, 1/8, \ldots\}$. Does it have a smallest element?

考虑集合 $\{1, 1/2, 1/4, 1/8, \ldots\}$。它有最小元素吗?

5. In the proof of Theorem 3.6.4 we weaseled out of showing that $d \mid b$. Fill in that part of the proof.

   在定理 3.6.4的证明中，我们回避了证明 $d|b$。请补全那部分证明。

   Yeah, I'm going to keep weaseling…

   是的，我将继续回避……

6. Give a proof of the unique existence of q and r in the division algorithm.

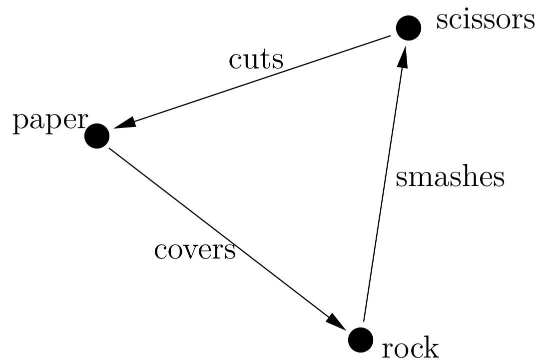
   给出除法算法中 q 和 r 唯一存在的证明。

   Unique existence proofs consist of two parts. First, just show existence. Then, show that if there were two of the things under consideration that they must in fact be equal.

   唯一存在性证明包括两部分。首先，只证明存在性。然后，证明如果存在两个所考虑的事物，它们实际上必须是相等的。

7. A *digraph* is a drawing containing a collection of points that are connected by arrows. The game known as *scissors-paper-rock* can be represented by a digraph that is *balanced* (each point has the same number of arrows going out as going in). Show that there is a balanced digraph having 5 points.

   一个有向图是一个包含由箭头连接的点集合的图画。被称为*剪刀-石头-布*的游戏可以用一个平衡的有向图来表示（每个点的出度和入度相同）。证明存在一个有 5 个点的平衡有向图。



If at first you don't succeed...

try googling "scissor paper rock lizard spock."

如果一开始不成功……

试试谷歌搜索“剪刀石头布蜥蜴斯波克”。

Chapter 4

Sets 集合

No more turkey, but I'd like some more of the bread it ate. –Hank Ketcham

不要火鸡了，但我想再来点它吃的面包。——汉克 · 凯查姆

4.1 Basic notions of set theory 集合论的基本概念

Exercises — 4.1

1. What is the power set of $\emptyset$? Hint: if you got the last exercise in the chapter you'd know that this power set has $2^0 = 1$ element.

   $\emptyset$ 的幂集是什么？提示：如果你做了本章的最后一个练习，你就会知道这个幂集有 $2^0 = 1$ 个元素。

   The power set of a set always includes the empty set as well as the whole set that we are forming the power set of. In this case they happen to coincide so $\mathcal{P}(\emptyset) = \{\emptyset\}$. Notice that $2^0 = 1$.

   一个集合的幂集总是包含空集以及我们正在构造其幂集的那个全集。在这种情况下，它们恰好重合，所以 $\mathcal{P}(\emptyset) = \{\emptyset\}$。注意 $2^0 = 1$。

2. Try iterating the power set operator. What is $\mathcal{P}(\mathcal{P}(\emptyset))$? What is $\mathcal{P}(\mathcal{P}(\mathcal{P}(\emptyset)))$?

   尝试迭代幂集运算符。$\mathcal{P}(\mathcal{P}(\emptyset))$ 是什么？$\mathcal{P}(\mathcal{P}(\mathcal{P}(\emptyset)))$ 是什么？

   I won't spoil you're fun, but as a check $\mathcal{P}(\mathcal{P}(\emptyset))$ should have 2 elements, and $\mathcal{P}(\mathcal{P}(\mathcal{P}(\emptyset)))$ should have 4.

   我不会剥夺你的乐趣，但作为检验，$\mathcal{P}(\mathcal{P}(\emptyset))$ 应该有 2 个元素，而 $\mathcal{P}(\mathcal{P}(\mathcal{P}(\emptyset)))$ 应该有 4 个。

3. Determine the following cardinalities.

   确定以下基数。

   (a) $A = \{1, 2, \{3, 4, 5\}\}$ $|A| =$______

   (b) $B = \{\{1, 2, 3, 4, 5\}\}$ $|B| =$______

   Three and one

   三和一

4. What, in Logic, corresponds the notion $\emptyset$ in Set theory?

   在逻辑学中，什么对应集合论中的概念 $\emptyset$?

   A contradiction.

   一个矛盾。

5. What, in Set theory, corresponds to the notion t (a tautology) in Logic?

   在集合论中，什么对应逻辑学中的概念 t（一个重言式)?

   The universe of discourse.

   论域。

6. What is the truth set of the proposition $P(x) =$ "3 divides x and 2 divides x"?

   命题 $P(x) =$ "3 整除 x 且 2 整除 x" 的真值集是什么?

The set of all multiples of 6.

所有 6 的倍数的集合。

7. Find a logical open sentence such that $\{0, 1, 4, 9, \ldots\}$ is its truth set.

   找一个逻辑开放句，使其真值集为 $\{0, 1, 4, 9, \ldots\}$。

   Many answers are possible. Perhaps the easiest is $\exists y \in \mathbb{Z}, x = y^2$.

   有很多可能的答案。也许最简单的是 $\exists y \in \mathbb{Z}, x = y^2$。

8. How many singleton sets are there in the power set of $\{a, b, c, d, e\}$? “Doubleton” sets?

   在 $\{a, b, c, d, e\}$ 的幂集中有多少个单元集？“双元集”呢？

   5, 10

9. How many 8 element subsets are there in

$$\mathcal{P}(\{a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p\})?$$

   在 $\mathcal{P}(\{a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p\})$ 中有多少个 8 元子集？

   $\binom{16}{8} = 12870$

10. How many singleton sets are there in the power set of $\{1, 2, 3, \ldots n\}$?

    在 $\{1, 2, 3, \ldots n\}$ 的幂集中有多少个单元集？

    n

4.2 Containment 包含关系

Exercises — 4.2

1. Insert either $\in$ or $\subseteq$ in the blanks in the following sentences (in order to produce true sentences).

   在下列句子的空白处填入 $\in$ 或 $\subseteq$（以产生真命题）。

   i) 1 ______ $\{3, 2, 1, \{a, b\}\}$ iii) $\{a, b\}$ ______ $\{3, 2, 1, \{a, b\}\}$

   ii) $\{a\}$ ______ $\{a, \{a, b\}\}$ iv) $\{\{a, b\}\}$ ______ $\{a, \{a, b\}\}$

   $\in, \subseteq, \in, \subseteq$

2. Suppose that p is a prime, for each n in $\mathbb{Z}^+$, define the set $P_n = \{x \in \mathbb{Z}^+ \mid p^n \mid x\}$. Conjecture and prove a statement about the containments between these sets.

   假设 p 是一个素数，对于 $\mathbb{Z}^+$ 中的每个 n，定义集合 $P_n = \{x \in \mathbb{Z}^+ \mid p^n \mid x\}$。猜想并证明一个关于这些集合之间包含关系的陈述。

   When $p = 2$ we have seen these sets. P_1 is the even numbers, P_2 is the doubly-even numbers, etc.

   当 $p = 2$ 时，我们见过这些集合。P_1 是偶数集，P_2 是双偶数集，等等。

3. Provide a counterexample to dispel the notion that a subset must have fewer elements than its superset.

   提供一个反例来反驳“子集的元素数量必须少于其父集”的观点。

   A subset is called *proper* if it is neither empty nor equal to the superset. If we are talking about finite sets then the proper subsets do indeed have fewer elements than the supersets. Among infinite sets it is possible to have proper subsets having the same number of elements as their superset, for example there are just as many even natural numbers as there are natural numbers all told.

如果一个子集既非空集也非等于其父集，则称之为真子集。如果我们讨论的是有限集，那么真子集的元素数量确实少于其父集。在无限集中，真子集可能与其父集拥有相同数量的元素，例如，偶数的数量与所有自然数的数量一样多。

4. We have seen that $A \subseteq B$ corresponds to $M_A \implies M_B$. What corresponds to the contrapositive statement?

   我们已经看到 $A \subseteq B$ 对应于 $M_A \implies M_B$。那么逆否命题对应什么？

   Turn “logical negation” into “set complement” and reverse the direction of the inclusion.

   将“逻辑否定”变为“集合补集”，并反转包含的方向。

5. Determine two sets A and B such that both of the sentences $A \in B$ and $A \subseteq B$ are true.

   确定两个集合 A 和 B，使得句子 $A \in B$ 和 $A \subseteq B$ 都为真。

   The smallest example I can think of would be $A = \emptyset$ and $B = \{\emptyset\}$. You should come up with a different example.

   我能想到的最小例子是 $A = \emptyset$ 和 $B = \{\emptyset\}$。你应该想出一个不同的例子。

6. Prove that the set of perfect fourth powers is contained in the set of perfect squares.

   证明四次方数集合包含于完全平方数集合中。

   It would probably be helpful to have precise definitions of the sets described in the problem. The fourth powers are

   对问题中描述的集合有精确的定义可能会很有帮助。四次方数是

$$F = \{x \mid \exists y \in \mathbb{Z}, x = y^4\}.$$

   The squares are

   平方数是

$$S = \{x \mid \exists z \in \mathbb{Z}, x = z^2\}.$$

   To show that one set is contained in another, we need to show that the first set's membership criterion implies that of the second set.

   要证明一个集合包含在另一个集合中，我们需要证明第一个集合的成员资格标准蕴涵了第二个集合的成员资格标准。

	Intersection version	Union version
Commutative laws	$A \cap B = B \cap A$	$A \cup B = B \cup A$
Associative laws	$A \cap (B \cap C) = (A \cap B) \cap C$	$A \cup (B \cup C) = (A \cup B) \cup C$
Distributive laws	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
DeMorgan's laws	$\overline{A \cap B} = \overline{A} \cup \overline{B}$	$\overline{A \cup B} = \overline{A} \cap \overline{B}$
Double complement	$\overline{\overline{A}} = A$	same
Complementarity	$A \cap \overline{A} = \emptyset$	$A \cup \overline{A} = U$
Identity laws	$A \cap U = A$	$A \cup \emptyset = A$
Domination	$A \cap \emptyset = \emptyset$	$A \cup U = U$
Idempotence	$A \cap A = A$	$A \cup A = A$
Absorption	$A \cap (A \cup B) = A$	$A \cup (A \cap B) = A$

4.3 Set operations 集合运算

Exercises — 4.3

1. Let $A = \{1, 2, \{1, 2\}, b\}$ and let $B = \{a, b, \{1, 2\}\}$. Find the following:

   令 $A = \{1, 2, \{1, 2\}, b\}$ 且 $B = \{a, b, \{1, 2\}\}$。求下列集合：

   (a) $A \cap B$

   $\{b, \{1, 2\}\}$

   (b) $A \cup B$

   $\{1, 2, a, b, \{1, 2\}\}$

   (c) $A \setminus B$

   $\{1, 2\}$

   (d) $B \setminus A$

   $\{a\}$

   (e) $A \triangle B$

   $\{1, 2, a\}$

2. In a standard deck of playing cards one can distinguish sets based on face-value and/or suit. Let $A, 2, \ldots 9, 10, J, Q$ and K represent the sets of cards having the various face-values. Also, let $\heartsuit$, $\spadesuit$, $\clubsuit$ and $\diamondsuit$ be the sets of cards having the possible suits. Find the following

   在一副标准扑克牌中，可以根据牌面值和/或花色来区分集合。令 $A, 2, \ldots 9, 10, J, Q$ 和 K 代表具有各种牌面值的牌的集合。另外，令 $\heartsuit$, $\spadesuit$, $\clubsuit$ 和 $\diamondsuit$ 为具有可能花色的牌的集合。求下列集合：

   (a) $A \cap \heartsuit$

   This is just the ace of hearts. 这就是红桃 A。

(b) $A \cup \heartsuit$

All of the hearts and the other three aces. 所有的红桃牌和其他三张 A。

(c) $J \cap (\spadesuit \cup \heartsuit)$

These two cards are known as the one-eyed jacks. 这两张牌被称为单眼 J。

(d) $K \cap \heartsuit$

The king of hearts, a.k.a. the suicide king. 红桃 K，又名自杀王。

(e) $A \cap K$

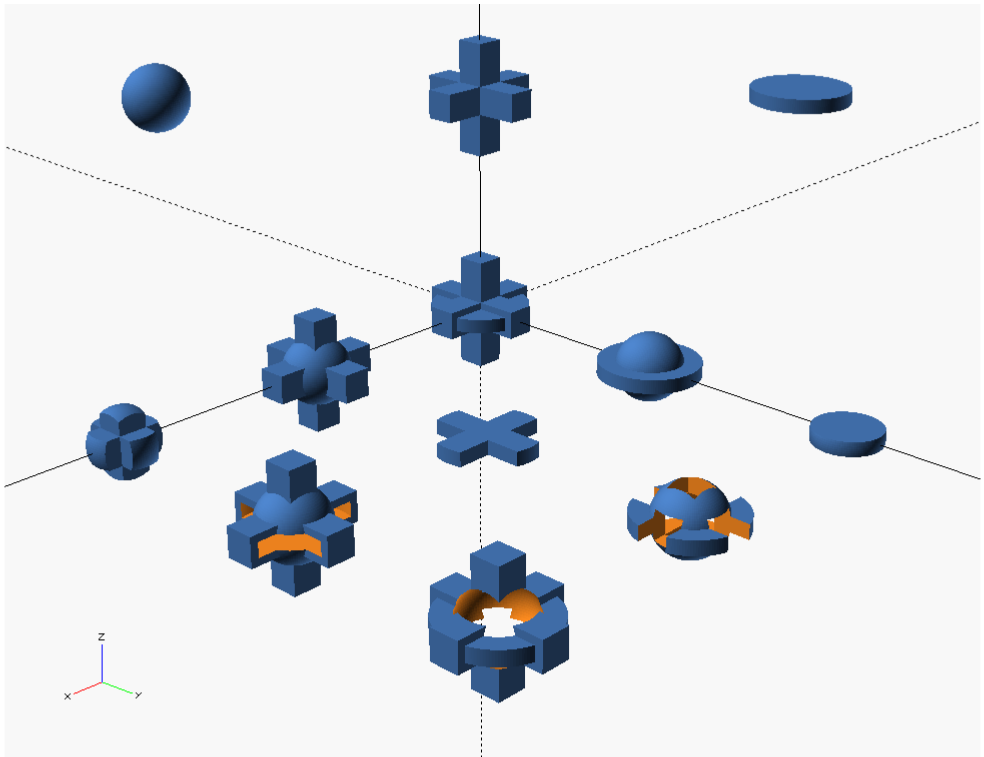
$\emptyset$

(f) $A \cup K$

Eight cards: all four kings and all four aces. 八张牌：所有四张 K 和所有四张 A。

3. The following is a screenshot from the computational geometry program OpenSCAD (very handy for making models for 3-d printing…) In computational geometry we use the basic set operations together with a few other types of transformations to create interesting models using simple components. Across the top of the image below we see 3 sets of points in $\mathbb{R}^3$, a ball, a sort of 3-dimensional plus sign, and a disk. Let's call the ball A, the plus sign B and the disk C. The nine shapes shown below them are made from A, B and C using union, intersection and set difference. Identify them!

下面是计算几何程序 OpenSCAD 的截图（对于制作 3D 打印模型非常方便……）。在计算几何中，我们使用基本的集合运算以及其他一些类型的变换，用简单的组件创建有趣的模型。在下图的顶部，我们看到 $\mathbb{R}^3$ 中的 3 个点集：一个球体、一个类似三维加号的形状和一个圆盘。我们称球体为 A，加号为 B，圆盘为 C。它们下方的九个形状是由 A, B, C 使用并集、交集和差集运算得到的。请识别它们！



4. Do element-chasing proofs (show that an element is in the left-hand side if and only if it is in the right-hand side) to prove each of the following set equalities.

   进行元素追踪证明（证明一个元素在左边当且仅当它在右边）来证明以下每个集合等式。

   (a) $\overline{A \cap B} \;=\; \overline{A} \cup \overline{B}$

   (b) $A \cup B \;=\; A \cup (\overline{A} \cap B)$

   (c) $A \triangle B \;=\; (A \cup B) \setminus (A \cap B)$

   (d) $(A \cup B) \setminus C \;=\; (A \setminus C) \cup (B \setminus C)$

   Here's the first one (although I'm omiting justifications for each step.

   这是第一个（尽管我省略了每一步的理由）。

$$
\begin{gathered}
x \in \overline{A \cap B} \\
\iff \neg(x \in A \cap B) \\
\iff \neg(x \in A \;\wedge\; x \in B) \\
\iff \neg(x \in A) \;\vee\; \neg(x \in B) \\
\iff x \in \overline{A} \;\vee\; x \in \overline{B} \\
\iff x \in \overline{A} \cup \overline{B}
\end{gathered}
$$

5. For each positive integer n, we'll define an interval I_n by

$$I_n = [-n, 1/n).$$

   对于每个正整数 n，我们定义一个区间 I_n 为

$$I_n = [-n, 1/n).$$

Find the union and intersection of all the intervals in this infinite family.

找出这个无限族中所有区间的并集和交集。

$$\bigcup_{n\in\mathbb{N}} I_n \quad =$$

$$\bigcap_{n\in\mathbb{N}} I_n \quad =$$

To better understand what is going on, first figure out what the first three or four intervals actually are.

为了更好地理解情况，首先弄清楚前三四个区间实际上是什么。

$$I_1 = \underline{\hspace{4cm}}$$

$$I_2 = \underline{\hspace{4cm}}$$

$$I_3 = \underline{\hspace{4cm}}$$

$$I_4 = \underline{\hspace{4cm}}$$

Any negative real number r will be in the intersection only if $r \geq -1$. Certainly 0 is in the intersection since it is in each of the intervals. Are there any positive numbers in the intersection?

任何负实数 r 只有在 $r \geq -1$ 时才会在交集中。当然，0 在交集中，因为它在每个区间里。交集中有正数吗?

In order to be in the union a real number just needs to be in *one* of the intervals.

一个实数只要在一个区间里，它就在并集中。

6. There is a set X such that, for all sets A, we have $X \triangle A = A$. What is X?

   存在一个集合 X，使得对于所有集合 A，都有 $X \triangle A = A$。X 是什么?

7. There is a set Y such that, for all sets A, we have $Y \triangle A = \overline{A}$. What is Y?

   存在一个集合 Y，使得对于所有集合 A，都有 $Y \triangle A = \overline{A}$。$Y$ 是什么?

   One of the answers to the last two questions is $\emptyset$ and the other is U. Decide which is which.

   最后两个问题的一个答案是 $\emptyset$，另一个是 U。请判断哪个是哪个。

8. In proving a set-theoretic identity, we are basically showing that two sets are equal. One reasonable way to proceed is to show that each is contained in the other. Prove that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ by showing that $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$ and $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$.

   在证明一个集合论恒等式时，我们基本上是在证明两个集合相等。一个合理的方法是证明它们互相包含。通过证明 $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$ 且 $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$ 来证明 $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$。

9. Prove that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ by showing that $A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$ and $(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C)$.

   通过证明 $A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$ 且 $(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C)$ 来证明 $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$。

   This exercise, as well as the previous one, is really just about converting set-theoretic statements into their logical equivalents, applying some rules of logic that we've already verified, and then returning to a set-theoretic version of things.

   这个练习，以及前一个练习，实际上只是关于将集合论的陈述转换为它们的逻辑等价物，应用一些我们已经验证过的逻辑规则，然后回到集合论的版本。

10. Prove the set-theoretic versions of DeMorgan's laws using the technique discussed in the previous problems.

    使用前面问题中讨论的技巧来证明集合论版本的德摩根定律。

11. The previous technique (showing that $A = B$ by arguing that $A \subseteq B \wedge B \subseteq A$) will have an outline something like

    前一个技巧（通过论证 $A \subseteq B \wedge B \subseteq A$ 来证明 $A = B$）的大纲大致如下：

    > *Proof:* First we will show that $A \subseteq B$.
    > Towards that end, suppose $x \in A$.
    > 首先我们将证明 $A \subseteq B$。
    > 为此，假设 $x \in A$。
    >
    > $$\vdots$$
    >
    > Thus $x \in B$.
    > 因此 $x \in B$。
    > Now, we will show that $B \subseteq A$.
    > Suppose that $x \in B$.
    > 现在，我们将证明 $B \subseteq A$。
    > 假设 $x \in B$。
    >
    > $$\vdots$$
    >
    > Thus $x \in A$.
    > 因此 $x \in A$。
    > Therefore $A \subseteq B \wedge B \subseteq A$ so we conclude that $A = B$.
    > 因此 $A \subseteq B \wedge B \subseteq A$，所以我们得出结论 $A = B$。
    > Q.E.D.

    Formulate a proof that $A \triangle B = (A \cup B) \setminus (A \cap B)$ that follows this outline.

构建一个遵循此大纲的证明，证明 $A \triangle B = (A \cup B) \setminus (A \cap B)$。

The definition of $A \triangle B$ is $(A \setminus B) \cup (B \setminus A)$. The definition of $X \setminus Y$ is $X \cap \overline{Y}$. Restating things in terms of $\cap$ and $\cup$ (and complementation) should help. So your first few lines should be:

$A \triangle B$ 的定义是 $(A \setminus B) \cup (B \setminus A)$。$X \setminus Y$ 的定义是 $X \cap \overline{Y}$。用 $\cap$ 和 $\cup$（以及补集）来重述问题应该会有帮助。所以你的前几行应该是：

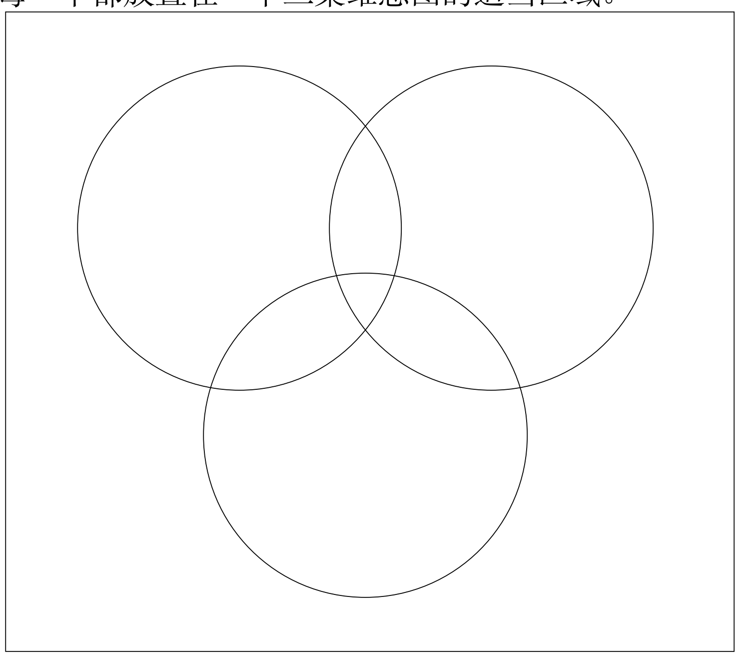
> Suppose $x \in A \triangle B$.
>
> 假设 $x \in A \triangle B$。Then, by definition, $x \in (A \setminus B) \cup (B \setminus A)$.
>
> 那么，根据定义，$x \in (A \setminus B) \cup (B \setminus A)$。
>
> So, $x \in (A \cap \overline{B}) \cup (B \cap \overline{A})$.
>
> 所以，$x \in (A \cap \overline{B}) \cup (B \cap \overline{A})$。
>
> $\vdots$

4.4 Venn diagrams 维恩图

Exercises — 4.4

1. Let $A = \{1, 2, 4, 5\}$, $B = \{2, 3, 4, 6\}$, and $C = \{1, 2, 3, 4\}$. Place each of the elements $1, \ldots, 6$ in the appropriate regions of a three-set Venn diagram.

   令 $A = \{1, 2, 4, 5\}$, $B = \{2, 3, 4, 6\}$, 和 $C = \{1, 2, 3, 4\}$。将元素 $1, \ldots, 6$ 中的每一个都放置在一个三集维恩图的适当区域。

   The center region contains 2 and 4.

   中心区域包含 2 和 4。

2. Prove or disprove:

   证明或证伪:

$$(A \cap C \subseteq B \cap C) \implies A \subseteq B$$

What will be the implications of the region $A \cap \overline{B} \cap \overline{C}$ being non-empty?

区域 $A \cap \overline{B} \cap \overline{C}$ 非空会有什么影响?

3. Venn diagrams are usually made using simple closed curves with no further restrictions. Try creating Venn diagrams for 3, 4 and 5 sets (in general position) using rectangular simple closed curves.

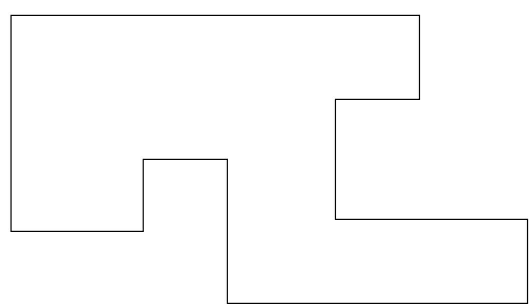
   维恩图通常是用没有进一步限制的简单闭合曲线制作的。尝试使用矩形简单闭合曲线为 3、4 和 5 个集合（处于一般位置）创建维恩图。

   I found it easier to experiment by making my drawings on graph paper. I never did manage to draw the 5 set Venn diagram with just rectangles…probably just a lack of persistence.

   我发现在方格纸上画图做实验更容易。我从来没能只用矩形画出 5 个集合的维恩图……可能只是缺乏毅力。

4. We call a curve *rectilinear* if it is made of line segments that meet at right angles. If you have ever played with an Etch-a-Sketch you'll know what we mean by the term "rectilinear." The following example of a rectilinear curve may also help to clarify this notion.

   如果一条曲线是由以直角相交的线段构成的，我们称之为直线型的。如果你玩过 Etch-a-Sketch，你就会明白我们所说的"直线型"是什么意思。下面这个直线型曲线的例子也可能有助于澄清这个概念。

   Use rectilinear simple closed curves to create a Venn diagram for 5 sets.

   使用直线型简单闭合曲线为 5 个集合创建一个维恩图。

   Of course, rectangles are rectilinear, so one could use the solution from the previous problem (if, unlike me, you were persistant enough to find it). Otherwise, start with the 4 set diagram made with rectangles and use your 5th (rectilinear) curve to split each region into 2 – don't forget to split the region on the outside too.

当然，矩形是直线型的，所以可以使用上一个问题的解（如果你不像我一样，有足够的毅力找到它的话）。否则，从用矩形制作的 4 个集合的图开始，用你的第 5 条（直线型）曲线将每个区域分成 2 个——别忘了也要分割外面的区域。

5. Argue as to why rectilinear curves will suffice to build any Venn diagram.

   论证为什么直线型曲线足以构建任何维恩图。

   Fortunately the instructions don't say to *prove* that rectilinear curves will always suffice, so we can be less rigorous. Try to argue as to why it will alway be possible to add one more rectilinear curve to an existing Venn diagram and split every region into two. One might also argue that any continuous curve can be approximated using rectilinear curves. So if a Venn diagram can be constructed using continuous curves we can also get the job done with rectilinear curves.

   幸运的是，说明没有要求证明直线型曲线总是足够的，所以我们可以不那么严谨。试着论证为什么总能在一个现有的维恩图中再添加一条直线型曲线，并将每个区域分成两个。也可以论证说，任何连续曲线都可以用直线型曲线来近似。所以如果一个维恩图可以用连续曲线构建，我们也可以用直线型曲线完成这项工作。

6. Find the disjunctive normal form of $A \cap (B \cup C)$.

   求 $A \cap (B \cup C)$ 的析取范式。

   $(A \cap B \cap \overline{C}) \cup (A \cap \overline{B} \cap C)$

7. Find the disjunctive normal form of $(A \triangle B) \triangle C$

   求 $(A \triangle B) \triangle C$ 的析取范式。

   It is $(A \cap \overline{B} \cap \overline{C}) \cup (\overline{A} \cap B \cap \overline{C}) \cup (\overline{A} \cap \overline{B} \cap C)$. Now find the disjunctive normal form of $A \triangle (B \triangle C)$.

   它是 $(A \cap \overline{B} \cap \overline{C}) \cup (\overline{A} \cap B \cap \overline{C}) \cup (\overline{A} \cap \overline{B} \cap C)$。现在求 $A \triangle (B \triangle C)$ 的析取范式。

8. The prototypes for the *modus ponens* and *modus tollens* argument forms are the following:

肯定前件式和否定后件式论证形式的原型如下：

All men are mortal. Socrates is a man. Therefore Socrates is mortal.	and (和)	All men are mortal. Zeus is not mortal. Therefore Zeus is not a man.
所有的人都会死。 苏格拉底是人。 因此苏格拉底会死。	and (和)	所有的人都会死。 宙斯不会死。 因此宙斯不是人。

Illustrate these arguments using Venn diagrams.

使用维恩图来说明这些论证。

The statement "All men are mortal" would be interpreted on a Venn diagram by showing the set of "All men" as being entirely contained within the set of "mortal beings." Socrates is an element of the inner set. Zeus, on the other hand, lies outside of the outer set.

"所有的人都会死"这个陈述在维恩图上会被解释为，"所有的人"的集合完全包含在"会死的生物"的集合之内。苏格拉底是内层集合的一个元素。而宙斯，则位于外层集合之外。

9. Use Venn diagrams to convince yourself of the validity of the following containment statement

   使用维恩图来说服自己以下包含陈述的有效性

   $$(A \cap B) \cup (C \cap D) \;\subseteq\; (A \cup C) \cap (B \cup D).$$

   Now prove it!

   现在证明它!

   Obviously we'll need one of the 4-set Venn diagrams.

   显然我们需要一个 4 集维恩图。

10. Use Venn diagrams to show that the following set equivalence is false.

    使用维恩图来证明以下集合等价式是错误的。

    $$(A \cup B) \cap (C \cup D) \;=\; (A \cup C) \cap (B \cup D)$$

    After constructing Venn diagrams for both sets you should be able to see that there are 4 regions where they differ. One is $A \cap B \cap \overline{C} \cap \overline{D}$. What are the other three?

    为两个集合构建维恩图后，你应该能看到它们有 4 个区域不同。一个是 $A \cap B \cap \overline{C} \cap \overline{D}$。另外三个是什么?

4.5 Russell's Paradox 罗素悖论

Exercises — 4.5

1. Verify that $(A \implies \neg A) \wedge (\neg A \implies A)$ is a logical contradiction in two ways: by filling out a truth table and using the laws of logical equivalence.

   通过两种方式验证 $(A \implies \neg A) \wedge (\neg A \implies A)$ 是一个逻辑矛盾：填写真值表和使用逻辑等价定律。

   In order to get started on this you'll need to convert the conditionals into equivalent disjunctions. Recall that $X \implies Y \equiv \neg X \vee Y$.

   为了开始这个问题，你需要将条件句转换为等价的析取式。回想一下 $X \implies Y \equiv \neg X \vee Y$。

2. One way out of Russell's paradox is to declare that the collection of sets that don't contain themselves as elements is not a set itself. Explain how this circumvents the paradox.

   摆脱罗素悖论的一种方法是宣称“所有不包含自身作为元素的集合”的集合本身不是一个集合。解释这如何规避了悖论。

   If it's not a set then it doesn't necessarily have to have the property that we can be *sure* whether an element is in it or not.

   如果它不是一个集合，那么它就不必一定具有我们能*确定*一个元素是否在其中的性质。

Chapter 5

Proof techniques II — Induction 证明技巧 II — 归纳法

5.1 The principle of mathematical induction 数学归纳法原理

Exercises — 5.1

1. Consider the sequence of numbers that are 1 greater than a multiple of 4. (Such numbers are of the form $4j + 1$.)

   考虑一个数列，其中的数比 4 的倍数大 1。(这样的数形式为 $4j + 1$。)

   $$1, 5, 9, 13, 17, 21, 25, 29, \ldots$$

   The sum of the first several numbers in this sequence can be expressed as a polynomial.

   这个序列前几个数的和可以用一个多项式表示。

   $$\sum_{j=0}^{n} 4j + 1 = 2n^2 + 3n + 1$$

Complete the following table in order to provide evidence that the formula above is correct.

完成下表，为上述公式的正确性提供证据。

n	$\sum_{j=0}^{n} 4j+1$	$2n^2+3n+1$
0	1	1
1	$1+5=6$	$2\cdot 1^2+3\cdot 1+1=6$
2	$1+5+9=$ 15	$2\cdot 2^2+3\cdot 2+1=15$
3	$1+5+9+13=28$	$2\cdot 3^2+3\cdot 3+1=28$
4		

I'm leaving the very last one for you to do.

我把最后一个留给你做。

2. What is wrong with the following inductive proof of "all horses are the same color."?

下面这个关于"所有马都是同一种颜色"的归纳证明错在哪里?

Theorem Let H be a set of n horses, all horses in H are the same color.

定理 设 H 是一个包含 n 匹马的集合，H 中所有的马都是同一种颜色。

Proof: We proceed by induction on n.

我们对 n 进行归纳。

Basis: Suppose H is a set containing 1 horse. Clearly this horse is the same color as itself.

基础步骤：假设 H 是一个只包含 1 匹马的集合。显然这匹马和它自己是同一种颜色。

Inductive step: Given a set of $k+1$ horses H we can construct two sets of k horses. Suppose $H=\{h_1,h_2,h_3,\ldots h_{k+1}\}$. Define $H_a=\{h_1,h_2,h_3,\ldots h_k\}$ (i.e. H_a contains just the first k horses) and $H_b=\{h_2,h_3,h_4,\ldots h_{k+1}\}$ (i.e. H_b contains the last k horses). By the inductive hypothesis both these sets contain horses that are "all the same color." Also, all the horses from h_2 to h_k are in both sets so both H_a and H_b contain only horses of this (same) color. Finally, we conclude that all the horses in H are the same color.

归纳步骤: 给定一个包含 $k+1$ 匹马的集合 H,我们可以构造出两个包含 k 匹马的集合。假设 $H = \{h_1, h_2, h_3, \ldots h_{k+1}\}$。定义 $H_a = \{h_1, h_2, h_3, \ldots h_k\}$（即 H_a 只包含前 k 匹马）和 $H_b = \{h_2, h_3, h_4, \ldots h_{k+1}\}$（即 H_b 包含后 k 匹马)。根据归纳假设，这两个集合中的马都"是同一种颜色"。并且，从 h_2 到 h_k 的所有马都在这两个集合中，所以 H_a 和 H_b 都只包含这种（相同）颜色的马。最后，我们得出结论，H 中所有的马都是同一种颜色。

Q.E.D.

Look carefully at the stage from $n = 2$ to $n = 3$.

仔细观察从 $n = 2$ 到 $n = 3$ 的阶段。

3. For each of the following theorems, write the statement that must be proved for the basis – then prove it, if you can!

   对于以下每个定理，写出基础步骤中必须证明的陈述——然后，如果可以的话，证明它！

   (a) The sum of the first n positive integers is $(n^2 + n)/2$.

   前 n 个正整数的和是 $(n^2 + n)/2$。

   The sum of the first 0 positive integers is $(0^2 + 0)/2$. Or, if you prefer to start with something rather than nothing: The sum of the first 1 positive integers is $(1^2 + 1)/2$.

   前 0 个正整数的和是 $(0^2 + 0)/2$。或者，如果你喜欢从有东西开始而不是没有：前 1 个正整数的和是 $(1^2 + 1)/2$。

   (b) The sum of the first n (positive) odd numbers is n^2.

   前 n 个（正）奇数的和是 n^2。

   The sum of the first 0 positive odd numbers is 0^2. Or, the sum of the first 1 positive odd numbers is 1^2.

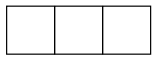
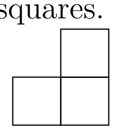
   前 0 个正奇数的和是 0^2。或者，前 1 个正奇数的和是 1^2。

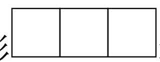
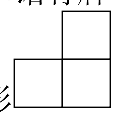
   (c) If n coins are flipped, the probability that all of them are "heads" is $1/2^n$.

   如果抛掷 n 枚硬币，它们全部为"正面"的概率是 $1/2^n$。

   If 1 coin is flipped, the the probability that it is "heads" is $1/2$. Or if we try it when $n = 0$, "If no coins are flipped the probability that all of them are heads is 1. Does that make sense to you? Is it reasonable that we would say it is 100% certain that all of the coins are heads in a set that doesn't contain *any* coins?

   如果抛掷 1 枚硬币，它为"正面"的概率是 $1/2$。或者如果我们尝试 $n = 0$ 的情况，"如果不抛掷硬币，它们全部为正面的概率是 1。"这对你来说有意义吗？在一个不包含任何硬币的集合中，我们说 100

(d) Every $2^n \times 2^n$ chessboard – with one square removed – can be tiled perfectly[1] by L-shaped trominoes. (A trominoe is like a domino but made up of 3 little squares. There are two kinds, straight  and L-shaped . This problem is only concerned with the L-shaped trominoes.)

每个 $2^n \times 2^n$ 的棋盘——移除一个方格后——都可以用 L 形的三格骨牌完美铺设[2]。(三格骨牌像多米诺骨牌，但由 3 个小方块组成。有两种：直形和 L 形。这个问题只关心 L 形的三格骨牌。)

If $n = 1$ we have: “Every 2×2 chessboard – with one square removed can be tiled perfectly by L-shaped trominoes. This version is trivial to prove. Try formulating the $n = 0$ case.

如果 $n = 1$，我们得到：“每个移除一个方格的 2×2 棋盘都可以用 L 形的三格骨牌完美铺设。” 这个版本证明起来很简单。尝试构思 $n = 0$ 的情况。

[1] Here, “perfectly tiled” means that every trominoe covers 3 squares of the chessboard (nothing hangs over the edge) and that every square of the chessboard is covered by some trominoe.

[2] 这里，“完美铺设” 意味着每个三格骨牌覆盖棋盘上的 3 个方格（没有任何部分悬在边缘之外），并且棋盘上的每个方格都被某个三格骨牌覆盖。

4. Suppose that the rules of the game for PMI were changed so that one did the following:

   - Basis. Prove that $P(0)$ is true.
   - Inductive step. Prove that for all k, P_k implies P_{k+2}

   假设数学归纳法的游戏规则被改变，变为如下操作：

   - 基础步骤。证明 $P(0)$ 为真。
   - 归纳步骤。证明对于所有 k，P_k 蕴涵 P_{k+2}。

   Explain why this would not constitute a valid proof that P_n holds for all natural numbers n. How could we change the basis in this outline to obtain a valid proof?

   解释为什么这不能构成一个证明 P_n 对所有自然数 n 成立的有效证明。我们如何改变这个大纲中的基础步骤来得到一个有效的证明？

   In this modified version, $P(0)$ is not going to imply $P(1)$, and in fact, none of the odd numbered statements will be proven. If we change the basis so that we prove both $P(0)$ and $P(1)$, all the even statements will be implied by $P(0)$ being true and all the odd statements get forced because $P(1)$ is true.

   在这个修改后的版本中，$P(0)$ 不会蕴涵 $P(1)$，事实上，所有奇数编号的陈述都不会被证明。如果我们改变基础步骤，使得我们同时证明 $P(0)$ 和 $P(1)$，那么所有偶数陈述将由 $P(0)$ 为真所蕴涵，而所有奇数陈述则因为 $P(1)$ 为真而被确定。

5. If we wanted to prove statements that were indexed by the integers,

$$\forall z \in \mathbb{Z}, \ P_z,$$

   what changes should be made to PMI?

如果我们想要证明由整数索引的陈述,

$$\forall z \in \mathbb{Z},\ P_z,$$

应该对数学归纳法做出什么改变?

A quick change would be to replace $\forall k,\ P_k \implies P_{k+1}$ in the inductive step with $\forall k,\ P_k \iff P_{k+1}$. While this would do the trick, a slight improvement is possible, if we treat the positive and negative cases for k separately.

一个快速的改变是在归纳步骤中用 $\forall k,\ P_k \iff P_{k+1}$ 替换 $\forall k,\ P_k \implies P_{k+1}$。虽然这能解决问题，但如果我们分开处理 k 的正负情况，可能会有轻微的改进。

6. In Calculus you learned the *power rule* for finding derivatives of powers of x.

$$(x^p)' = p \cdot x^{p-1}.$$

在微积分中，你学习了求 x 幂次导数的幂法则。

$$(x^p)' = p \cdot x^{p-1}.$$

The power rule can be proved using mathematical induction. You will need to use the Liebniz rule (a.k.a. the product rule) which states that

幂法则可以用数学归纳法证明。你将需要使用莱布尼茨法则（也叫乘法法则），该法则陈述为

$$(f(x) \cdot g(x))' = f'(x) \cdot g(x) + f(x) \cdot g'(x)$$

and a few other basic facts from Calculus.

以及一些其他微积分的基本事实。

To get started, notice that $(x^0)' = 0$, since x^0 is really, really close to just being the constant 1 (the problem is that it's undefined when

$x = 0$, but other than that one value for x, the zero-th power is 1.) Does this value (0) agree with what the power rule gives us? You'll use the Leibniz rule in the inductive step, by noting that x^{k+1} can be broken up into the product $x^1 \cdot x^k$.

首先，注意到 $(x^0)' = 0$，因为 x^0 非常非常接近于常数 1（问题在于当 $x = 0$ 时它未定义，但除了那个 x 值，零次幂就是 1）。这个值（0）与幂法则给出的结果一致吗？你将在归纳步骤中使用莱布尼茨法则，通过注意到 x^{k+1} 可以分解为乘积 $x^1 \cdot x^k$。

5.2 Formulas for sums and products 和与积的公式

Exercises — 5.2

1. Write an inductive proof of the formula for the sum of the first n cubes.

   写出前 n 个立方数之和公式的归纳证明。

Theorem.

$$\forall n \in \mathbb{N}, \sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2$$

Proof: (By mathematical induction)

(通过数学归纳法)

Base case: $(n = 1)$

基础情形: $(n = 1)$ For the base case, note that when $n = 1$ we have

对于基础情形，注意到当 $n = 1$ 时我们有

$$\sum_{k=1}^{n} k^3 = 1$$

and

且

$$\left(\frac{n(n+1)}{2}\right)^2 = 1.$$

Inductive step:

归纳步骤:

Suppose that $m > 1$ is an integer such that

假设 $m > 1$ 是一个整数，使得

$$\sum_{k=1}^{m} k^3 = \left(\frac{m(m+1)}{2}\right)^2$$

Add $(m+1)^3$ to both sides to obtain

两边同时加上 $(m+1)^3$ 得到

$$(m+1)^3 + \sum_{k=1}^{m} k^3 = \left(\frac{m(m+1)}{2}\right)^2 + (m+1)^3.$$

Thus

因此

$$\begin{aligned}
\sum_{k=1}^{m+1} k^3 &= \left(\frac{m^2(m+1)^2}{4}\right) + \frac{4(m+1)^3}{4} \\
&= \left(\frac{m^2(m+1)^2 + 4(m+1)^3}{4}\right) \\
&= \left(\frac{(m+1)^2(m^2 + 4(m+1))}{4}\right) \\
&= \left(\frac{(m+1)^2(m^2 + 4m + 4)}{4}\right) \\
&= \left(\frac{(m+1)^2(m+2)^2}{4}\right) \\
&= \left(\frac{(m+1)(m+2)}{2}\right)^2
\end{aligned}$$

Q.E.D.

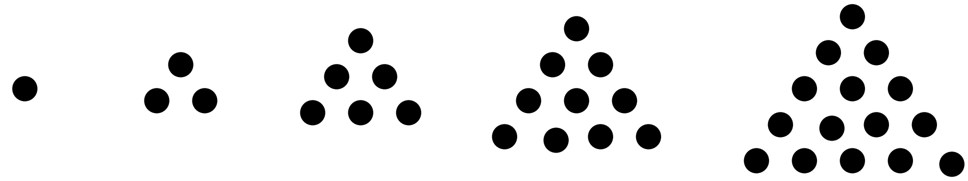
2. Find a formula for the sum of the first n fourth powers.

   找出前 n 个四次方数之和的公式。

$$\frac{n \cdot (n+1) \cdot (2n+1) \cdot (3n^2+3n-1)}{30}$$

3. The sum of the first n natural numbers is sometimes called the n-th triangular number T_n. Triangular numbers are so-named because one can represent them with triangular shaped arrangements of dots.

   前 n 个自然数的和有时被称为第 n 个三角数T_n。三角数因此得名，因为可以用三角形排列的点来表示它们。

   The first several triangular numbers are 1, 3, 6, 10, 15, et cetera. Determine a formula for the sum of the first n triangular numbers $\left(\sum_{i=1}^{n} T_i\right)$ and prove it using PMI.

   前几个三角数是 1，3，6，10，15，等等。确定前 n 个三角数之和 $\left(\sum_{i=1}^{n} T_i\right)$ 的公式，并用数学归纳法证明它。

   The formula is $\frac{n(n+1)(n+2)}{6}$.

   公式是 $\frac{n(n+1)(n+2)}{6}$。

4. Consider the alternating sum of squares:

考虑平方数的交错和：

$$1$$
$$1-4=-3$$
$$1-4+9=6$$
$$1-4+9-16=-10$$
$$\text{et cetera}$$

Guess a general formula for $\sum_{i=1}^{n}(-1)^{i-1}i^2$, and prove it using PMI.

猜测 $\sum_{i=1}^{n}(-1)^{i-1}i^2$ 的通用公式，并用数学归纳法证明它。

Theorem.

$$\forall n \in \mathbb{N},\ \sum_{i=1}^{n}(-1)^{i-1}i^2 \;=\; (-1)^{n-1}\frac{n(n+1)}{2}$$

Proof: (By mathematical induction)

(通过数学归纳法)

Base case: $(n=1)$

基础情形： $(n=1)$ For the base case, note that when $n=1$ we have

对于基础情形，注意到当 $n=1$ 时我们有

$$\sum_{i=1}^{n}(-1)^{i-1}i^2 \;=\; 1$$

and also

并且

$$(-1)^{n-1}\frac{n(n+1)}{2} \;=\; 1.$$

Inductive step:

归纳步骤：

Suppose that $k > 1$ is an integer such that

假设 $k > 1$ 是一个整数，使得

$$\sum_{i=1}^{k}(-1)^{i-1}i^2 \;=\; (-1)^{k-1}\frac{k(k+1)}{2}.$$

Adding $(-1)^k(k+1)^2$ to both sides gives

两边同时加上 $(-1)^k(k+1)^2$ 得到

$$\begin{aligned}
\sum_{i=1}^{k+1}(-1)^{i-1}i^2 \;&=\; (-1)^{k-1}\frac{k(k+1)}{2} + (-1)^k(k+1)^2 \\
&=\; (-1)^{k-1}\frac{k(k+1)}{2} - (-1)^{k-1}(k+1)^2 \\
&=\; (-1)^{k-1}\left(\frac{k(k+1)}{2} - \frac{2(k+1)^2}{2}\right) \\
&=\; (-1)^{k}\left(\frac{2(k+1)^2}{2} - \frac{k(k+1)}{2}\right) \\
&=\; (-1)^{k}\frac{(k+1)(2(k+1)-k)}{2} \\
&=\; (-1)^{k}\frac{(k+1)(k+2)}{2}
\end{aligned}$$

Q.E.D.

5. Prove the following formula for a product.

   证明以下乘积公式。

$$\prod_{i=2}^{n}\left(1-\frac{1}{i}\right) = \frac{1}{n}$$

Notice that the problem statement didn't specify the domain – but the smallest value of n that gives a non-empty product on the left-hand side is $n = 2$.

注意到题目陈述没有指定定义域——但使左边乘积非空的最小 n 值是 $n = 2$。

Proof: (By mathematical induction)

(通过数学归纳法)

Base case: $(n = 2)$

基础情形： $(n = 2)$ For the base case, note that when $n = 2$ we have

对于基础情形，注意到当 $n = 2$ 时我们有

$$\prod_{i=2}^{2}\left(1-\frac{1}{i}\right) \quad = \quad \left(1-\frac{1}{2}\right) \quad = \quad 1/2$$

and, when $n = 2$, the right-hand side $(1/n)$ also evaluates to $1/2$.

并且，当 $n = 2$ 时，右边 $(1/n)$ 的值也是 $1/2$。**Inductive step:**

归纳步骤：

Suppose that $k \geq 2$ is an integer such that

假设 $k \geq 2$ 是一个整数，使得

$$\prod_{i=2}^{k}\left(1-\frac{1}{i}\right) = \frac{1}{k}.$$

Then,

那么，

$$\begin{aligned}
&\prod_{i=2}^{k+1}\left(1-\frac{1}{i}\right) \\
&= \left(1-\frac{1}{k+1}\right) \cdot \prod_{i=2}^{k}\left(1-\frac{1}{i}\right) \\
&= \left(1-\frac{1}{k+1}\right) \cdot \frac{1}{k} \\
&= \frac{1}{k+1}.
\end{aligned}$$

Q.E.D.

The final line skips over a tiny bit of algebraic detail. You may feel more comfortable if you fill in those steps.

最后一行跳过了一点代数细节。如果你补上这些步骤，可能会感觉更舒服。

6. Prove $\sum_{j=0}^{n}(4j+1) = 2n^2+3n+1$ for all integers $n \geq 0$.

证明对于所有整数 $n \geq 0$，$\sum_{j=0}^{n}(4j+1) = 2n^2+3n+1$ 成立。

Proof: (By mathematical induction)

(通过数学归纳法)

Base case: $(n=0)$

基础情形： $(n=0)$ For the base case, note that when $n=0$ we have

对于基础情形，注意到当 $n=0$ 时我们有

$$\sum_{j=0}^{n}(4j+1) = (4\cdot 0+1 = 1$$

also, when $n=0$,

同样，当 $n=0$ 时，

$$2n^2+3n+1 = 2\cdot 0^2+3\cdot 0+1 = 1.$$

Inductive step:

归纳步骤：

Suppose that $k \geq 0$ is an integer such that

假设 $k \geq 0$ 是一个整数，使得

$$\sum_{j=0}^{k}(4j+1) = 2k^2+3k+1.$$

(We want to show that $\sum_{j=0}^{k+1}(4j+1) = 2(k+1)^2+3(k+1)+1.$)

（我们想要证明 $\sum_{j=0}^{k+1}(4j+1) \;=\; 2(k+1)^2+3(k+1)+1$。）

So consider the sum $\sum_{j=0}^{k+1}(4j+1)$:

那么考虑和 $\sum_{j=0}^{k+1}(4j+1)$：

$$\begin{aligned}
&\sum_{j=0}^{k+1}(4j+1)\\
&= \; 4(k+1)+1 \;+\; \sum_{j=0}^{k}(4j+1)\\
&= \; 4(k+1)+1 \;+\; 2k^2+3k+1\\
&=\\
&=\\
&=
\end{aligned}$$

Q.E.D.

Notice that the last line given in the proof is where the inductive hypothesis gets used. The actual last line of the proof is fairly easy to determine (hint: it is given in the ”We want to show” sentence.) So now you just have to fill in the gaps...

注意到证明中给出的倒数第二行是使用归纳假设的地方。证明的实际最后一行是相当容易确定的（提示：它在“我们想要证明”的句子中给出）。所以现在你只需要填补空白⋯⋯

7. Prove $\sum_{i=1}^{n} \frac{1}{(2i-1)(2i+1)} = \frac{n}{2n+1}$ for all natural numbers n.

证明对于所有自然数 n，$\sum_{i=1}^{n} \frac{1}{(2i-1)(2i+1)} = \frac{n}{2n+1}$ 成立。

Proof: (By mathematical induction)

(通过数学归纳法)

Base case: $(n = 0)$

基础情形： $(n = 0)$ For the base case, note that when $n = 0$

对于基础情形，注意到当 $n = 0$ 时

$$\sum_{j=0}^{n} \frac{1}{(2i-1)(2i+1)}$$

contains no terms. Thus its value is 0.

不包含任何项。因此其值为 0。

And, $\frac{n}{2n+1}$ also evaluates to 0 when $n = 0$.

并且，当 $n = 0$ 时 $\frac{n}{2n+1}$ 的值也为 0。**Inductive step:**

归纳步骤：

By the inductive hypothesis we may write

根据归纳假设，我们可以写出

$$\sum_{i=1}^{k} \frac{1}{(2i-1)(2i+1)} = \frac{k}{2k+1}.$$

Adding $\frac{1}{(2(k+1)-1)(2(k+1)+1)}$ to both side of this gives

两边同时加上 $\frac{1}{(2(k+1)-1)(2(k+1)+1)}$ 得到

$$\sum_{i=1}^{k+1} \frac{1}{(2i-1)(2i+1)} = \frac{k}{2k+1} + \frac{1}{(2(k+1)-1)(2(k+1)+1)}.$$

To complete the proof we must verify that

为了完成证明，我们必须验证

$$\frac{k}{2k+1} + \frac{1}{(2(k+1)-1)(2(k+1)+1)} = \frac{k+1}{2(k+1)+1}.$$

Note that

注意到

$$\begin{aligned}
&\frac{k}{2k+1} + \frac{1}{(2(k+1)-1)(2(k+1)+1)} \\
&= \frac{k}{2k+1} + \frac{1}{(2k+1)(2k+3)} \\
&= \frac{k(2k+3)}{(2k+1)(2k+3)} + \frac{1}{(2k+1)(2k+3)} \\
&= \frac{k(2k+3)+1}{(2k+1)(2k+3)} \\
&= \frac{2k^2+3k+1}{(2k+1)(2k+3)} \\
&= \frac{(2k+1)(k+1)}{(2k+1)(2k+3)} \\
&= \frac{k+1}{2k+3} = \frac{k+1}{2(k+1)+1}
\end{aligned}$$

as desired.

如所愿。

Q.E.D.

8. The *Fibonacci numbers* are a sequence of integers defined by the rule that a number in the sequence is the sum of the two that precede it.

斐波那契数是一个整数序列，其定义规则是序列中的一个数是它前面两个数的和。

$$F_{n+2} = F_n + F_{n+1}$$

The first two Fibonacci numbers (actually the zeroth and the first) are both 1.

前两个斐波那契数（实际上是第零个和第一个）都是 1。

Thus, the first several Fibonacci numbers are

因此，前几个斐波那契数是

$$F_0 = 1, F_1 = 1, F_2 = 2, F_3 = 3, F_4 = 5, F_5 = 8, F_6 = 13, F_7 = 21, \text{ et cetera}$$

Use mathematical induction to prove the following formula involving Fibonacci numbers.

使用数学归纳法证明以下涉及斐波那契数的公式。

$$\sum_{i=0}^{n} (F_i)^2 \;=\; F_n \cdot F_{n+1}$$

Proof: (by induction)

(通过归纳法)

Base case: $(n = 0)$

基础情形： $(n = 0)$

For the base case, note that when $n = 0$

对于基础情形，注意到当 $n = 0$ 时

$$\sum_{i=0}^{n} (F_i)^2 \;=\; 1.$$

And, $F_n \cdot F_{n+1} \;=\; F_0 \cdot F_1 \;=\; 1 \cdot 1 \;=\; 1.$

且，$F_n \cdot F_{n+1} \;=\; F_0 \cdot F_1 \;=\; 1 \cdot 1 \;=\; 1$。

Inductive step:

归纳步骤：

By the inductive hypothesis we may write

根据归纳假设，我们可以写出

$$\sum_{i=0}^{k} (F_i)^2 \;=\; F_k \cdot F_{k+1}.$$

Adding $(F_{k+1})^2$ to both sides gives

两边同时加上 $(F_{k+1})^2$ 得到

$$\sum_{i=0}^{k+1} (F_i)^2 \;=\; F_k \cdot F_{k+1} + (F_{k+1})^2.$$

Finally, note that (using factoring and the defining property of the Fibonacci numbers) we can show that

最后，注意到（使用因式分解和斐波那契数的定义属性）我们可以证明

$$\begin{aligned} & F_k \cdot F_{k+1} + (F_{k+1})^2 \\ = \;& F_{k+1} \cdot (F_k + F_{k+1}) \\ = \;& F_{k+1} \cdot F_{k+2} \end{aligned}$$

So the inductive step has been proved and the result follows by PMI.

所以归纳步骤已经证明，结果由数学归纳法原理得出。

Q.E.D.

5.3 Divisibility statements and other proofs using PMI 整除性陈述及其他使用数学归纳法的证明

Exercises — 5.3

Give inductive proofs of the following

为以下各项给出归纳证明

1. $\forall x \in \mathbb{N},\ 3 \,|\, x^3 - x$

Proof: (by induction)

(通过归纳法)

Base case: $(x = 0)$

基础情形： $(x = 0)$

The base case is trivial since, when $x = 0$, $x^3 - x$ evaluates to 0 and every natural number is a divisor of 0.

基础情形是平凡的，因为当 $x = 0$ 时，$x^3 - x$ 的值为 0，而任何自然数都是 0 的约数。**Inductive step:**

归纳步骤：

In the inductive step we want to show that $\forall k \in \mathbb{N},\ 3 \,|\, k^3 - k \implies 3 \,|\, (k+1)^3 - (k+1)$. So, let k be an arbitrary natural number such that $3 \,|\, k^3 - k$. Consider the quantity $(k+1)^3 - (k+1)$,

在归纳步骤中，我们想要证明 $\forall k \in \mathbb{N},\ 3 \,|\, k^3 - k \implies 3 \,|\, (k+1)^3 - (k+1)$。所以，设 k 是一个任意的自然数，使得 $3 \,|\, k^3 - k$。考虑量 $(k+1)^3 - (k+1)$，

$$
\begin{aligned}
& (k+1)^3 - (k+1) \\
= \; & (k^3 + 3k^2 + 3k + 1) - (k+1) \\
= \; & (k^3 - k) + (3k^2 + 3k).
\end{aligned}
$$

By the inductive hypothesis (and the definition of divisibility), we know that there is an integer q such that $k^3 - k = 3q$. So, by substitution, we obtain

根据归纳假设（以及整除的定义），我们知道存在一个整数 q 使得 $k^3 - k = 3q$。所以，通过代换，我们得到

$$(k+1)^3 - (k+1) \; = \; 3q + 3k^2 + 3k \; = 3 \cdot (q + k^2 + k).$$

Finally, $q + k^2 - k$ (being the sum of integers) is certainly an integer, so by the definition of divisibility, it follows that $(k+1)^3 - (k+1)$ is divisible by 3.

最后，$q + k^2 - k$（作为整数之和）当然是一个整数，所以根据整除的定义，可知 $(k+1)^3 - (k+1)$ 能被 3 整除。

Q.E.D.

2. $\forall x \in \mathbb{N}, \; 3 \,|\, x^3 + 5x$

3. $\forall x \in \mathbb{N}, \; 11 \,|\, x^{11} + 10x$

4. $\forall n \in \mathbb{N}, \; 3 \,|\, 4^n - 1$

5. $\forall n \in \mathbb{N}, \; 6 \,|\, (3n^2 + 3n - 12)$

6. $\forall n \in \mathbb{N}, \; 5 \,|\, (n^5 - 5n^3 + 14n)$

7. $\forall n \in \mathbb{N}, \; 4 \,|\, (13^n + 4n - 1)$

8. $\forall n \in \mathbb{N},\ 7 \,|\, 8^n + 6$

9. $\forall n \in \mathbb{N},\ 6 \,|\, 2n^3 - 2n - 12$

10. $\forall n \geq 3 \in \mathbb{N},\ 3n^2 + 3n + 1 < 2n^3$

11. $\forall n > 3 \in \mathbb{N},\ n^3 < 3^n$

12. $\forall n \geq 3 \in \mathbb{N},\ n^3 + 3 > n^2 + 3n + 1$

13. $\forall x \geq 4 \in \mathbb{N},\ x^2 2^x \leq 4^x$

5.4 The strong form of mathematical induction 数学归纳法的强形式

Exercises — 5.4

Give inductive proofs of the following

为以下各项给出归纳证明

1. A "postage stamp problem" is a problem that (typically) asks us to determine what total postage values can be produced using two sorts of stamps. Suppose that you have 3¢ stamps and 7¢ stamps, show (using strong induction) that any postage value 12¢ or higher can be achieved. That is,

   "邮票问题" 是一个（通常）要求我们确定使用两种邮票可以凑出哪些总邮资的问题。假设你有 3 美分和 7 美分的邮票，请（使用强归纳法）证明任何 12 美分或更高的邮资都可以实现。也就是说，

$$\forall n \in \mathbb{N}, n \geq 12 \implies \exists x, y \in \mathbb{N}, n = 3x + 7y.$$

2. Show that any integer postage of 12¢ or more can be made using only 4¢ and 5¢ stamps.

   证明任何 12 美分或更多的整数邮资都可以仅用 4 美分和 5 美分的邮票凑出。

3. The polynomial equation $x^2 = x + 1$ has two solutions, $\alpha = \frac{1+\sqrt{5}}{2}$ and $\beta = \frac{1-\sqrt{5}}{2}$. Show that the Fibonacci number F_n is less than or equal to α^n for all $n \geq 0$.

   多项式方程 $x^2 = x + 1$ 有两个解，$\alpha = \frac{1+\sqrt{5}}{2}$ 和 $\beta = \frac{1-\sqrt{5}}{2}$。证明对于所有 $n \geq 0$，斐波那契数 F_n 小于或等于 α^n。

Chapter 6

Relations and functions 关系与函数

6.1 Relations 关系

Exercises — 6.1
练习 — 6.1

1. The *lexicographic order*, $<_{\text{lex}}$, is a relation on the set of all words, where $x <_{\text{lex}} y$ means that x would come before y in the dictionary. Consider just the three letter words like "iff", "fig", "the", et cetera. Come up with a usable definition for $x_1x_2x_3 <_{\text{lex}} y_1y_2y_3$.

   字典序，$<_{\text{lex}}$，是所有单词集合上的一个关系，其中 $x <_{\text{lex}} y$ 意味着 x 在字典中会排在 y 之前。只考虑像 "iff"、"fig"、"the" 等三个字母的单词。为 $x_1x_2x_3 <_{\text{lex}} y_1y_2y_3$ 给出一个可用的定义。

2. What is the graph of "=" in $\mathbb{R} \times \mathbb{R}$?

   在 $\mathbb{R} \times \mathbb{R}$ 中，"=" 的图像是什么？

3. The *inverse* of a relation R is denoted R^{-1}. It contains exactly the same ordered pairs as R but with the order switched. (So technically, they

aren't *exactly* the same ordered pairs ...)

$$\mathsf{R}^{-1} = \{(b,a) \,|\, (a,b) \in \mathsf{R}\}$$

Define a relation S on $\mathbb{R} \times \mathbb{R}$ by $\mathsf{S} = \{(x,y) \,|\, y = \sin x\}$. What is S^{-1}? Draw a single graph containing S and S^{-1}.

一个关系 R 的逆关系记作 R^{-1}。它包含与 R 完全相同的有序对，但顺序相反。（所以技术上讲，它们并非完全相同的有序对⋯⋯）

$$\mathsf{R}^{-1} = \{(b,a) \,|\, (a,b) \in \mathsf{R}\}$$

在 $\mathbb{R} \times \mathbb{R}$ 上定义一个关系 S 为 $\mathsf{S} = \{(x,y) \,|\, y = \sin x\}$。$\mathsf{S}^{-1}$ 是什么？在同一个图中画出 S 和 S^{-1}。

4. The "socks and shoes" rule is a very silly little mnemonic for remembering how to invert a composition. If we think of undoing the process of putting on our socks and shoes (that's socks first, then shoes) we have to first remove our shoes, *then* take off our socks. The socks and shoes rule is valid for relations as well.

   Prove that $(\mathsf{S} \circ \mathsf{R})^{-1} = \mathsf{R}^{-1} \circ \mathsf{S}^{-1}$.

   "袜子和鞋子"规则是一个非常傻的记忆法，用来记住如何对复合求逆。如果我们考虑撤销穿袜子和鞋子的过程（即先穿袜子，再穿鞋子），我们必须先脱掉鞋子，然后再脱掉袜子。"袜子和鞋子"规则对关系也同样有效。

   证明 $(\mathsf{S} \circ \mathsf{R})^{-1} = \mathsf{R}^{-1} \circ \mathsf{S}^{-1}$。

6.2 Properties of relations 关系的性质

Exercises — 6.2
练习 — 6.2

1. Consider the relation S defined by

   $$\mathsf{S} = \{(x, y) \mid x \text{ is smarter than } y\}.$$

   Is S symmetric or anti-symmetric? Explain.

   考虑由以下方式定义的关系 S

   $$\mathsf{S} = \{(x, y) \mid x \text{ 比 } y \text{ 更聪明}\}.$$

   S 是对称的还是反对称的？请解释。

2. Consider the relation A defined by

   $$\mathsf{A} = \{(x, y) \mid x \text{ has the same astrological sign as } y\}.$$

   Is A symmetric or anti-symmetric? Explain.

   考虑由以下方式定义的关系 A

   $$\mathsf{A} = \{(x, y) \mid x \text{ 和 } y \text{ 有相同的星座}\}.$$

   A 是对称的还是反对称的？请解释。

3. Explain why both of the relations just described (in problems 1 and 2) have the transitive property.

   解释为什么刚才描述的两个关系（在问题 1 和 2 中）都具有传递性。

4. For each of the five properties, name a relation that has it and a relation that doesn't.

   对这五个性质中的每一个，各举一个具有该性质的关系和一个不具有该性质的关系。

5. Show by counterexample that "|" (divisibility) is not symmetric as a relation on $\mathbb{Z}$.

   通过反例证明，"|"（整除性）作为 $\mathbb{Z}$ 上的关系不是对称的。

6. Prove that "|" is an ordering relation (you must verify that it is reflexive, anti-symmetric and transitive).

   证明"|"是一个序关系（你必须验证它是自反的、反对称的和传递的）。

6.3 Equivalence relations 等价关系

Exercises — 6.3
练习 — 6.3

1. Consider the relation A defined by

$$\mathsf{A} = \{(x, y) \mid x \text{ has the same astrological sign as } y\}.$$

   Show that A is an equivalence relation. What equivalence class under A do you belong to?

   考虑由以下方式定义的关系 A

$$\mathsf{A} = \{(x, y) \mid x \text{和} y \text{有相同的星座}\}.$$

   证明 A 是一个等价关系。在 A 关系下，你属于哪个等价类?

2. Define a relation $\square$ on the integers by $x \square y \iff x^2 = y^2$. Show that $\square$ is an equivalence relation. List the equivalence classes $x/\square$ for $0 \leq x \leq 5$.

   在整数上定义一个关系 $\square$ 为 $x \square y \iff x^2 = y^2$。证明 $\square$ 是一个等价关系。列出 $0 \leq x \leq 5$ 的等价类 $x/\square$。

3. Define a relation A on the set of all words by

$$w_1 \mathsf{A} w_2 \iff w_1 \text{ is an anagram of } w_2.$$

   Show that A is an equivalence relation. (Words are anagrams if the letters of one can be re-arranged to form the other. For example, 'ART' and 'RAT' are anagrams.)

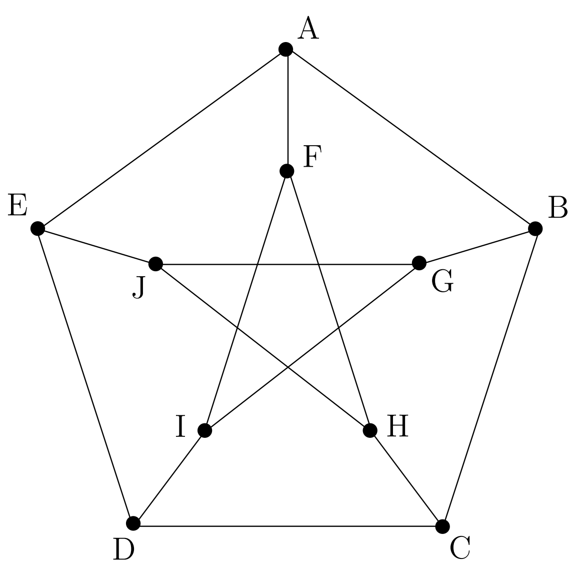
   在所有单词的集合上定义一个关系 A

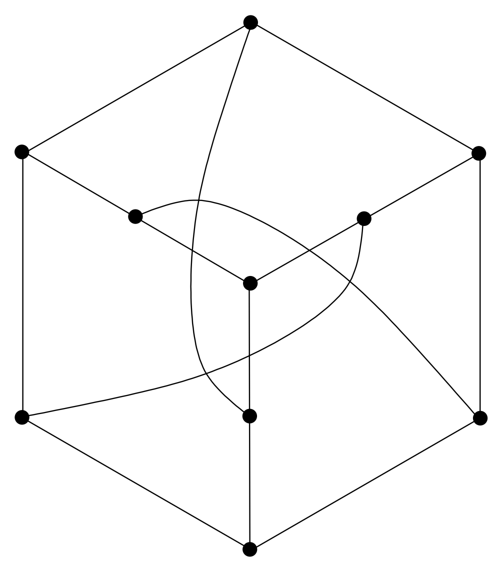
$$w_1 \mathsf{A} w_2 \iff w_1 \text{ 是} w_2 \text{ 的字谜}$$

证明 A 是一个等价关系。(如果一个词的字母可以重新排列形成另一个词，那么这两个词就是字谜。例如，'ART' 和 'RAT' 是字谜。)

4. The two diagrams below both show a famous graph known as the Petersen graph. The picture on the left is the usual representation which emphasizes its five-fold symmetry. The picture on the right highlights the fact that the Petersen graph also has a three-fold symmetry. Label the right-hand diagram using the same letters (A through J) in order to show that these two representations are truly isomorphic.

   下面的两个图都展示了一个著名的图，称为 彼得森图。左边的图是通常的表示法，强调了它的五重对称性。右边的图突出了彼得森图也具有三重对称性的事实。请使用相同的字母（A 到 J）标记右边的图，以证明这两种表示是真正同构的。





5. We will use the symbol $\mathbb{Z}^*$ to refer to the set of all integers *except* 0. Define a relation Q on the set of all pairs in $\mathbb{Z} \times \mathbb{Z}^*$ (pairs of integers where the second coordinate is non-zero) by $(a,b)\mathsf{Q}(c,d) \iff ad = bc$. Show that Q is an equivalence relation.

我们将使用符号 $\mathbb{Z}^*$ 来指代除 0 之外的所有整数集合。在 $\mathbb{Z} \times \mathbb{Z}^*$（第二个坐标非零的整数对）中的所有对的集合上定义一个关系 Q，通过 $(a, b)\mathsf{Q}(c, d) \iff ad = bc$。证明 Q 是一个等价关系。

6. The relation Q defined in the previous problem partitions the set of all pairs of integers into an interesting set of equivalence classes. Explain why

$$\mathbb{Q} \quad = \quad (\mathbb{Z} \times \mathbb{Z}^*)/\mathsf{Q}.$$

Ultimately, this is the "right" definition of the set of rational numbers!

在前一个问题中定义的关系 Q 将所有整数对的集合划分为一个有趣的等价类集合。解释为什么

$$\mathbb{Q} \quad = \quad (\mathbb{Z} \times \mathbb{Z}^*)/\mathsf{Q}.$$

最终，这才是对有理数集合的"正确"定义！

7. Reflect back on the proof in problem 5. Note that we were fairly careful in assuring that the second coordinate in the ordered pairs is non-zero. (This was the whole reason for introducing the $\mathbb{Z}^*$ notation.) At what point in the argument did you use this hypothesis?

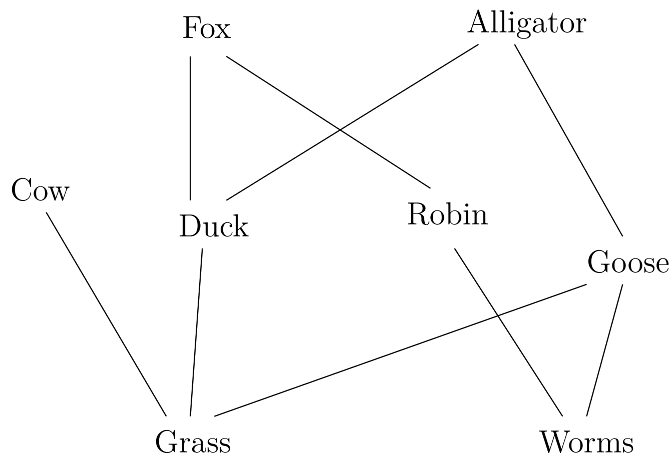
回顾问题 5 中的证明。注意我们相当小心地确保了有序对中的第二个坐标是非零的。（这就是引入 $\mathbb{Z}^*$ 符号的全部原因。）在论证的哪一点你使用了这个假设?

6.4 Ordering relations 序关系

Exercises — 6.4
练习 — 6.4

1. In population ecology there is a partial order “predates” which basically means that one organism feeds upon another. Strictly speaking this relation is not transitive; however, if we take the point of view that when a wolf eats a sheep, it is also eating some of the grass that the sheep has fed upon, we see that in a certain sense it is transitive. A chain in this partial order is called a “food chain” and so-called apex predators are said to “sit atop the food chain”. Thus “apex predator” is a term for a maximal element in this poset. When poisons such as mercury and PCBs are introduced into an ecosystem, they tend to collect disproportionately in the apex predators – which is why pregnant women and young children should not eat shark or tuna but sardines are fine. Below is a small example of an ecology partially ordered by “predates”

   在种群生态学中，有一个偏序关系“捕食”，基本上意味着一个生物体以另一个生物体为食。严格来说，这个关系不是传递的；但是，如果我们采取这样的观点，即当狼吃羊时，它也吃了一些羊吃过的草，我们看到在某种意义上它是传递的。这个偏序中的一个链被称为“食物链”，而所谓的顶级捕食者据说“位于食物链的顶端”。因此，“顶级捕食者”是这个偏序集（poset）中极大元的一个术语。当像汞和多氯联苯这样的毒物被引入生态系统时，它们往往不成比例地在顶级捕食者体内积聚——这就是为什么孕妇和幼儿不应该吃鲨鱼或金枪鱼，而沙丁鱼则没问题的原因。下面是一个由“捕食”关系偏序的小型生态系统示例。



Find the largest antichain in this poset.

找出这个偏序集中的最大反链。

2. Referring to the poset given in exercise 1, match the following.

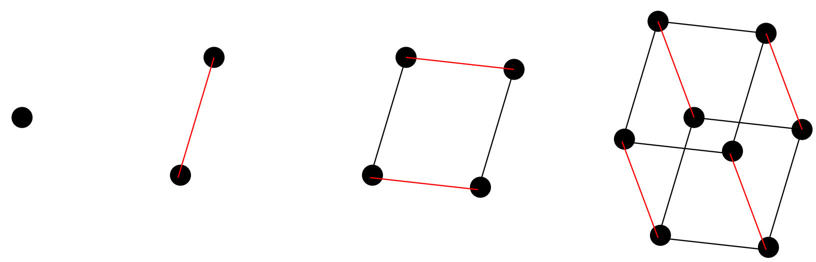
   参考练习 1 中给出的偏序集，匹配下列各项。

1. An (non-maximal) antichain	a. Grass
1. 一个（非极大的）反链	a. 草
2. A maximal antichain	b. Goose
2. 一个极大反链	b. 鹅
3. A maximal element	c. Fox
3. 一个极大元	c. 狐狸
4. A (non-maximal) chain	d. $\{\text{Grass}, \text{Duck}\}$
4. 一个（非极大的）链	d. $\{\text{草}, \text{鸭子}\}$
5. A maximal chain	e. There isn't one!
5. 一个极大链	e. 没有！
6. A cover for "Worms"	f. $\{\text{Fox}, \text{Alligator}, \text{Cow}\}$
6.“蠕虫”的一个覆盖	f. $\{\text{狐狸}, \text{短吻鳄}, \text{牛}\}$
7. A least element	g. $\{\text{Cow}, \text{Duck}, \text{Goose}\}$
7. 一个最小元	g. $\{\text{牛}, \text{鸭子}, \text{鹅}\}$
8. A minimal element	h. $\{\text{Worms}, \text{Robin}, \text{Fox}\}$
8. 一个极小元	h. $\{\text{蠕虫}, \text{知更鸟}, \text{狐狸}\}$

3. The graph of the edges of a cube is one in an infinite sequence of graphs. These graphs are defined recursively by "Make two copies of

the previous graph then join corresponding nodes in the two copies with edges.” The 0-dimensional ‘cube’ is just a single point. The 1-dimensional cube is a single edge with a node at either end. The 2-dimensional cube is actually a square and the 3-dimensional cube is what we usually mean when we say “cube.”

立方体边的图是一个无限图序列中的一个。这些图是递归定义的：“制作前一个图的两个副本，然后用边连接两个副本中对应的节点。” 0 维“立方体” 只是一个单点。1 维立方体是一条两端各有一个节点的边。2 维立方体实际上是一个正方形，而 3 维立方体就是我们通常所说的“立方体”。



Make a careful drawing of a *hypercube* – which is the name of the graph that follows the ordinary cube in this sequence.

仔细画一个*超立方体*——这是这个序列中紧随普通立方体之后的图的名称。

4. Label the nodes of a hypercube with the divisors of 210 in order to produce a Hasse diagram of the poset determined by the divisibility relation.

   用 210 的因子标记一个超立方体的节点，以产生由整除关系确定的偏序集的哈斯图。

5. Label the nodes of a hypercube with the subsets of $\{a, b, c, d\}$ in order to produce a Hasse diagram of the poset determined by the subset containment relation.

用 $\{a, b, c, d\}$ 的子集标记一个超立方体的节点，以产生由子集包含关系确定的偏序集的哈斯图。

6. Complete a Hasse diagram for the poset of divisors of 11025 (partially ordered by divisibility).

   完成 11025 的因子偏序集（按整除性偏序）的哈斯图。

7. Find a collection of sets so that, when they are partially ordered by $\subseteq$, we obtain the same Hasse diagram as in the previous problem.

   找一个集合的集合，使得当它们按 $\subseteq$ 偏序时，我们得到与前一个问题中相同的哈斯图。

6.5 Functions 函数

Exercises — 6.5
练习 — 6.5

1. For each of the following functions, give its domain, range and a possible codomain.

   对于下列每个函数，给出其定义域、值域和一个可能的上域。

   (a) $f(x) = \sin(x)$

   (b) $g(x) = e^x$

   (c) $h(x) = x^2$

   (d) $m(x) = \frac{x^2+1}{x^2-1}$

   (e) $n(x) = \lfloor x \rfloor$

   (f) $p(x) = \langle \cos(x), \sin(x) \rangle$

2. Find a bijection from the set of odd squares, $\{1, 9, 25, 49, \ldots\}$, to the non-negative integers, $\mathbb{Z}^{\text{noneg}} = \{0, 1, 2, 3, \ldots\}$. Prove that the function you just determined is both injective and surjective. Find the inverse function of the bijection above.

   找出一个从奇数平方数集合 $\{1, 9, 25, 49, \ldots\}$ 到非负整数集合 $\mathbb{Z}^{\text{noneg}} = \{0, 1, 2, 3, \ldots\}$ 的双射。证明你刚刚确定的函数既是单射的也是满射的。找出上述双射的逆函数。

3. The natural logarithm function $\ln(x)$ is defined by a definite integral with the variable x in the upper limit.

   自然对数函数 $\ln(x)$ 是通过一个定积分定义的，其中变量 x 在积分上限。

$$\ln(x) = \int_{t=1}^{x} \frac{1}{t}\,\mathrm{d}t.$$

From this definition we can deduce that $\ln(x)$ is strictly increasing on its entire domain, $(0, \infty)$. Why is this true?

从这个定义我们可以推断出 $\ln(x)$ 在其整个定义域 $(0, \infty)$ 上是严格递增的。为什么这是真的?

We can use the above definition with $x = 2$ to find the value of $\ln(2) \approx .693$. We will also take as given the following rule (which is valid for all logarithmic functions).

我们可以使用上述定义，当 $x = 2$ 时，求得 $\ln(2) \approx .693$ 的值。我们也将以下法则视为已知（该法则对所有对数函数都有效）。

$$\ln(a^b) = b \ln(a)$$

Use the above information to show that there is neither an upper bound nor a lower bound for the values of the natural logarithm. These facts together with the information that ln is strictly increasing show that $\text{Rng}(\ln) = \mathbb{R}$.

使用以上信息证明自然对数的值既没有上界也没有下界。这些事实，加上 ln 是严格递增的信息，共同证明了 $\text{Rng}(\ln) = \mathbb{R}$。

4. Georg Cantor developed a systematic way of listing the rational numbers. By "listing" a set one is actually developing a bijection from $\mathbb{N}$ to that set. The method known as "Cantor's Snake" creates a bijection from the naturals to the non-negative rationals. First we create an infinite table whose rows are indexed by positive integers and whose columns are indexed by non-negative integers – the entries in this table are rational numbers of the form "column index" / "row index." We then follow a snake-like path that zig-zags across this table – whenever we encounter a rational number that we haven't seen before (in lower terms) we write it down. This is indicated in the diagram below by circling the entries.

格奥尔格 · 康托尔发明了一种系统地列出有理数的方法。通过“列出”一个集合，实际上是在建立一个从 $\mathbb{N}$ 到该集合的双射。被称为“康托尔的蛇”的方法创建了一个从自然数到非负有理数的双射。首先，我们创建一个无限的表格，其行由正整数索引，其列由非负整数索引——此表中的条目是形式为“列索引”/“行索引”的有理数。然后我们沿着一条蛇形路径在该表上曲折前行——每当遇到一个我们之前没有见过的有理数（以最简形式）时，我们就把它写下来。这在下面的图表中通过圈出条目来表示。

	0	1	2	3	4	5	6	7	8
1	0/1	1/1	2/1	3/1	4/1	5/1	6/1	7/1	8/1
2	0/2	1/2	2/2	3/2	4/2	5/2	6/2	7/2	8/2
3	0/3	1/3	2/3	3/3	4/3	5/3	6/3	7/3	8/3
4	0/4	1/4	2/4	3/4	4/4	5/4	6/4	7/4	8/4
5	0/5	1/5	2/5	3/5	4/5	5/5	6/5	7/5	8/5
6	0/6	1/6	2/6	3/6	4/6	5/6	6/6	7/6	8/6
7	0/7	1/7	2/7	3/7	4/7	5/7	6/7	7/7	8/7
8	0/8	1/8	2/8	3/8	4/8	5/8	6/8	7/8	8/8

Effectively this gives us a function f which produces the rational number that would be found in a given position in this list. For example $f(1) = 0/1, f(2) = 1/1$ and $f(5) = 1/3$.

What is $f(26)$? What is $f(30)$? What is $f^{-1}(3/4)$? What is $f^{-1}(6/7)$?

实际上，这给了我们一个函数 f，它能产生在这个列表给定位置上找到的有理数。例如，$f(1) = 0/1, f(2) = 1/1$ 以及 $f(5) = 1/3$。

$f(26)$ 是什么？$f(30)$ 是什么？$f^{-1}(3/4)$ 是什么？$f^{-1}(6/7)$ 是什么？

6.6 Special functions 特殊函数

Exercises — 6.6
练习 — 6.6

1. The n-th triangular number, denoted $T(n)$, is given by the formula $T(n) = (n^2 + n)/2$.

   第 n 个三角数，记为 $T(n)$，由公式 $T(n) = (n^2 + n)/2$ 给出。

   If we regard this formula as a function from $\mathbb{R}$ to $\mathbb{R}$, it fails the horizontal line test and so it is not invertible.

   如果我们将此公式视为从 $\mathbb{R}$ 到 $\mathbb{R}$ 的函数，它通不过水平线测试，因此是不可逆的。

   Find a suitable restriction so that T is invertible.

   找出一个合适的限制，使得 T 是可逆的。

2. The usual algebraic procedure for inverting $T(x) = (x^2 + x)/2$ fails.

   对 $T(x) = (x^2 + x)/2$ 求逆的常规代数过程会失败。

   Use your knowledge of the geometry of functions and their inverses to find a formula for the inverse.

   利用你关于函数及其逆函数的几何知识来找出一个逆函数的公式。

   (Hint: it may be instructive to first invert the simpler formula $S(x) = x^2/2$ — this will get you the right vertical scaling factor.)

   （提示：先对更简单的公式 $S(x) = x^2/2$ 求逆可能会有启发——这将帮助你找到正确的垂直缩放因子。）

3. What is $\pi_2(W(t))$?

   $\pi_2(W(t))$ 是什么？

4. Find a right inverse for $f(x) = |x|$.

   找出 $f(x) = |x|$ 的一个右逆。

5. In three-dimensional space we have projection functions that go onto the three coordinate axes (π_1, π_2 and π_3) and we also have projections onto coordinate planes.

   在三维空间中，我们有投影到三个坐标轴的投影函数（π_1, π_2 和 π_3），我们也有投影到坐标平面的投影。

   For example, $\pi_{12} : \mathbb{R} \times \mathbb{R} \times \mathbb{R} \longrightarrow \mathbb{R} \times \mathbb{R}$, defined by

   例如，$\pi_{12} : \mathbb{R} \times \mathbb{R} \times \mathbb{R} \longrightarrow \mathbb{R} \times \mathbb{R}$，定义为

$$\pi_{12}((x, y, z)) = (x, y)$$

   is the projection onto the x–y coordinate plane.

   是到 x–y 坐标平面的投影。

   The triple of functions $(\cos t, \sin t, t)$ is a parametric expression for a helix.

   函数三元组 $(\cos t, \sin t, t)$ 是螺旋线的参数表达式。

   Let $H = \{(\cos t, \sin t, t) \,|\, t \in \mathbb{R}\}$ be the set of all points on the helix.

   令 $H = \{(\cos t, \sin t, t) \,|\, t \in \mathbb{R}\}$ 为螺旋线上所有点的集合。

   What is the set $\pi_{12}(H)$? What are the sets $\pi_{13}(H)$ and $\pi_{23}(H)$?

   集合 $\pi_{12}(H)$ 是什么？集合 $\pi_{13}(H)$ 和 $\pi_{23}(H)$ 又是什么？

6. Consider the set $\{1, 2, 3, \ldots, 10\}$. Express the characteristic function of the subset $S = \{1, 2, 3\}$ as a set of ordered pairs.

   考虑集合 $\{1, 2, 3, \ldots, 10\}$。将子集 $S = \{1, 2, 3\}$ 的特征函数表示为有序对的集合。

7. If S and T are subsets of a set D, what is the product of their characteristic functions $1_S \cdot 1_T$?

   如果 S 和 T 是集合 D 的子集，它们的特征函数 $1_S \cdot 1_T$ 的乘积是什么？

8. Evaluate the sum

   计算这个和

$$\sum_{i=1}^{10} \frac{1}{i} \cdot [i \text{ is prime}].$$

Chapter 7

Proof techniques III — Combinatorics 证明技巧 III — 组合数学

7.1 Counting 计数

1. Determine the number of entries in the following sequences.

   确定以下序列中的项数。

   (a) $(999, 1000, 1001, \ldots 2006)$

   (b) $(13, 15, 17, \ldots 199)$

   (c) $(13, 19, 25, \ldots 601)$

   (d) $(5, 10, 17, 26, 37, \ldots 122)$

   (e) $(27, 64, 125, 216, \ldots 8000)$

   (f) $(7, 11, 19, 35, 67, \ldots 131075)$

2. How many "full houses" are there in Yahtzee? (A full house is a pair together with a three-of-a-kind.)

Yahtzee 骰子游戏中有多少种“葫芦”?(“葫芦”是指一对和三条的组合。)

3. In how many ways can you get “two pairs” in Yahtzee?

   在 Yahtzee 骰子游戏中有多少种方式可以得到“两对”?

4. Prove that the binomial coefficients $\binom{n+k-1}{k}$ and $\binom{n+k-1}{n-1}$ are equal.

   证明二项式系数 $\binom{n+k-1}{k}$ 和 $\binom{n+k-1}{n-1}$ 相等。

5. The “Cryptographer’s alphabet” is used to supply small examples in coding and cryptography. It consists of the first 6 letters, $\{a, b, c, d, e, f\}$. How many “words” of length up to 6 can be made with this alphabet? (A word need not actually be a word in English, for example both “fed” and “dfe” would be words in the sense we are using the term.)

   “密码学家字母表”用于提供编码和密码学中的小例子。它由前 6 个字母组成，即 $\{a, b, c, d, e, f\}$。用这个字母表可以组成多少个长度不超过 6 的“单词”?(一个“单词”不一定非得是英语单词，例如，“fed”和“dfe”在我们使用的意义上都是单词。)

6. How many “words” are there of length 4, with distinct letters from the Cryptographer’s alphabet, in which the letters appear in increasing order alphabetically? (“Acef” would be one such word, but “cafe” would not.)

   从密码学家字母表中选取不同的字母，可以组成多少个长度为 4 且字母按字母顺序递增的“单词”?(例如，“acef”是这样的一个词，但“cafe”不是。)

7. How many “words” are there of length 4 from the Cryptographer’s alphabet, with repeated letters allowed, in which the letters appear in non-decreasing order alphabetically?

从密码学家字母表中（允许重复字母）可以组成多少个长度为 4 且字母按字母顺序非递减排列的“单词”？

8. How many subsets does a finite set have?

   一个有限集有多少个子集？

9. How many handshakes will transpire when n people first meet?

   当 n 个人初次见面时，会发生多少次握手？

10. How many functions are there from a set of size n to a set of size m?

    从一个大小为 n 的集合到一个大小为 m 的集合，有多少个函数？

11. How many relations are there from a set of size n to a set of size m?

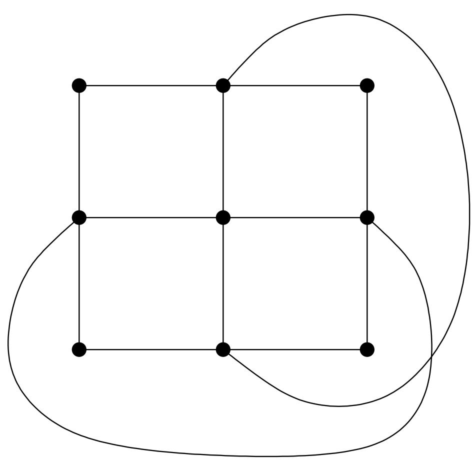
    从一个大小为 n 的集合到一个大小为 m 的集合，有多少个关系？

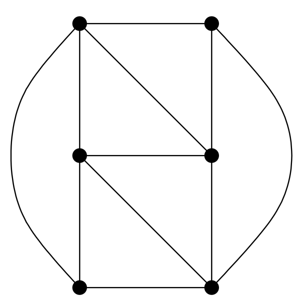
7.2 Parity and Counting arguments 奇偶性与计数论证

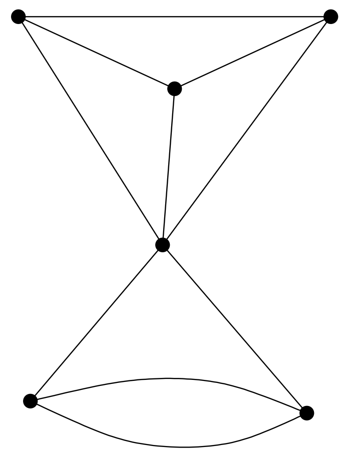
Exercises — 7.2
练习 — 7.2

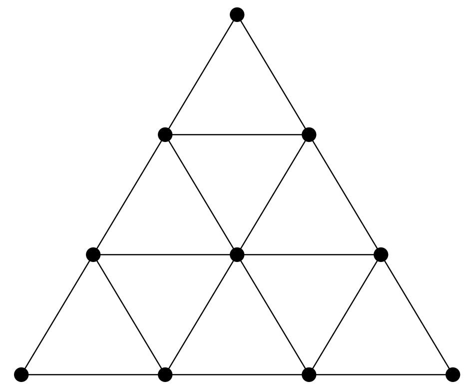
1. A walking tour of Königsberg such as is described in this section, or more generally, a circuit through an arbitrary graph that crosses each edge precisely once and begins and ends at the same node is known as an *Eulerian circuit*. An *Eulerian path* also crosses every edge of a graph exactly once but it begins and ends at distinct nodes. For each of the following graphs determine whether an Eulerian circuit or path is possible, and if so, draw it.

   如本节所述的柯尼斯堡漫步，或者更一般地，在一个任意图中，精确地穿过每条边一次并起止于同一节点的回路，被称为欧拉回路。而欧拉路径也精确地穿过图的每条边一次，但它的起点和终点是不同的节点。对于下面的每一个图，判断是否存在欧拉回路或欧拉路径，如果存在，请画出它。









2. Complete the proof of the fact that "Every graph has an even number of odd nodes."

   完成"每个图都有偶数个奇数度节点"这一事实的证明。

3. Provide an argument as to why an 8×8 chessboard with two squares pruned from diagonally opposite corners cannot be tiled with dominoes.

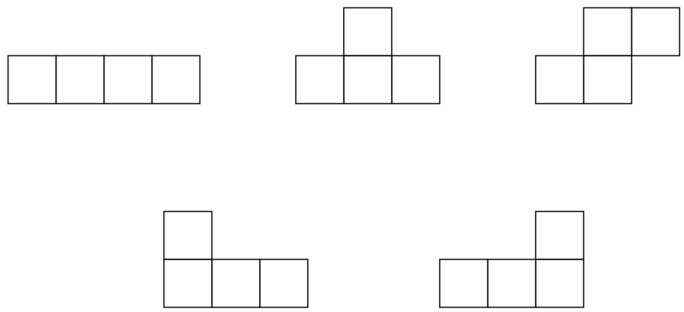
   请给出一个论证，说明为什么一个 8×8 的棋盘，在对角线两端各去掉一个方格后，无法用多米诺骨牌完全覆盖。

4. Prove that, if n is odd, any $n \times n$ chessboard with a square the same color as one of its corners pruned can be tiled by dominoes.

   证明，如果 n 是奇数，任何 $n \times n$ 的棋盘，在去掉一个与其角格颜色相同的方格后，可以用多米诺骨牌完全覆盖。

5. The five tetrominoes (familiar to players of the video game Tetris) are relatives of dominoes made up of four small squares.

   五个四格骨牌（俄罗斯方块玩家很熟悉）是多米诺骨牌的亲戚，由四个小方格组成。



All together these five tetrominoes contain 20 squares so it is conceivable that they could be used to tile a 4×5 chessboard. Prove that this is actually impossible.

这五个四格骨牌总共包含 20 个方格，所以可以想象它们可以用来铺满一个 4×5 的棋盘。证明这实际上是不可能的。

6. State necessary and sufficient conditions for the existence of an Eulerian circuit in a graph.

   陈述图中存在欧拉回路的充要条件。

7. State necessary and sufficient conditions for the existence of an Eulerian path in a graph.

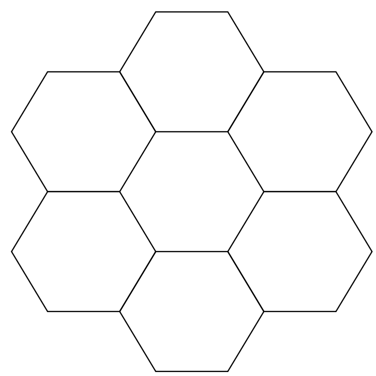
   陈述图中存在欧拉路径的充要条件。

8. Construct magic squares of order 4 and 5.

   构建 4 阶和 5 阶的幻方。

9. A magic hexagon of order 2 would consist of filling-in the numbers from 1 to 7 in the hexagonal array below. The magic condition means that each of the 9 "lines" of adjacent hexagons would have the same sum. Is this possible?

   一个 2 阶的幻六边形需要将数字 1 到 7 填入下面的六边形阵列中。幻方的条件意味着每 9 条相邻六边形的"线"上的数字之和都相同。这可能吗?



10. Is there a magic hexagon of order 3?

    存在 3 阶的幻六边形吗?

7.3 The pigeonhole principle 鸽巢原理

Exercises — 7.3
练习 — 7.3

1. The statement that there are two non-bald New Yorkers with the same number of hairs on their heads requires some careful estimates to justify it. Please justify it.

   “纽约市有两个非秃头的人头发数量相同”这一说法需要一些仔细的估计来证明。请证明之。

2. A mathematician, who always rises earlier than her spouse, has developed a scheme – using the pigeonhole principle – to ensure that she always has a matching pair of socks. She keeps only blue socks, green socks and black socks in her sock drawer – 10 of each. So as not to wake her husband she must select some number of socks from her drawer in the early morning dark and take them with her to the adjacent bathroom where she dresses. What number of socks does she choose?

   一位总比配偶早起的数学家，设计了一个方案——利用鸽巢原理——来确保她总能拿到一双配对的袜子。她的袜筒里只有蓝色、绿色和黑色的袜子——每种颜色 10 只。为了不吵醒丈夫，她必须在清晨的黑暗中从抽屉里拿出一定数量的袜子，带到隔壁的浴室里穿。她需要选择多少只袜子?

3. If we select 1001 numbers from the set $\{1, 2, 3, \ldots, 2000\}$ it is certain that there will be two numbers selected such that one divides the other. We can prove this fact by noting that every number in the given set can be expressed in the form $2^k \cdot m$ where m is an odd number and using the pigeonhole principle. Write-up this proof.

如果我们从集合 $\{1, 2, 3, \ldots, 2000\}$ 中选取 1001 个数，那么可以肯定，选出的数中必有两个数，其中一个能整除另一个。我们可以通过指出给定集合中的每个数都可以表示为 $2^k \cdot m$ 的形式（其中 m 是一个奇数）并利用鸽巢原理来证明这一事实。请写出这个证明。

4. Given any set of 53 integers, show that there are two of them having the property that either their sum or their difference is evenly divisible by 103.

   给定任意 53 个整数的集合，证明其中必有两个整数，它们的和或差能被 103 整除。

5. Prove that if 10 points are placed inside a square of side length 3, there will be 2 points within $\sqrt{2}$ of one another.

   证明：如果在边长为 3 的正方形内放置 10 个点，则必有两个点之间的距离在 $\sqrt{2}$ 以内。

6. Prove that if 10 points are placed inside an equilateral triangle of side length 3, there will be 2 points within 1 of one another.

   证明：如果在边长为 3 的等边三角形内放置 10 个点，则必有两个点之间的距离在 1 以内。

7. Prove that in a simple graph (an undirected graph with no loops or parallel edges) having n nodes, there must be two nodes having the same degree.

   证明在一个有 n 个节点的简单图（无环或平行边的无向图）中，必定有两个节点具有相同的度。

7.4 The algebra of combinations 组合代数

Exercises — 7.4

练习 — 7.4

1. Use the binomial theorem (with $x = 1000$ and $y = 1$) to calculate 1001^6.

   使用二项式定理（设 $x = 1000$ 和 $y = 1$）计算 1001^6。

2. Find $(2x + 3)^5$.

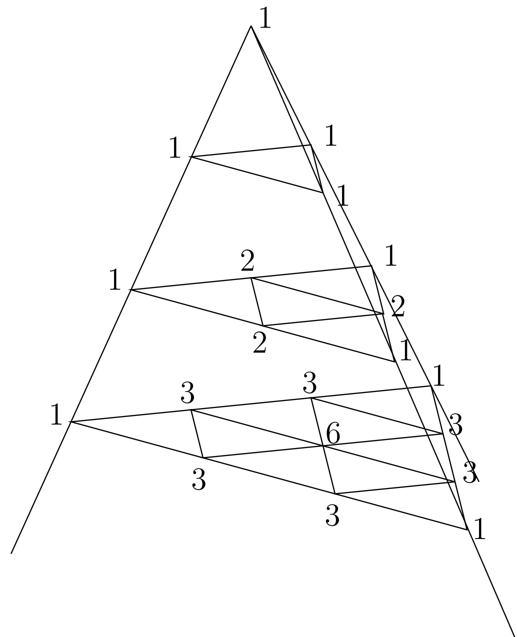
   求 $(2x + 3)^5$ 的展开式。

3. Find $(x^2 + y^2)^6$.

   求 $(x^2 + y^2)^6$ 的展开式。

4. The following diagram contains a 3-dimensional analog of Pascal's triangle that we might call "Pascal's tetrahedron." What would the next layer look like?

   下图包含一个帕斯卡三角形的三维模拟，我们可以称之为 "帕斯卡四面体"。下一层会是什么样子?



5. The student government at Lagrange High consists of 24 members chosen from amongst the general student body of 210. Additionally, there is a steering committee of 5 members chosen from amongst those in student government. Use the multiplication rule to determine two different formulas for the total number of possible governance structures.

   拉格朗日高中的学生会由从 210 名普通学生中选出的 24 名成员组成。此外，还有一个由学生会成员中选出的 5 名成员组成的指导委员会。请使用乘法法则确定两种不同的公式，来计算可能的治理结构总数。

6. Prove the identity

$$\binom{n}{k} \cdot \binom{k}{r} = \binom{n}{r} \cdot \binom{n-r}{k-r}$$

   combinatorially.

   用组合方法证明恒等式

$$\binom{n}{k} \cdot \binom{k}{r} = \binom{n}{r} \cdot \binom{n-r}{k-r}$$

7. Prove the binomial theorem.

$$\forall n \in \mathbb{N},\ \forall x, y \in \mathbb{R},\ (x+y)^n \ = \ \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k$$

证明二项式定理。

$$\forall n \in \mathbb{N},\ \forall x, y \in \mathbb{R},\ (x+y)^n \ = \ \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k$$

Chapter 8

Cardinality 基数

8.1 Equivalent sets 等价集

Exercises — 8.1

1. Name four sets in the equivalence class of $\{1, 2, 3\}$.

   说出与 $\{1, 2, 3\}$ 等价类中的四个集合。

2. Prove that set equivalence is an equivalence relation.

   证明集合等价是一种等价关系。

3. Construct a Venn diagram showing the relationships between the sets of sets which are finite, infinite, countable, denumerable and uncountable.

   构造一个维恩图，显示有限集、无限集、可数集、可列集和不可数集这些集合的集合之间的关系。

4. Place the sets $\mathbb{N}$, $\mathbb{R}$, $\mathbb{Q}$, $\mathbb{Z}$, $\mathbb{Z} \times \mathbb{Z}$, $\mathbb{C}$, $\mathbb{N}_{2007}$ and $\emptyset$; somewhere on the Venn diagram above. (Note to students (and graders): there are no wrong answers to this question, the point is to see what your intuition about these sets says at this point.)

将集合 $\mathbb{N}$, $\mathbb{R}$, $\mathbb{Q}$, $\mathbb{Z}$, $\mathbb{Z} \times \mathbb{Z}$, $\mathbb{C}$, $\mathbb{N}_{2007}$ 和 $\emptyset$ 放置在上面的维恩图中的某个位置。（给学生（和评分者）的注释：这个问题没有错误答案，关键是看看你目前对这些集合的直觉是什么。）

8.2 Examples of set equivalence 集合等价的例子

Exercises — 8.2

1. Prove that positive numbers of the form $3k+1$ are equinumerous with positive numbers of the form $4k+2$.

   证明形式为 $3k+1$ 的正数与形式为 $4k+2$ 的正数等势。

2. Prove that $f(x) = c + \dfrac{(x-a)(d-c)}{(b-a)}$ provides a bijection from the interval $[a, b]$ to the interval $[c, d]$.

   证明 $f(x) = c + \dfrac{(x-a)(d-c)}{(b-a)}$ 提供了从区间 $[a, b]$ 到区间 $[c, d]$ 的一个双射。

3. Prove that any two circles are equinumerous (as sets of points).

   证明任意两个圆（作为点的集合）是等势的。

4. Determine a formula for the bijection from $(-1, 1)$ to the line $y = 1$ determined by vertical projection onto the upper half of the unit circle, followed by projection from the point $(0, 0)$.

   确定一个从 $(-1, 1)$ 到直线 $y = 1$ 的双射公式，该双射由垂直投影到单位圆的上半部分，然后从点 $(0, 0)$ 投影得到。

5. It is possible to generalize the argument that shows a line segment is equivalent to a line to higher dimensions. In two dimensions we would show that the unit disk (the interior of the unit circle) is equinumerous with the entire plane $\mathbb{R} \times \mathbb{R}$. In three dimensions we would show that the unit ball (the interior of the unit sphere) is equinumerous with the entire space $\mathbb{R}^3 = \mathbb{R} \times \mathbb{R} \times \mathbb{R}$. Here we would like you to prove the two-dimensional case.

证明线段与直线等价的论证可以推广到更高维度。在二维中，我们将证明单位圆盘（单位圆的内部）与整个平面 $\mathbb{R} \times \mathbb{R}$ 等势。在三维中，我们将证明单位球体（单位球的内部）与整个空间 $\mathbb{R}^3 = \mathbb{R} \times \mathbb{R} \times \mathbb{R}$ 等势。在这里，我们希望你证明二维的情况。Gnomonic projection is a style of map rendering in which a portion of a sphere is projected onto a plane that is tangent to the sphere. The sphere's center is used as the point to project from. Combine vertical projection from the unit disk in the x–y plane to the upper half of the unit sphere $x^2 + y^2 + z^2 = 1$, with gnomonic projection from the unit sphere to the plane z = 1, to deduce a bijection between the unit disk and the (infinite) plane.

球心投影是一种地图绘制风格，其中球体的一部分被投影到一个与球体相切的平面上。球心被用作投影点。将从 x-y 平面上的单位圆盘到单位球体 $x^2 + y^2 + z^2 = 1$ 上半部分的垂直投影，与从单位球体到平面 z=1 的球心投影相结合，来推导出一个单位圆盘和（无限）平面之间的双射。

8.3 Cantor's theorem 康托尔定理

Exercises — 8.3

1. Determine a substitution rule – a consistent way of replacing one digit with another along the diagonal so that a diagonalization proof showing that the interval (0, 1) is uncountable will work in decimal. Write up the proof.

   确定一个替换规则——一种沿着对角线用一个数字替换另一个数字的一致方法，使得证明区间 (0, 1) 不可数的对角化证明在十进制中成立。写出该证明。

2. Can a diagonalization proof showing that the interval (0, 1) is uncountable be made workable in base-3 (ternary) notation?

   一个证明区间 (0, 1) 不可数的对角化证明能否在三进制（三进制）表示法中可行?

3. In the proof of Cantor's theorem we construct a set S that cannot be in the image of a presumed bijection from A to $\mathcal{P}(A)$. Suppose $A = \{1, 2, 3\}$ and f determines the following correspondences: $1 \longleftrightarrow \emptyset$, $2 \longleftrightarrow \{1, 3\}$ and $3 \longleftrightarrow \{1, 2, 3\}$. What is S?

   在康托尔定理的证明中，我们构造了一个集合 S，该集合不能位于一个假定的从 A 到 $\mathcal{P}(A)$ 的双射的像中。假设 $A = \{1, 2, 3\}$ 并且 f 确定了以下对应关系：$1 \longleftrightarrow \emptyset$，$2 \longleftrightarrow \{1, 3\}$ 和 $3 \longleftrightarrow \{1, 2, 3\}$。那么 S 是什么?

4. An argument very similar to the one embodied in the proof of Cantor's theorem is found in the Barber's paradox. This paradox was originally introduced in the popular press in order to give laypeople an understanding of Cantor's theorem and Russell's paradox. It sounds somewhat sexist to modern ears. (For example, it is presumed without comment that the Barber is male.)

一个与康托尔定理证明中所体现的论证非常相似的论证可以在理发师悖论中找到。这个悖论最初是在大众媒体中引入的，目的是让外行了解康托尔定理和罗素悖论。对于现代人来说，这听起来有些性别歧视。(例如，它不加评论地假定理发师是男性。)

> In a small town there is a Barber who shaves those men (and only those men) who do not shave themselves. Who shaves the Barber?
>
> 在一个小镇上，有一位理发师，他给那些不自己刮胡子的男人（且仅给那些男人）刮胡子。谁给这位理发师刮胡子?

Explain the similarity to the proof of Cantor's theorem.

解释其与康托尔定理证明的相似之处。

5. Cantor's theorem, applied to the set of all sets leads to an interesting paradox. The power set of the set of all sets is a collection of sets, so it must be contained in the set of all sets. Discuss the paradox and determine a way of resolving it.

   康托尔定理应用于所有集合的集合时，会引出一个有趣的悖论。所有集合的集合的幂集是一个集合的搜集，所以它必须包含在所有集合的集合中。讨论这个悖论并确定一种解决方法。

6. Verify that the final deduction in the proof of Cantor's theorem, "$(y \in S \implies y \notin S) \wedge (y \notin S \implies y \in S)$," is truly a contradiction.

   验证康托尔定理证明中的最终推论，"$(y \in S \implies y \notin S) \wedge (y \notin S \implies y \in S)$"，确实是一个矛盾。

8.4 Dominance 支配关系

Exercises — 8.4

1. How could the clerk at the Hilbert Hotel accommodate a countable number of new guests?

   希尔伯特旅馆的店员如何容纳可数个新客人?

2. Let F be the collection of all real-valued functions defined on the real line. Find an injection from $\mathbb{R}$ to F. Do you think it is possible to find an injection going the other way? In other words, do you think that F and $\mathbb{R}$ are equivalent? Explain.

   设 F 是定义在实数线上的所有实值函数的集合。找一个从 $\mathbb{R}$ 到 F 的单射。你认为是否可能找到一个反向的单射? 换句话说，你认为 F 和 $\mathbb{R}$ 是等价的吗? 请解释。

3. Fill in the details of the proof that dominance is an ordering relation. (You may simply cite the C-B-S theorem in proving anti-symmetry.)

   填写支配关系是一个序关系的证明细节。(在证明反对称性时，你可以简单地引用 C-B-S 定理。)

4. We can inject $\mathbb{Q}$ into $\mathbb{Z}$ by sending $\pm\frac{a}{b}$ to $\pm 2^a 3^b$. Use this and another obvious injection to (in light of the C-B-S theorem) reaffirm the equivalence of these sets.

   我们可以通过将 $\pm\frac{a}{b}$ 映射到 $\pm 2^a 3^b$ 来将 $\mathbb{Q}$ 单射到 $\mathbb{Z}$ 中。使用这个单射和另一个明显的单射来（根据 C-B-S 定理）再次确认这些集合的等价性。

Chapter 9

Proof techniques IV — Magic 证明技巧 IV — “魔术”

9.1 Morley's miracle 莫雷奇迹

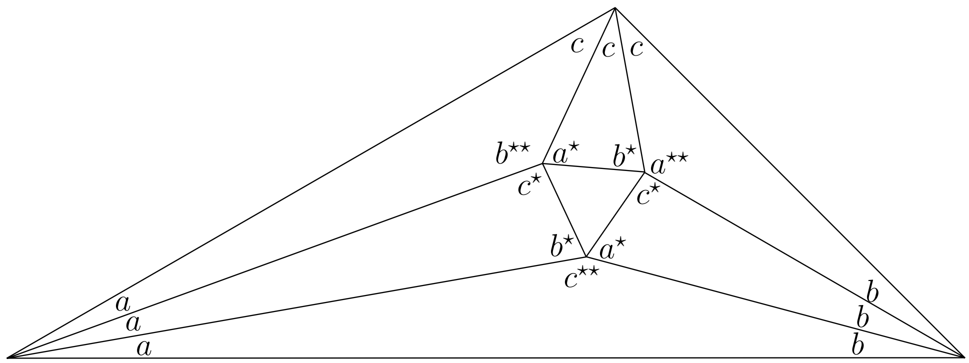
Exercises — 9.1 练习 — 9.1

1. What value should we get if we sum all of the angles that appear around one of the interior vertices in the finished diagram?

   如果我们将完成的图表中某个内部顶点周围出现的所有角度相加，应该得到什么值?

   Verify that all three have the correct sum.

   验证所有三个顶点的角度和都是正确的。



2. In this section we talked about similarity.

   Two figures in the plane are similar if it is possible to turn one into the other by a sequence of mappings: a translation, a rotation and a scaling.

   本节中，我们讨论了相似性。平面上的两个图形是相似的，如果可以通过一系列映射将一个图形变成另一个：平移、旋转和缩放。

   Geometric similarity is an equivalence relation. To fix our notation, let $T(x, y)$ represent a generic translation, $R(x, y)$ a rotation and $S(x, y)$ a scaling – thus a generic similarity is a function from $\mathbb{R}^2$ to $\mathbb{R}^2$ that can be written in the form $S(R(T(x, y)))$.

   几何相似是一种等价关系。为了确定我们的记法，让 $T(x, y)$ 代表一个通用的平移，$R(x, y)$ 代表一个旋转，$S(x, y)$ 代表一个缩放——因此一个通用的相似变换是一个从 $\mathbb{R}^2$ 到 $\mathbb{R}^2$ 的函数，可以写成 $S(R(T(x, y)))$ 的形式。

   Discuss the three properties of an equivalence relation (reflexivity, symmetry and transitivity) in terms of geometric similarity.

   请根据几何相似性，讨论等价关系的三个性质（自反性、对称性和传递性）。

9.2 Five steps into the void 深入虚空的五步

Exercises — 9.2
练习 — 9.2

1. Do the algebra (and show all your work!) to prove that invariant defined in this section actually has the value 1 for the set of all the men occupying the x-axis and the lower half-plane.

   请进行代数运算（并展示所有步骤！）来证明本节中定义的不变量对于占据 x 轴和下半平面的所有男性集合确实具有值 1。

2. “Escape of the clones” is a nice puzzle, originally proposed by Maxim Kontsevich.

   “克隆人逃脱”是一个很好的谜题，最初由马克西姆 · 康采维奇提出。
   The game is played on an infinite checkerboard restricted to the first quadrant – that is the squares may be identified with points having integer coordinates (x, y) with $x > 0$ and $y > 0$.

   游戏在一个仅限于第一象限的无限棋盘上进行——也就是说，棋盘格可以被识别为具有整数坐标 (x, y) 且 $x > 0, y > 0$ 的点。

   The “clones” are markers (checkers, coins, small rocks, whatever…) that can move in only one fashion – if the squares immediately above and to the right of a clone are empty, then it can make a “clone move.” The clone moves one space up and a copy is placed in the cell one to the right.

   “克隆人”是标记物（棋子、硬币、小石子等等……），它们只能以一种方式移动——如果一个克隆人正上方和正右方的格子是空的，那么它就可以进行一次“克隆移动”。克隆人向上移动一格，同时一个复制品被放置在右边一格的单元格中。

   We begin with three clones occupying cells $(1, 1), (2, 1)$ and $(1, 2)$ – we'll refer to those three checkerboard squares as “the prison.” The question is this: can these three clones escape the prison?

我们开始时有三个克隆人占据着单元格 $(1,1),(2,1)$ 和 $(1,2)$ ——我们将这三个棋盘格称为“监狱”。问题是：这三个克隆人能逃出监狱吗?

You must either demonstrate a sequence of moves that frees all three clones or provide an argument that the task is impossible.

你必须要么展示一个能让所有三个克隆人都获得自由的移动序列，要么提供一个论证说明这个任务是不可能的。

9.3 Monge's circle theorem 蒙日圆定理

Exercises — 9.3
练习 — 9.3

1. There is a scenario where the proof we have sketched for Monge's circle theorem doesn't really work. Can you envision it?

   几何相似是一种等价关系。在我们为蒙日圆定理勾勒的证明中，存在一种情况，该证明实际上并不成立。几何相似是一种等价关系。在我们为蒙日圆定理勾勒的证明中，存在一种情况，该证明实际上并不成立。你能想象出来吗?

   Hint: consider two relatively large spheres and one that is quite small.

   提示：考虑两个相对较大的球体和一个非常小的球体。